

JAPAN RADIO TEST REPORT

Applicant:	ZHUHAI TAIXIN SEMICONDUCTOR CO.,LTD		
Applicant add.:	Gang building 11, 3rd floor, Harbor 1 Science and Technology Park, High-tech Zone, Zhuhai, Guangdong, China		
Product Name:	Taixin AH module		
Model No.:	TX-AH-R900PNR		
Derivative model No.:	TX-AH-R900P, TX-AH-R900PTR		
Brand Name:	N/A		
Report No.:	AIT23072103TW1		
Standards:	MIC Certification Rule, Article 2 Paragraph 1 Item 8; MIC Notice No.88 Appendix No.22		
Prepared By:			
Dongguan Yaxu (AiT) Technology Limited			
Add. : No.22, Jinqianling Third Street, Jitigang, Huangjiang, Dongguan, Guangdong, China			
Date of Receipt:	July. 22, 2023	Date of Test:	July. 22, 2023~ Aug. 09, 2023
Date of Issue:	Aug. 09, 2023	Test Result:	Pass
This device has been tested and found to comply with the stated standard(s) above, which is (are) applicable only to the tested sample identified in the report. Note: This report shall not be reproduced except in full, without the written approval of Dongguan Yaxu (AiT) Technology Limited, this document may be altered or revised by Dongguan Yaxu (AiT) Technology Limited, personal only, and shall be noted in the revision of the document. This test report must not be used by the client to claim product endorsement.			
Reviewed by:	 Simba Huang		
Approved by:	  Seal Chen		

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2 Test Summary

MIC Notice No.88 Appendix No.22

Report Clause	Ref. Std. Article	Test Item	Judgment	Remark
1.1.1	RLE:6	Modulation Method	PASS	--
1.1.1	RLE:6	Frequency Band	PASS	--
3.1	ORE:5	Frequency Error	PASS	--
3.2	ORE:6	Occupied Bandwidth	PASS	--
3.3	ORE:6	Antenna Power	PASS	--
3.3	ORE:14	Antenna Power Error	PASS	--
1.1.2	ORE:49.14	Antenna Gain	PASS	--
3.4	ORE:49.14	Adjacent Channel Power	PASS	--
3.5	ORE:7, Table 3	Transmitter Spurious Emissions	PASS	--
3.6	ORE:24	Receiver Spurious Emissions	PASS	--
3.7	ORE:49.14	Transmission Time	PASS	--
3.8	ORE:49.14	Carrier Sense	PASS	--
3.9	-	Interference Prevention Function	PASS	--
3.10	ORE:49.14	EUT Construction Protection	PASS	--

Remark:

EUT: In this whole report EUT means Equipment Under Test.

Tx: In this whole report Tx (or tx) means Transmitter.

Rx: In this whole report Rx (or rx) means Receiver.

RF: In this whole report RF means Radio Frequency.

2.1 Measurement Uncertainty

The report uncertainty of measurement $y \pm U$, where expended uncertainty U is based on a standard uncertainty Multiplied by a coverage factor of $k=2$, providing a level of confidence of approximately 95%.

No.	Item	Uncertainty
1	Frequency Error / 99% & 90% Bandwidth	$\pm 0.85 \times 10^{-7}$
2	Antenna Power	± 0.70 dB
3	Spurious Emissions	± 0.80 dB
4	DC / AC Power Source	$\pm 1.4\%$

3 Test Facility

The test facility is recognized, certified or accredited by the following organizations:

.CNAS- Registration No: L6177

Dongguan Yaxu (AiT) technology Limited is accredited to ISO/IEC 17025:2017 general Requirements for the competence of testing and calibration laboratories (CNAS-CL01 Accreditation Criteria for the competence of testing and calibration laboratories) on Aug.04, 2020

FCC-Registration No.: 703111 Designation Number: CN1313

Dongguan Yaxu (AiT) technology Limited has been registered and fully described in a report filed with the (FCC) Federal Communications Commission. The acceptance letter from the FCC is maintained in our files.

IC —Registration No.: 6819A CAB identifier: CN0122

The 3m Semi-anechoic chamber of Dongguan Yaxu (AiT) technology Limited has been registered by Certification and Engineering Bureau of Industry Canada for radio equipment testing with Registration No.: 6819A

A2LA-Lab Cert. No.: 6317.01

Dongguan Yaxu (AiT) technology Limited has been accredited by A2LA for technical competence in the field of electrical testing, and proved to be in compliance with ISO/IEC 17025: 2017 General Requirements for the Competence of Testing and Calibration Laboratories and any additional program requirements in the identified field of testing.

3.1 Deviation from standard

None

3.2 Abnormalities from standard conditions

None

3.3 Test Location

Dongguan Yaxu (AiT) Technology Limited

Address: No.22, Jinqianling 3rd Street, Jitigang, Huangjiang,Dongguan, Guangdong, China

Tel.: +86-769-8202 0499

Fax.: +86-769-8202 0495

4 General Information

Manufacturer:	ZHUHAI TAIXIN SEMICONDUCTOR CO.,LTD
Manufacturer Address:	Gang building 11, 3rd floor, Harbor 1 Science and Technology Park, High-tech Zone, Zhuhai, Guangdong,China
Product Name:	Taixin AH module
Model No.:	TX-AH-R900PNR
Brand Name:	N/A
Derivative model No.:	TX-AH-R900P, TX-AH-R900PTR
Operating Frequency	920.5-928.1MHz
Type of Modulation:	OFDM
Number of Channels	7 Channels
Channel Separation:	/
Antenna Type	FPC antenna
Antenna gain:	0 dBi
Normal antenna power:	20 mW
Power Supply Range:	DC 5V
Normal Test Voltage:	The same as above
Hard Ware Version:	N/A
Soft Ware Version:	N/A
Model difference:	The model name is different, everything else is the same

4.1 Test Location

All tests were performed at:

Dongguan Yaxu (AiT) Technology Limited
No.22, Jinqianling Third Street, Jitigang, Huangjiang, Dongguan, Guangdong, China.

Tel.: +86.769.82020499 Fax.: +86.769.82020495

4.2 EUT Peripheral List

No.	Equipment	Manufacturer	Model No.	Serial No.	Power cord	Signal cord
1	N/A	N/A	N/A	N/A	N/A	N/A

4.3 Test Peripheral List

No.	Equipment	Manufacturer	EMC Compliance	Model No.	Serial No.	Power cord	Signal cord
1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

5 Equipment Used during Test

No	Test Equipment	Manufacturer	Model No	Serial No	Cal. Date	Cal. Due Date	Cal. Lab
1	SIGNAL Analyzer	R&S	FSV40	101470	2022.09.02	2023.09.01	Guangzhou Lisai
2	EMI Measuring Receiver	R&S	ESR	101660	2022.09.02	2023.09.01	Guangzhou Lisai
3	Mobile phone	Samsung	GALAXY S4	R33D20 SQYNW	N/A	N/A	N/A
4	DC Power supply	Manson	HCS-3604	G521100 129	2022.09.02	2023.09.01	Guangzhou Lisai
5	Digital Phosphor Oscilloscope	Tektronix	TDS3012	B021220	2022.09.02	2023.09.01	Guangzhou Lisai
6	Signal Generator	Agilent	N5182A	MY50143 009	2022.09.02	2023.09.01	Guangzhou Lisai

5.1 Antenna Power

5.1.1 Limits of Antenna Power

Limit	20 mW
Tolerance	+20%, -80%

5.1.2 TEST SETUP LAYOUT



5.1.3 Test procedure

- (1) Connected the EUT's antenna port to the Power Meter.
- (2) The cable loss and attenuator loss have been put into the result.

5.1.4 Environmental Conditions

Temperature:	22 °C
Relative Humidity:	64%
ATM Pressure:	1010 mbar

5.1.5 TEST RESULT

Test Voltage (V)	Mode	Frequency (MHz)	Power (dBm)	Antenna Power (mW)	Declared Power (mW)	Tolerance (%)	Upper Limit (%)	Lower Limit (%)	Verdict
NV	SRD 1 MHz	922	8.81	7.603	8	-5.222	20	-80	Pass
NV	SRD 1 MHz	924	8.85	7.674	8	-4.248	20	-80	Pass
NV	SRD 1 MHz	926	8.97	7.889	8	-1.407	20	-80	Pass
NV	SRD 2 MHz	922.1	7.98	6.281	7	-11.45	20	-80	Pass
NV	SRD 2MHz	924.3	8.03	6.353	7	-10.185	20	-80	Pass
NV	SRD 2 MHz	926.5	8.06	6.397	7	-9.426	20	-80	Pass
NV	SRD 4 MHz	924.5	6.23	4.198	4.5	-7.194	20	-80	Pass
LV	SRD 1 MHz	922	8.73	7.46	8	-7.24	20	-80	Pass
LV	SRD 1 MHz	924	8.83	7.64	8	-4.712	20	-80	Pass
LV	SRD 1 MHz	926	8.76	7.52	8	-6.38	20	-80	Pass
LV	SRD 2 MHz	922.1	7.89	6.15	7	-13.82	20	-80	Pass
LV	SRD 2MHz	924.3	7.82	6.05	7	-15.70	20	-80	Pass
LV	SRD 2 MHz	926.5	7.85	6.10	7	-14.75	20	-80	Pass
LV	SRD 4 MHz	924.5	5.41	3.48	4.5	-29.31	20	-80	Pass
HV	SRD 1 MHz	922	8.54	7.14	8	-12.045	20	-80	Pass
HV	SRD 1 MHz	924	8.56	7.18	8	-11.421	20	-80	Pass
HV	SRD 1 MHz	926	8.63	7.29	8	-9.739	20	-80	Pass
HV	SRD 2 MHz	922.1	7.82	6.05	7	-15.702	20	-80	Pass
HV	SRD 2MHz	924.3	7.83	6.07	7	-15.321	20	-80	Pass
HV	SRD 2 MHz	926.5	7.81	6.04	7	-15.894	20	-80	Pass
HV	SRD 4 MHz	924.5	6.03	4.00	4.5	-12.500	20	-80	Pass

6 Test Results

6.1 E.U.T. Test Conditions

Power supply:	DC 5V.
	The measurement result of the voltage fluctuation at RF circuit when DC 5V +/- 10%
	DC 5V
	5V
	5.5V
	4.5V
	Pre-test the EUT in all voltage mode at the DC 5.5V, DC 5V and DC4.5V and conducted to determine the worst-case mode, only the worst-case results (DC 5V) are recorded in this report.
Temperature:	5.0 -35.0 °C
Humidity:	45-85 % RH
Atmospheric Pressure:	1000 -1010 mbar
Test frequencies:	If the EUT can be set to 3 of more different (carrier) frequencies in 1 allocated band, testing shall be performed using the Lowest, Middle and the Highest frequency (L, M and H). If there are 2 or fewer frequencies, testing shall be performed with the available frequencies.
Parameters of test software setting	During testing channel & power controlling software provided by the manufacturer was used to control the operating channel as well as the output power level.

EUT Channels and frequencies list:

Mode	Power Level Setting
920.5-928.1 MHz SRD 1 MHz Nss1,(3Mbps)1TX	-
922MHz	9
924MHz	9
926MHz	9
920.5-928.1MHz SRD 2MHz Nss1,(6.5Mbps)1TX	-
922.1MHz	8
924.3MHz	8
926.5MHz	8
920.5-928.1 MHz_SRD_4MHz_Nss1 ,(13.5Mbps)_1TX	-
924.5MHz	7

RF General Information					
Frequency Range (MHz)	Modulation	Ch. Frequency (MHz)	Channel Number	n	Antenna Power (mW)
920.5-928.1	OFDM	921.0-927.6	40-46 [7]	5	
920.5-928.1	OFDM	921.5-927.1	36-39 [4]	10	
920.5-928.1	OFDM	922.5-926.1	47-48 [2]	20	
“n” is number of units channel simultaneously used.					

Mode Test Duty Cycle

Condition	Mode	Frequency (MHz)	Duty Cycle (%)	Correction Factor (dB)	1/T (kHz)
NVNT	SRD 1 MHz	922	100	0	0
NVNT	SRD 1 MHz	924	100	0	0
NVNT	SRD 1 MHz	926	100	0	0
NVNT	SRD 2 MHz	922.1	100	0	0
NVNT	SRD 2 MHz	924.3	100	0	0
NVNT	SRD 2 MHz	926.5	100	0	0
NVNT	SRD 4 MHz	924.5	100	0	0

6.2 Antenna Requirement

Standard requirement

Applicable for equipment with an antenna terminal, including testing terminals) If an antenna connector is available, all relevant tests will be carried out conducted. If not, tests will be carried out in an anechoic room or with a suitable test-fixture.

EUT Antenna

Module certification, no fixed antenna, the final antenna is developed by the OEM.

Result: All relevant tests will be carried out conducted.

6.3 Tolerance Of Occupied Bandwidth

6.3.1 Limits of Occupied Bandwidth Measurement

It shall be $(200 \times n)$ kHz or less. (n is a number of unit radio channels constituting the radio channel and is an integer from 1 to 5.)

6.3.2 Test Procedure

(1) Connected the EUT's antenna port to the Spectrum Analyzer by suitable attenuator, set the Spectrum Analyzer as below:

Centre Frequency: The centre frequency of the channel under test.

Span: 3000KHz

Resolution BW: 300 kHz

Video BW: 300 kHz

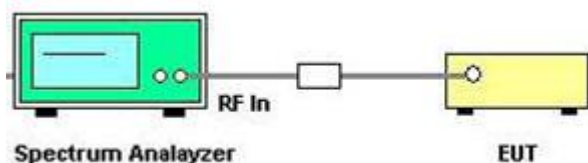
Sweep Time: Auto

Detector mode: Positive peak

Indication mode: Max hold

(2) When the trace is complete, measure the occupied bandwidth (99% bandwidth) with spectrum analyzer's bandwidth measure function.

6.3.3 TEST SETUP LAYOUT



6.3.4 Environmental Conditions

Temperature:	22 °C
Relative Humidity:	64%
ATM Pressure:	1010 mbar

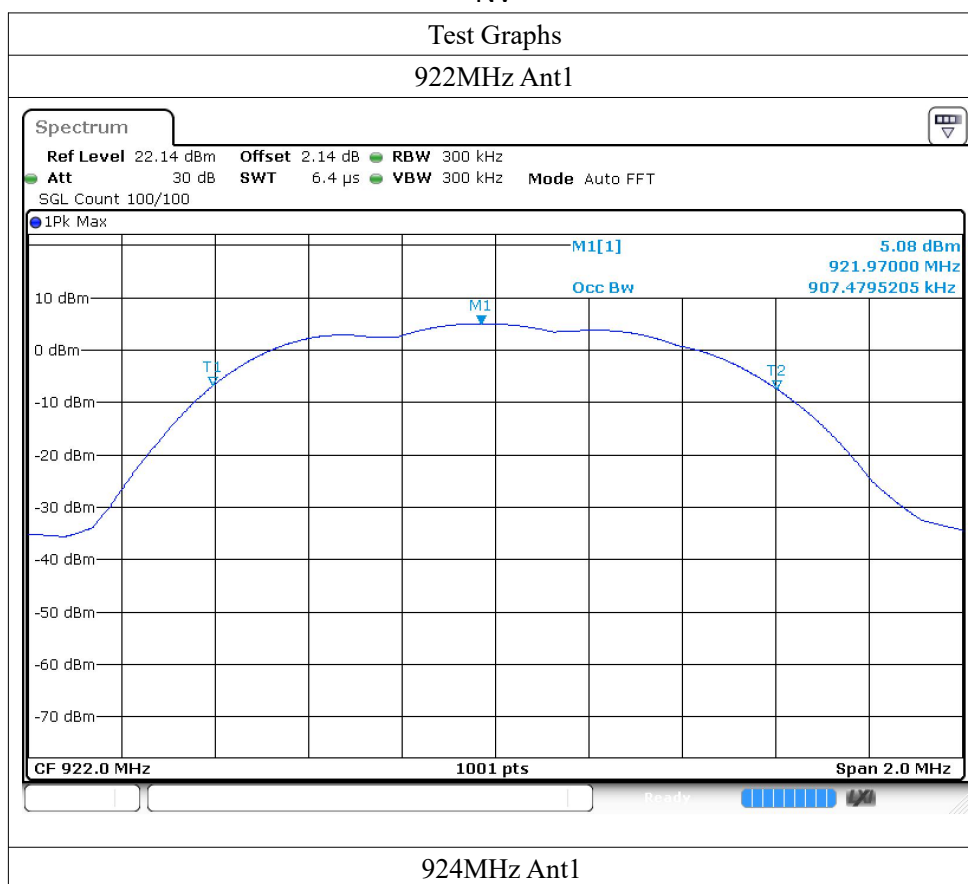
6.3.5 TEST RESULT

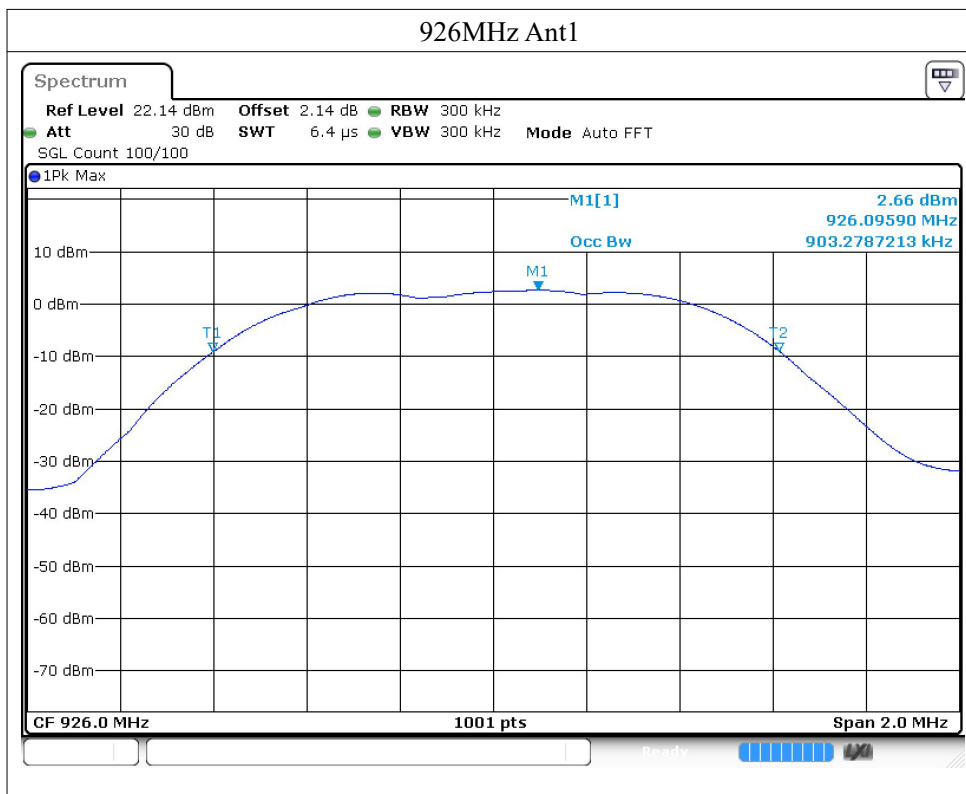
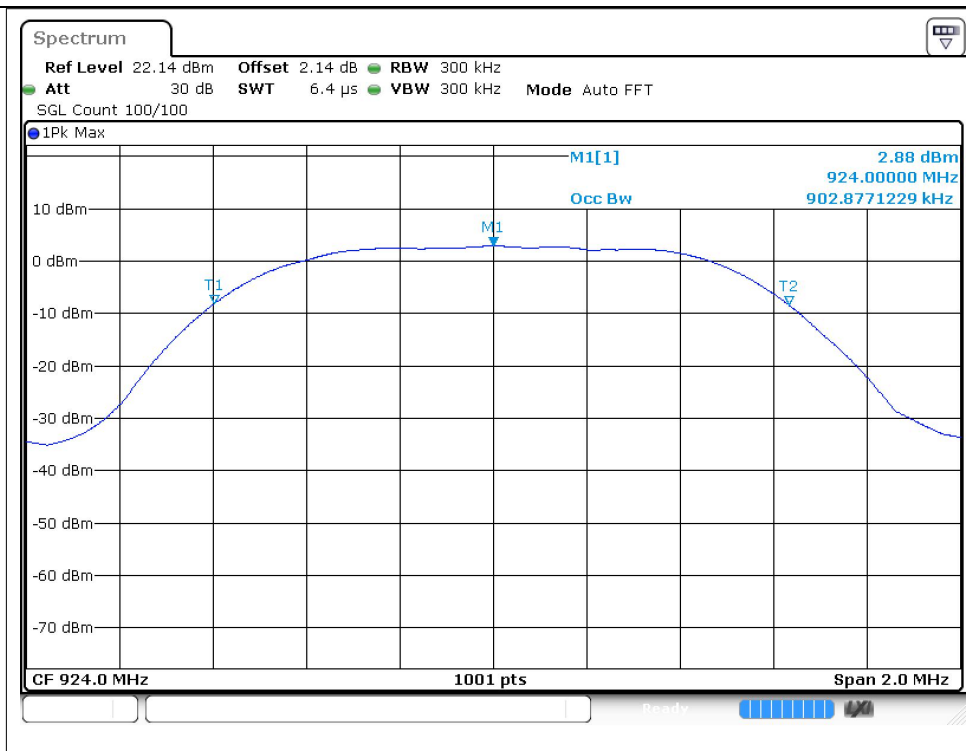
Note: The report shows the worst test diagram for NV testing.

SRD 1 MHz Nss1,(3Mbps)1TX

Test Voltage (V)	Test Frequency MHz	99% Bandwidth kHz	Limit kHz	Result
NV	922	907.48	200*5	Pass
NV	924	902.88	200*5	Pass
NV	926	903.28	200*5	Pass
LV	922	907.35	200*5	Pass
LV	924	902.28	200*5	Pass
LV	926	903.23	200*5	Pass
HV	922	907.39	200*5	Pass
HV	924	902.66	200*5	Pass
HV	926	903.25	200*5	Pass

NV

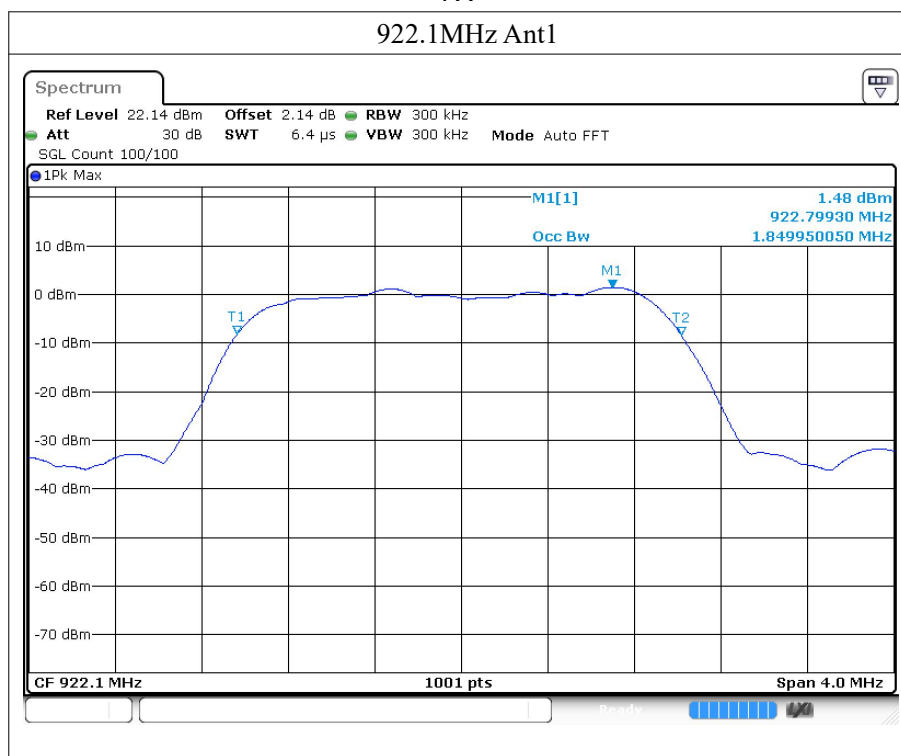


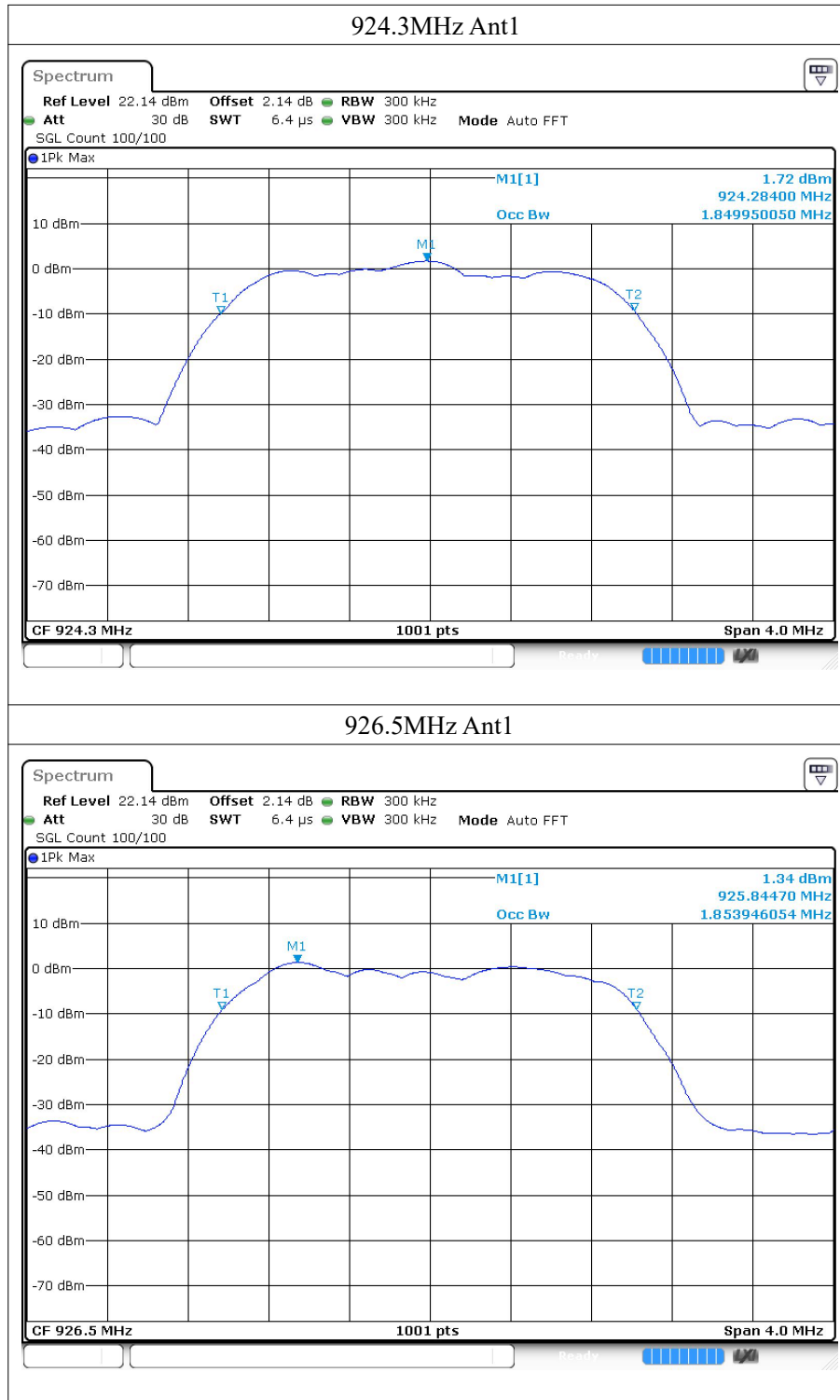


SRD 2MHz Nss1,(6.5Mbps)1TX

Test Voltage (V)	Test Frequency MHz	99% Bandwidth MHz	Limit kHz	Result
NV	922.1	1.85	200*10	Pass
NV	924.3	1.85	200*10	Pass
NV	926.5	1.854	200*10	Pass
LV	922.1	1.83	200*10	Pass
LV	924.3	1.84	200*10	Pass
LV	926.5	1.852	200*10	Pass
HV	922.1	1.84	200*10	Pass
HV	924.3	1.84	200*10	Pass
HV	926.5	1.853	200*10	Pass

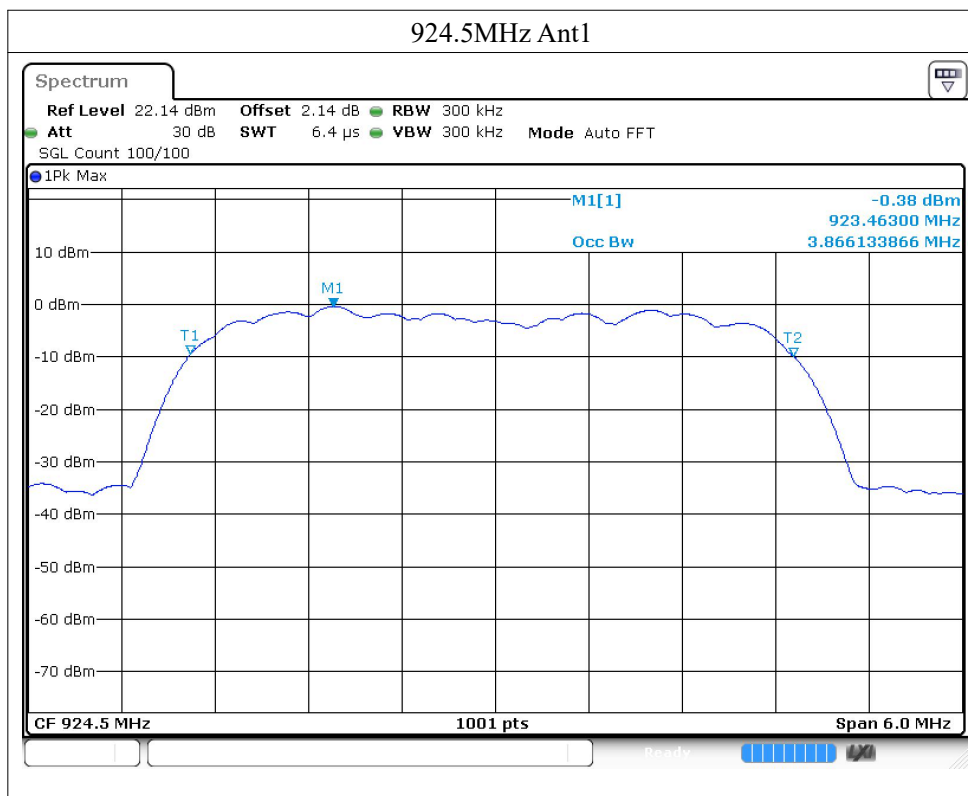
NV





Test Voltage (V)	Test Frequency MHz	99% Bandwidth MHz	Limit kHz	Result
NV	924.5	3.866	200*20	Pass
LV	924.5	3.863	200*20	Pass
HV	924.5	3.861	200*20	Pass

NV



6.4 . Spurious Emissions for Transmitter Measurement

6.4.1 Limit

Frequencies (MHz)	Limit (dBm)	RBW
$f \leq 710 \text{ MHz}$	-36	100kHz
$710 \text{ MHz} < f \leq 900 \text{ MHz}$	-55	1MHz
$900 \text{ MHz} < f \leq 915 \text{ MHz}$	-55	100kHz
$915 \text{ MHz} < f \leq 930 \text{ MHz}^*$ (Except for $ f-f_c \leq (200+100 \times n) \text{ kHz}$ if bandwidth of unit radio channel is 200 kHz, except for $ f-f_c \leq (100+50 \times n) \text{ kHz}$ if bandwidth of unit radio channel is 100 kHz. Except for $ f-f_c \leq (100+100 \times n) \text{ kHz}$ If frequency band is $915.9 \text{ MHz} \leq f \leq 916.9 \text{ MHz}$ and $920.5 \text{ MHz} \leq f \leq 922.3 \text{ MHz}$. Where n is a number of unit radio channels constituting the radio channel and is an integer from 1 to 5)	-36	100kHz
$930 \text{ MHz} < f \leq 1000 \text{ MHz}$	-55	100kHz
$1000 \text{ MHz} < f \leq 1,215 \text{ MHz}$	-45	1MHz
$1,215 \text{ MHz} < f$	-30	1MHz

6.4.2 Test Procedure

(1) Connected the EUT's antenna port to the Spectrum Analyzer by suitable attenuator, set the Spectrum Analyzer as below:

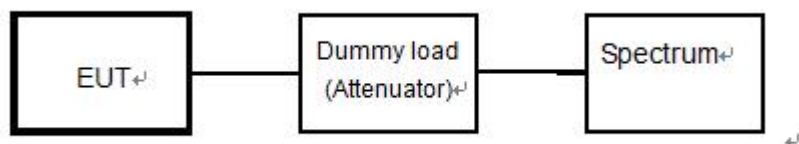
Detector: Peak.

Trace Mode: Max Hold.

(2) When the trace is complete, search the frequency of peak power

(3) All the emissions from 30 MHz to 5 GHz were measured and record.

6.4.3 TEST SETUP LAYOUT



6.4.4 Environmental Conditions

Temperature:	22 °C
Relative Humidity:	64%
ATM Pressure:	1010mbar

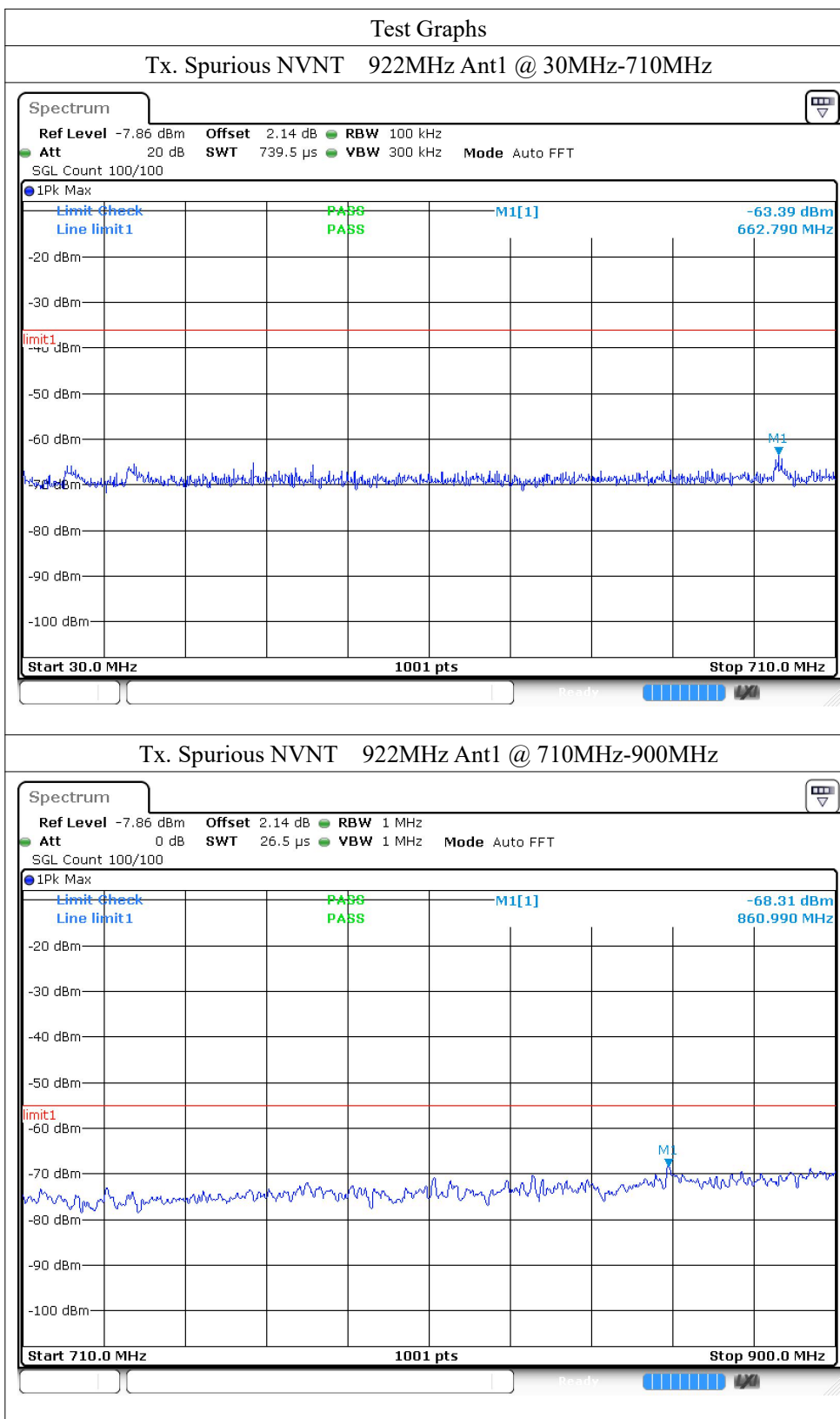
6.4.5 TEST RESULT

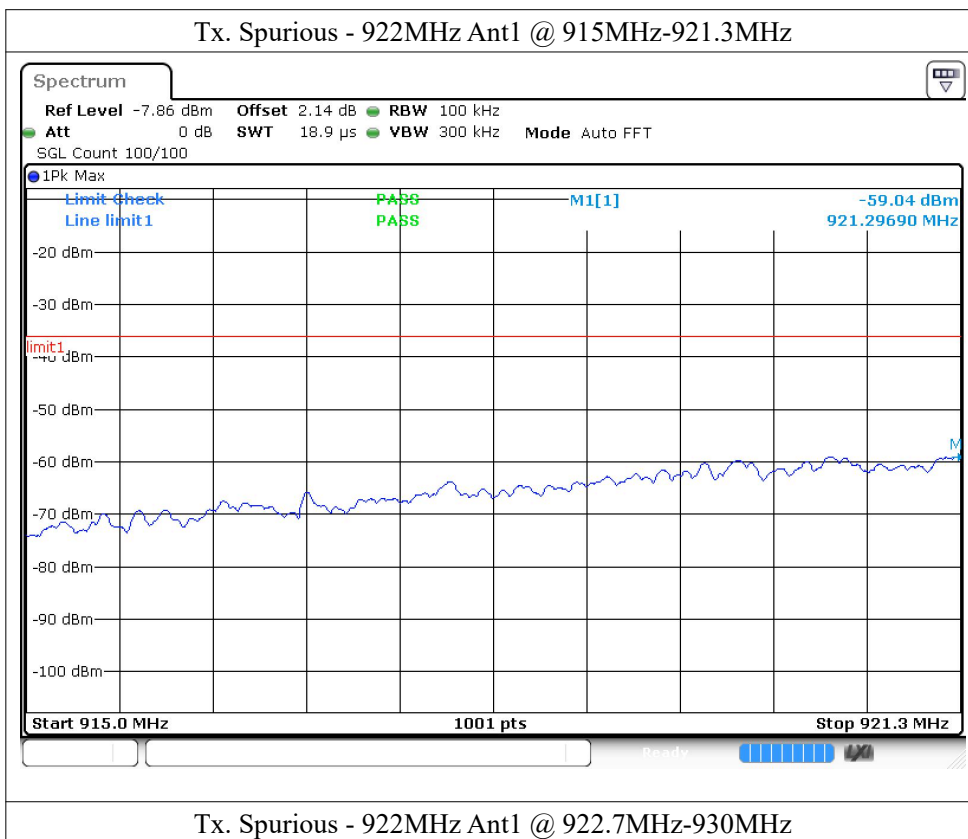
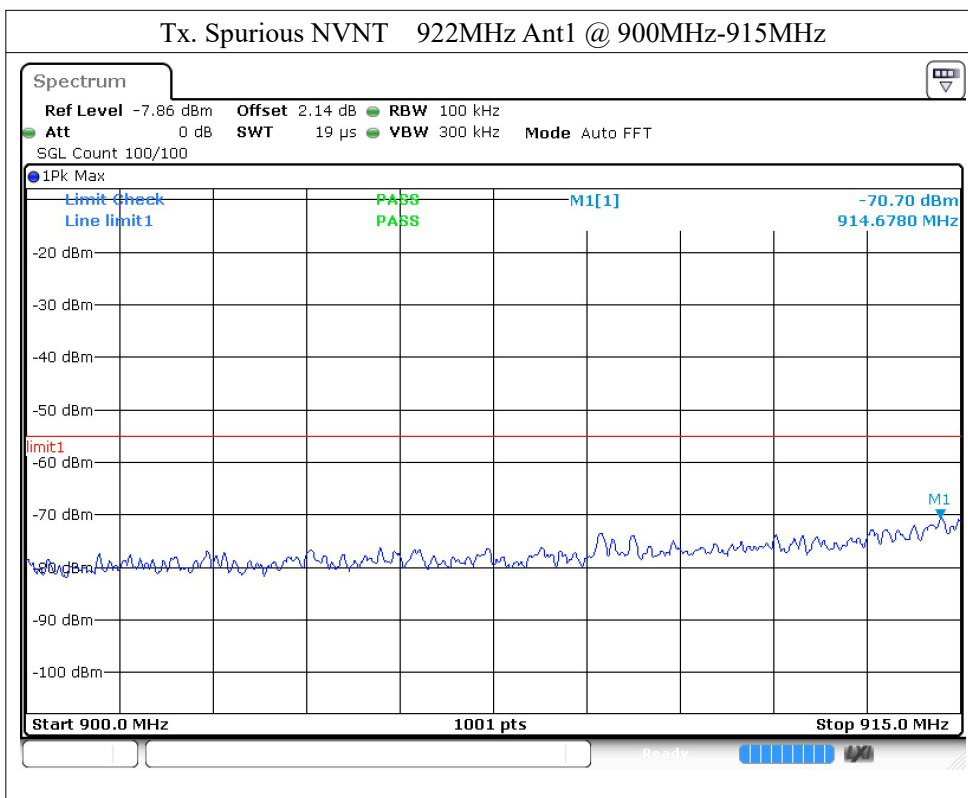
Note: The report shows the worst test diagram for NV testing .

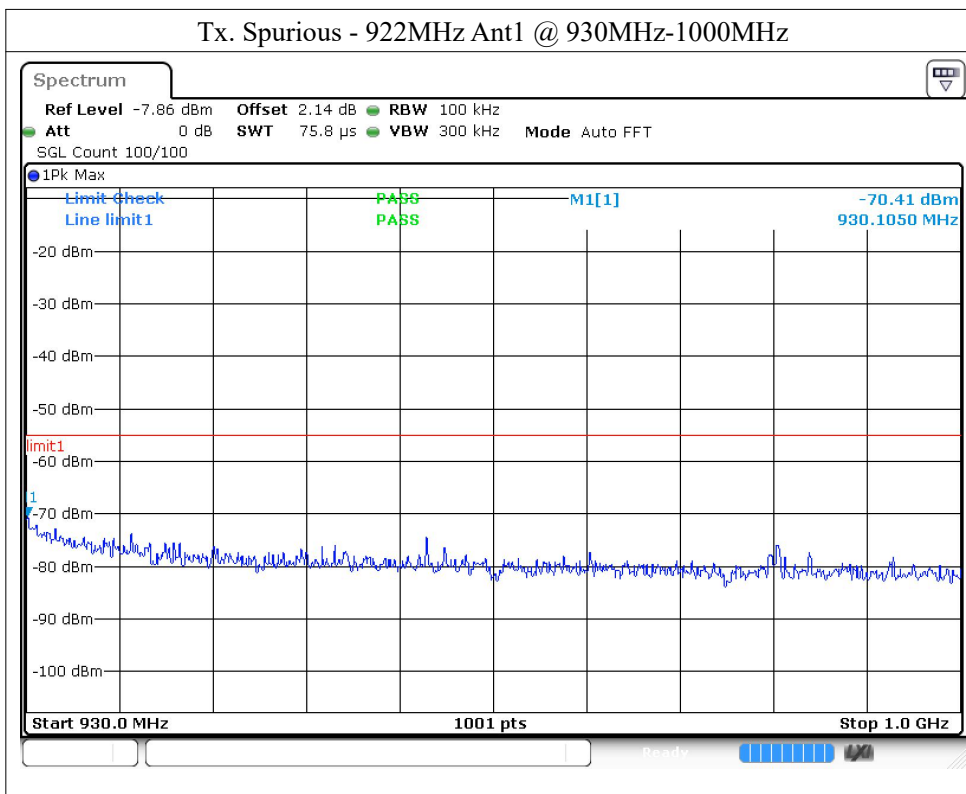
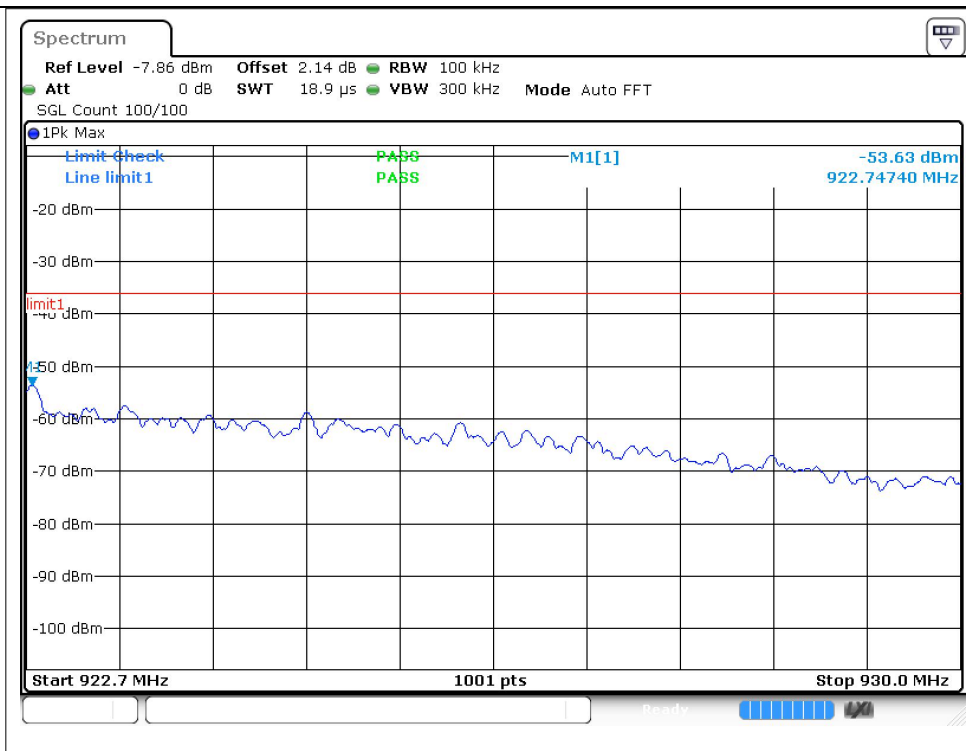
SRD 1 MHz Nss1,(3Mbps)1TX

Condition	Frequency (MHz)	Range (MHz)	Emission Frequency (MHz)	Max Value (dBm)	Limit Start(dBm)	Limit Stop (dBm)	Verdict
NVNT	922	30-710	662.79	-63.39	-36	-36	Pass
NVNT	922	710-900	860.99	-68.31	-55	-55	Pass
NVNT	922	900-915	914.68	-60.7	-55	-55	Pass
NVNT	922	915-921.3	921.3	-49.04	-36	-36	Pass
NVNT	922	922.7-930	922.75	-43.63	-36	-36	Pass
NVNT	922	930-1000	930.1	-60.41	-55	-55	Pass
NVNT	922	1000-1215	1020.5	-50.54	-45	-45	Pass
NVNT	922	1215-5000	4945.2	-52.85	-30	-30	Pass
NVNT	924	30-710	659.39	-64.50	-36	-36	Pass
NVNT	924	710-900	895.73	-68.25	-55	-55	Pass
NVNT	924	900-915	914.35	-63.99	-55	-55	Pass
NVNT	924	915-923.3	923.22	-48.37	-36	-36	Pass
NVNT	924	924.7-930	925.74	-46.3	-36	-36	Pass
NVNT	924	930-1000	930.25	-54.05	-55	-55	Pass
NVNT	924	1000-1215	1187.19	-57.41	-45	-45	Pass
NVNT	924	1215-5000	4188.9	-52.83	-30	-30	Pass
NVNT	926	30-710	658.71	-64.81	-36	-36	Pass
NVNT	926	710-900	899.15	-67.78	-55	-55	Pass
NVNT	926	900-915	914.96	-64.13	-55	-55	Pass
NVNT	926	915-925.3	925.1	-47.82	-36	-36	Pass
NVNT	926	926.7-930	926.84	-47.13	-36	-36	Pass
NVNT	926	930-1000	930.39	-52.29	-55	-55	Pass
NVNT	926	1000-1215	1159.69	-58.12	-45	-45	Pass
NVNT	926	1215-5000	4302.4	-52.19	-30	-30	Pass
LVNT	922	30-710	662.69	-63.98	-36	-36	Pass
LVNT	922	710-900	860.06	-68.43	-55	-55	Pass
LVNT	922	900-915	913.72	-61.66	-55	-55	Pass
LVNT	922	915-921.3	920.75	-49.79	-36	-36	Pass
LVNT	922	922.7-930	922.10	-43.82	-36	-36	Pass
LVNT	922	930-1000	929.29	-61.16	-55	-55	Pass
LVNT	922	1000-1215	1020.07	-50.90	-45	-45	Pass
LVNT	922	1215-5000	4944.53	-53.20	-30	-30	Pass
LVNT	924	30-710	659.29	-64.95	-36	-36	Pass
LVNT	924	710-900	894.84	-68.54	-55	-55	Pass
LVNT	924	900-915	913.97	-64.73	-55	-55	Pass
LVNT	924	915-923.3	922.92	-48.45	-36	-36	Pass
LVNT	924	924.7-930	924.90	-46.67	-36	-36	Pass
LVNT	924	930-1000	930.10	-54.94	-55	-55	Pass
LVNT	924	1000-1215	1187.18	-58.18	-45	-45	Pass

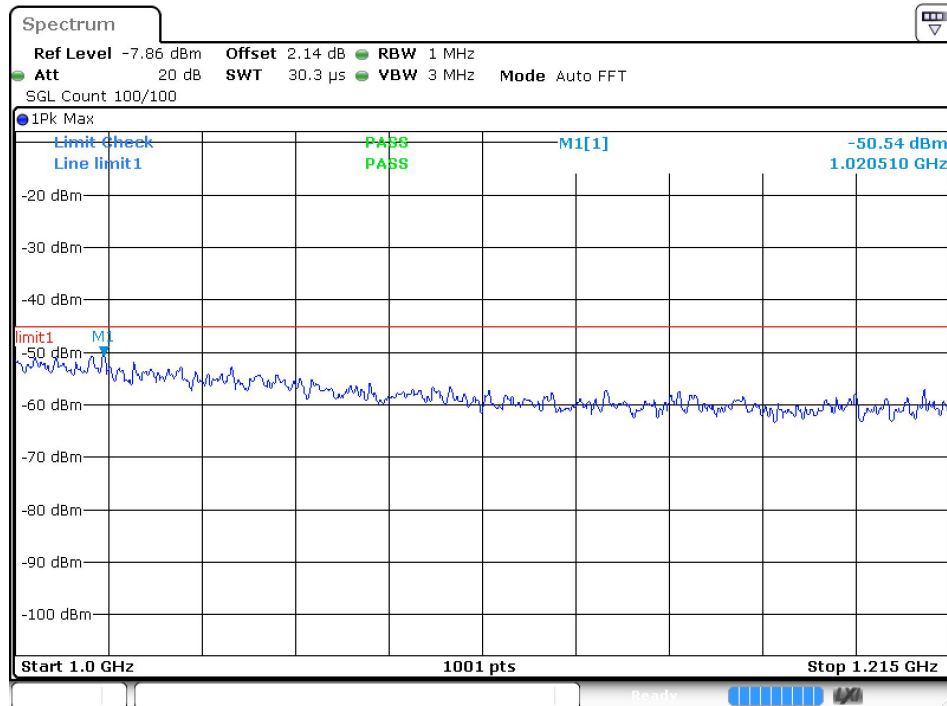
LVNT	924	1215-5000	4188.41	-53.79	-30	-30	Pass
LVNT	926	30-710	658.20	-64.86	-36	-36	Pass
LVNT	926	710-900	898.70	-68.50	-55	-55	Pass
LVNT	926	900-915	914.59	-65.03	-55	-55	Pass
LVNT	926	915-925.3	924.14	-48.53	-36	-36	Pass
LVNT	926	926.7-930	926.69	-47.23	-36	-36	Pass
LVNT	926	930-1000	929.70	-52.43	-55	-55	Pass
LVNT	926	1000-1215	1159.18	-58.39	-45	-45	Pass
LVNT	926	1215-5000	4302.18	-52.92	-30	-30	Pass
HVNT	922	30-710	662.63	-63.71	-36	-36	Pass
HVNT	922	710-900	860.60	-69.01	-55	-55	Pass
HVNT	922	900-915	914.11	-60.91	-55	-55	Pass
HVNT	922	915-921.3	920.65	-49.80	-36	-36	Pass
HVNT	922	922.7-930	921.78	-43.87	-36	-36	Pass
HVNT	922	930-1000	929.75	-60.71	-55	-55	Pass
HVNT	922	1000-1215	1019.79	-51.54	-45	-45	Pass
HVNT	922	1215-5000	4944.47	-53.46	-30	-30	Pass
HVNT	924	30-710	658.73	-64.96	-36	-36	Pass
HVNT	924	710-900	894.91	-69.00	-55	-55	Pass
HVNT	924	900-915	913.55	-64.69	-55	-55	Pass
HVNT	924	915-923.3	922.37	-48.37	-36	-36	Pass
HVNT	924	924.7-930	925.65	-46.53	-36	-36	Pass
HVNT	924	930-1000	929.78	-54.87	-55	-55	Pass
HVNT	924	1000-1215	1186.88	-57.66	-45	-45	Pass
HVNT	924	1215-5000	4188.30	-53.47	-30	-30	Pass
HVNT	926	30-710	658.59	-64.92	-36	-36	Pass
HVNT	926	710-900	898.71	-67.86	-55	-55	Pass
HVNT	926	900-915	914.83	-64.91	-55	-55	Pass
HVNT	926	915-925.3	924.33	-47.91	-36	-36	Pass
HVNT	926	926.7-930	926.16	-47.65	-36	-36	Pass
HVNT	926	930-1000	929.81	-52.96	-55	-55	Pass
HVNT	926	1000-1215	1158.77	-58.54	-45	-45	Pass
HVNT	926	1215-5000	4302.12	-52.91	-30	-30	Pass



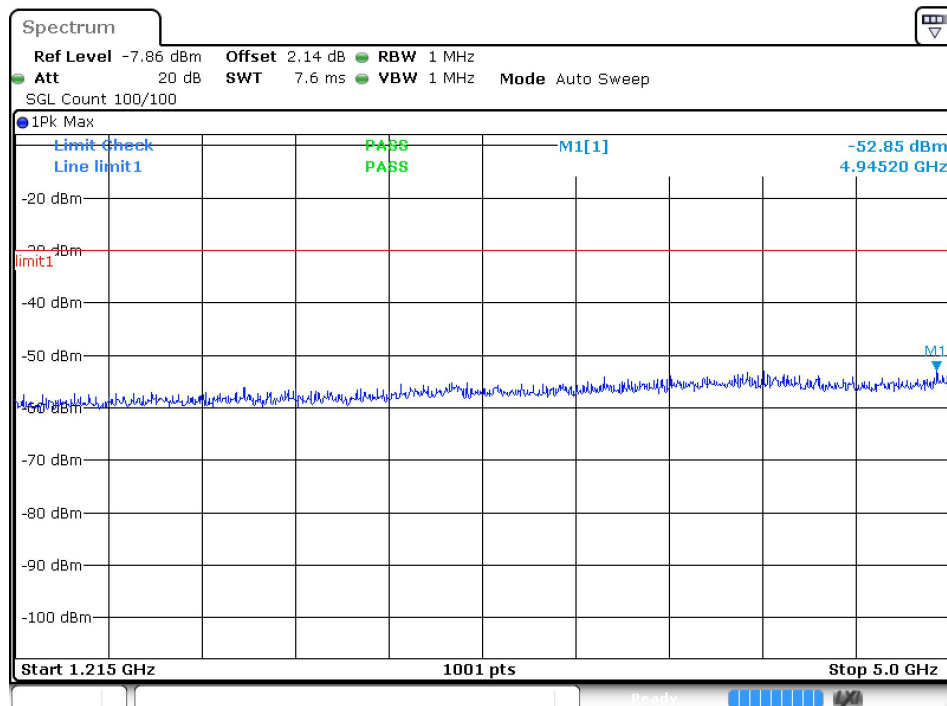




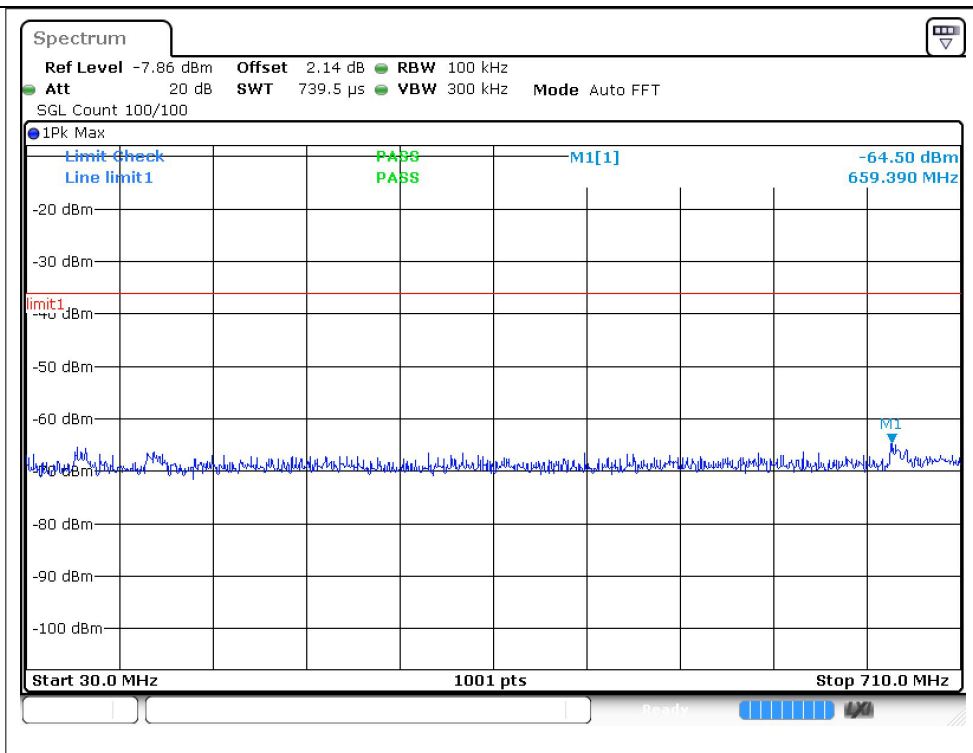
Tx. Spurious - 922MHz Ant1 @ 1000MHz-1215MHz

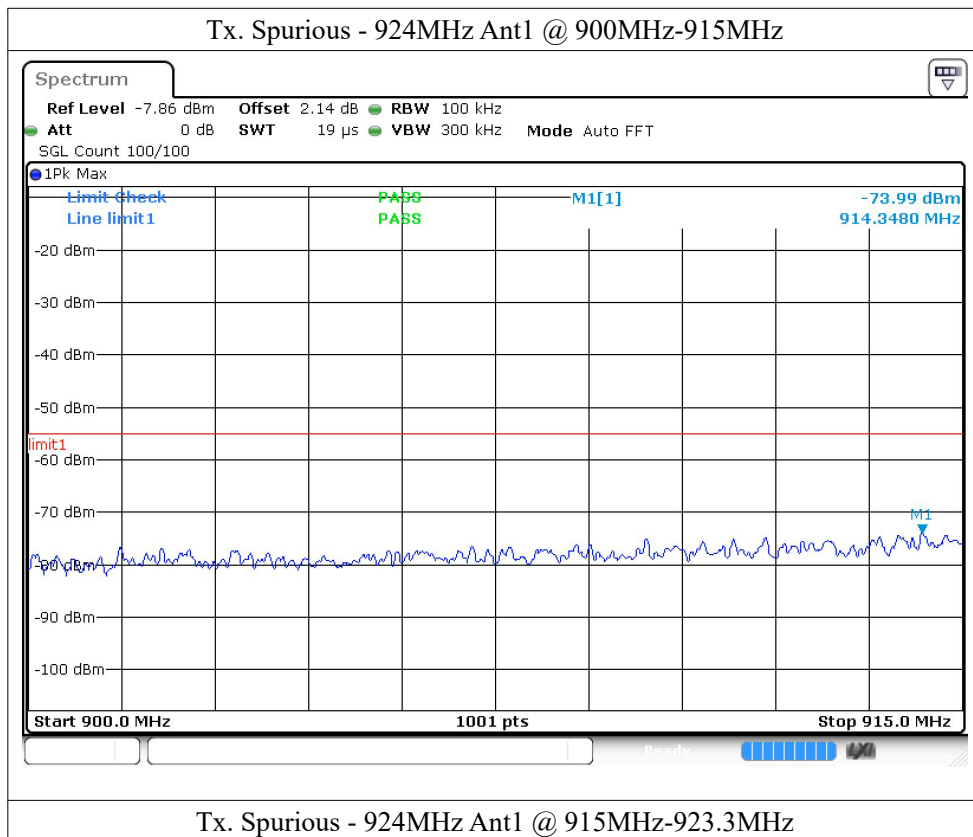
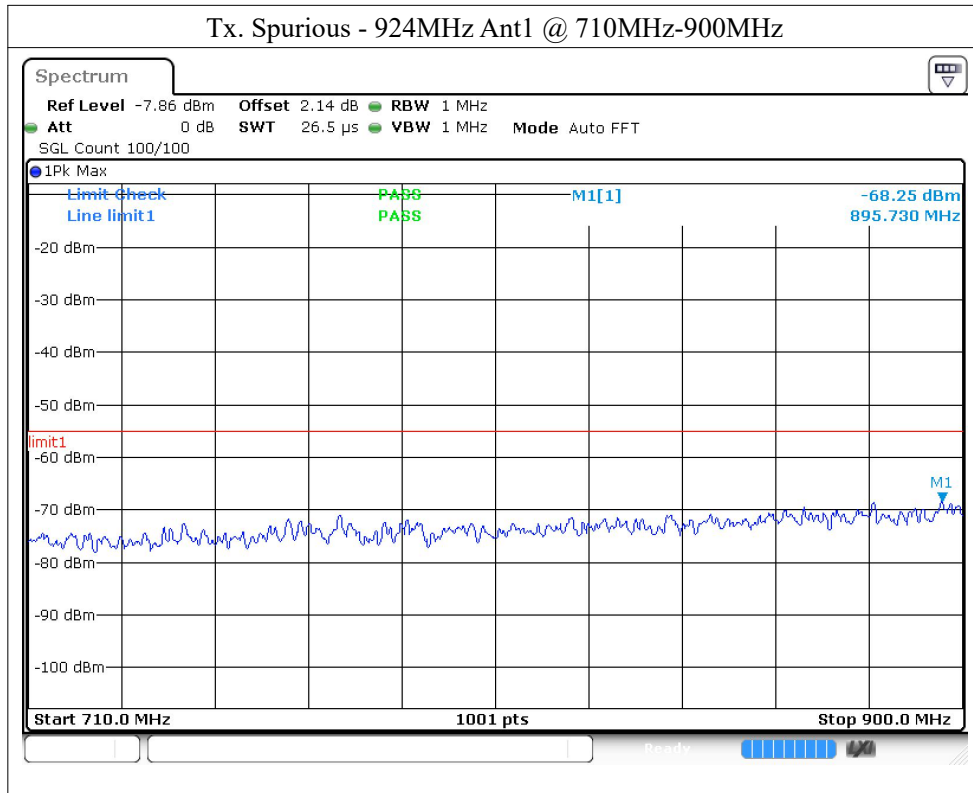


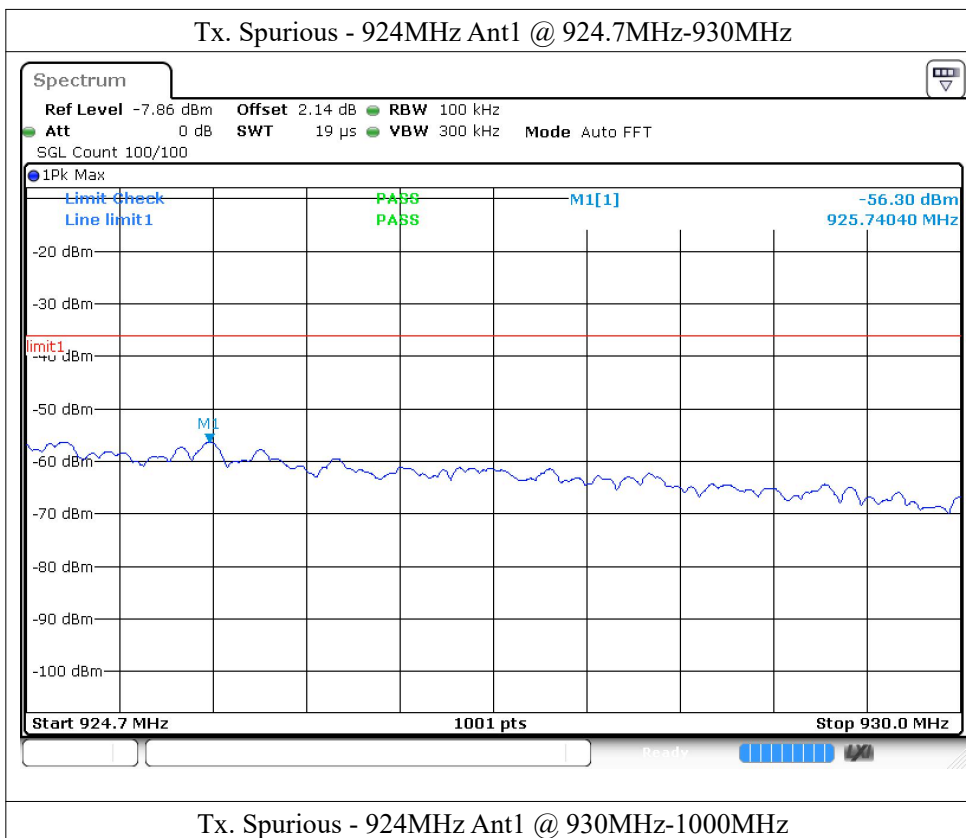
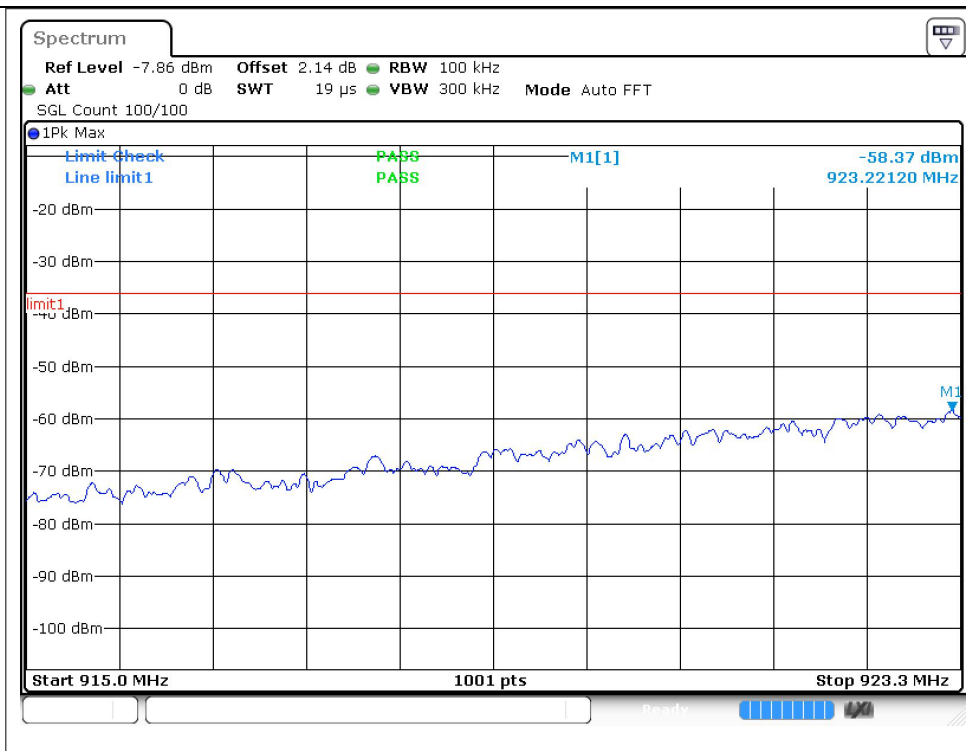
Tx. Spurious NVNT 922MHz Ant1 @ 1215MHz-5000MHz

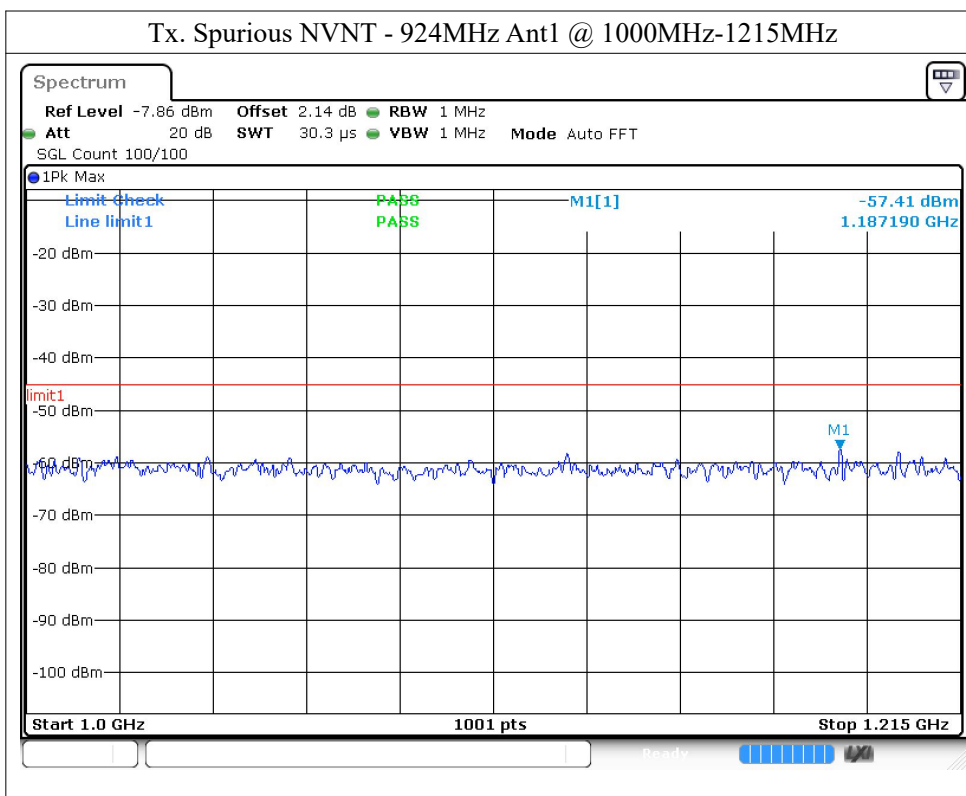
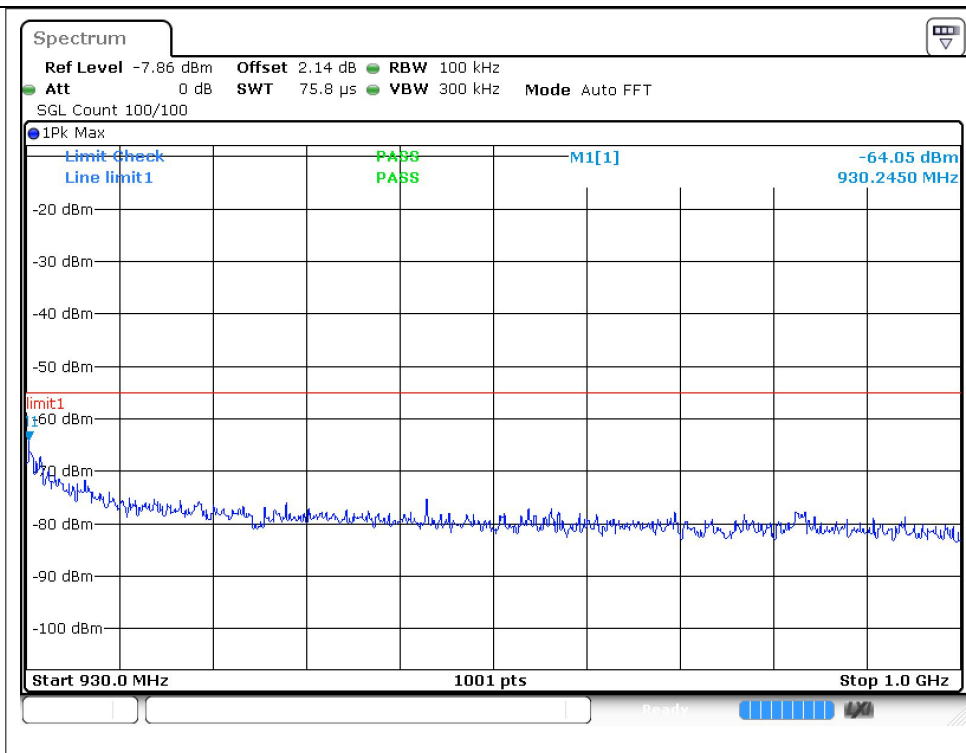


Tx. Spurious NVNT - 924MHz Ant1 @ 30MHz-710MHz

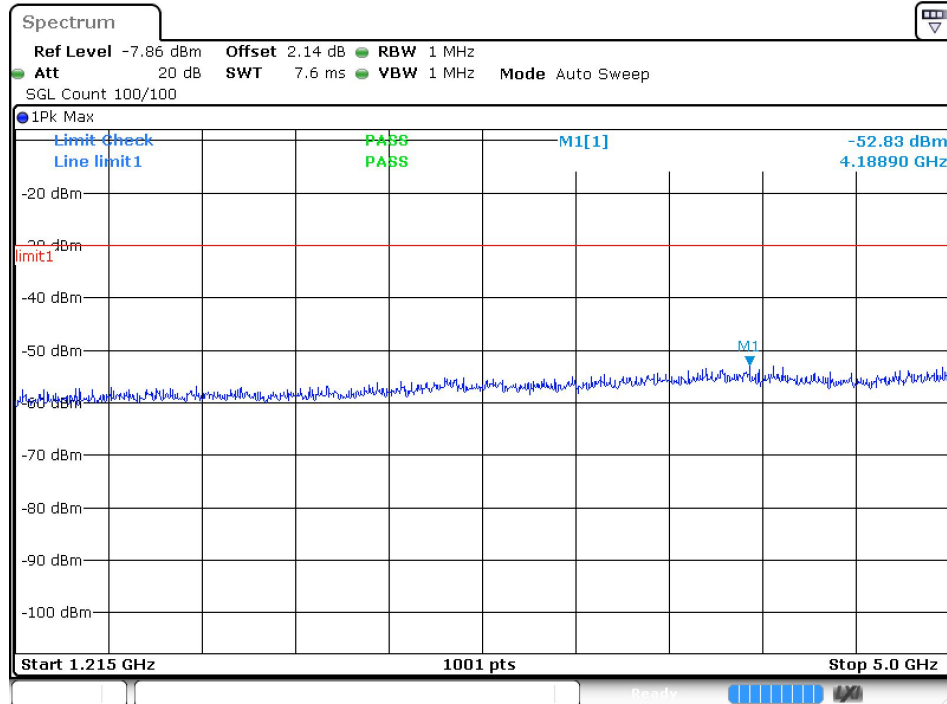




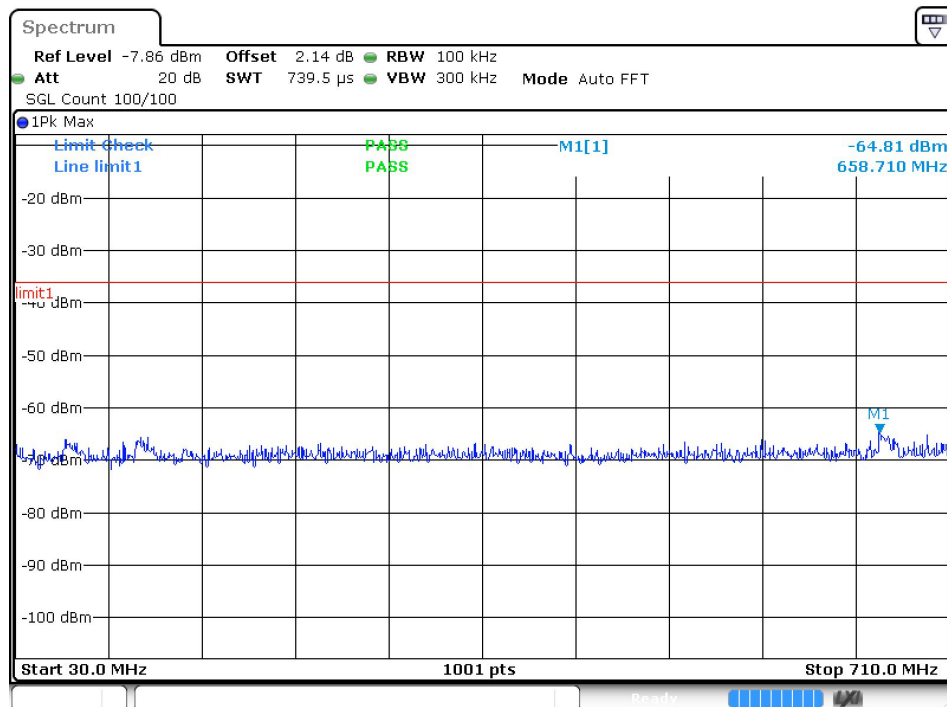




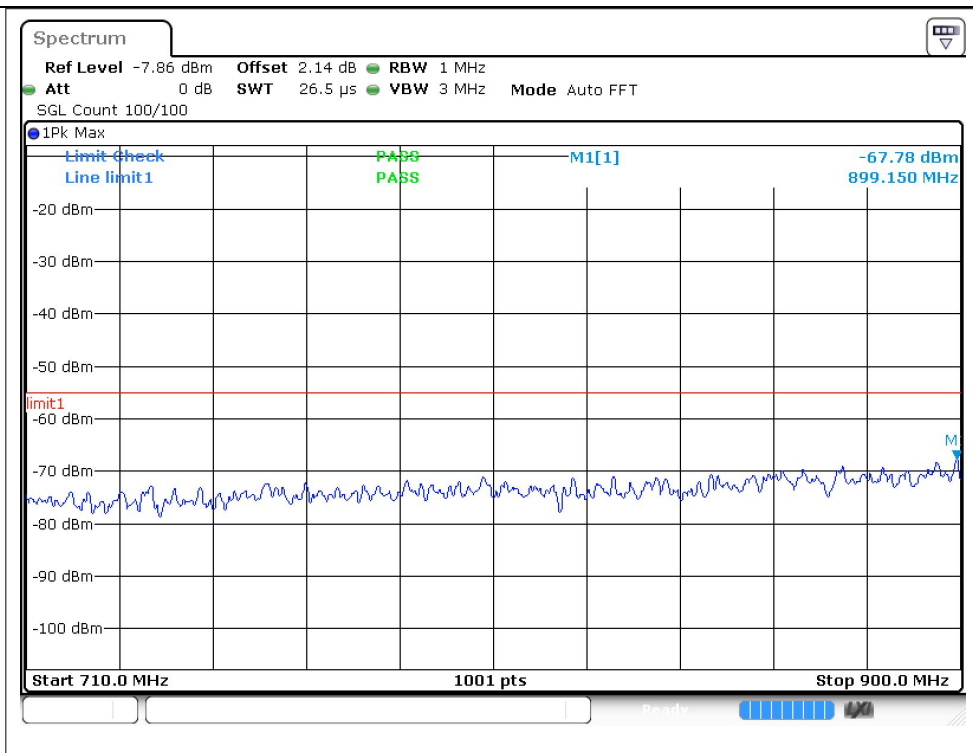
Tx. Spurious NVNT - 924MHz Ant1 @ 1215MHz-5000MHz



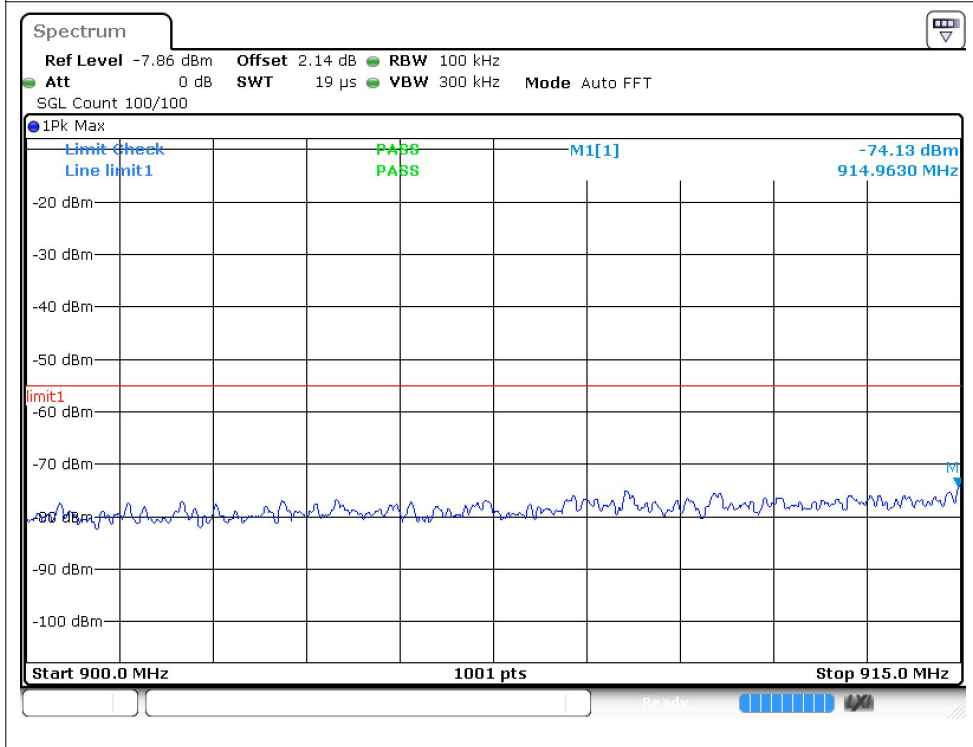
Tx. Spurious NVNT - 926MHz Ant1 @ 30MHz-710MHz

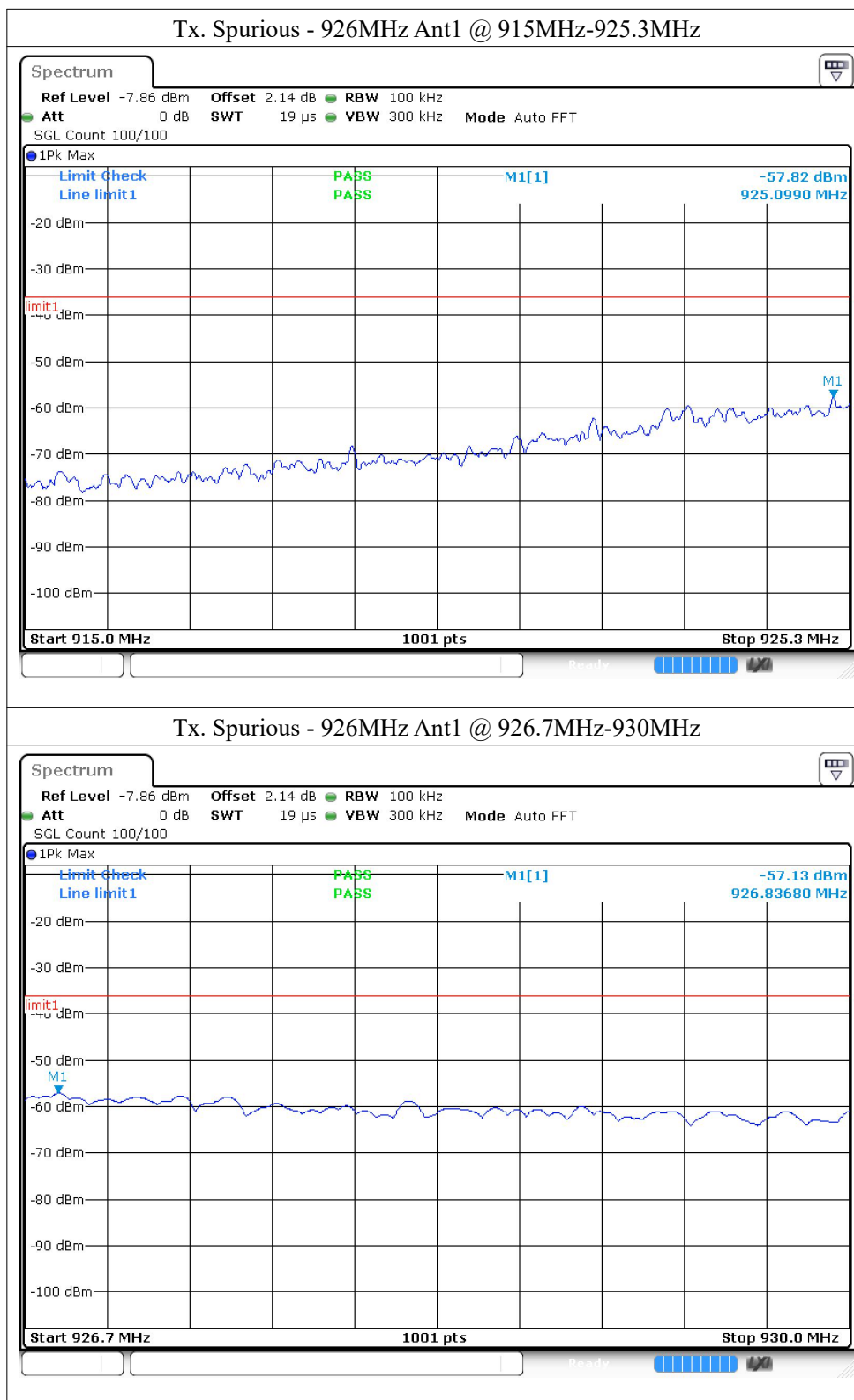


Tx. Spurious - 926MHz Ant1 @ 710MHz-900MHz

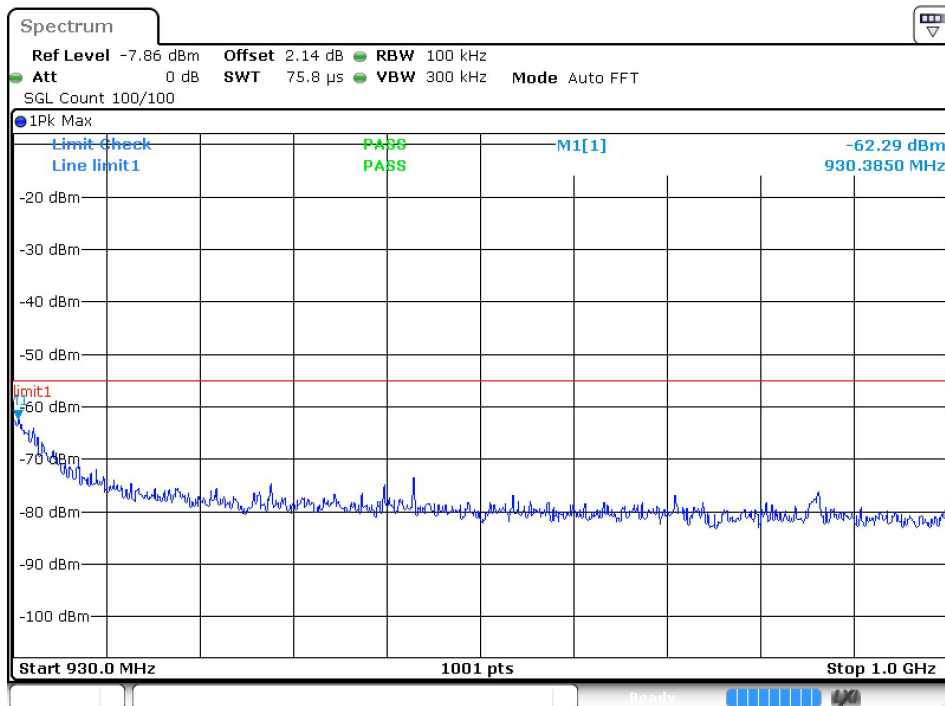


Tx. Spurious - 926MHz Ant1 @ 900MHz-915MHz

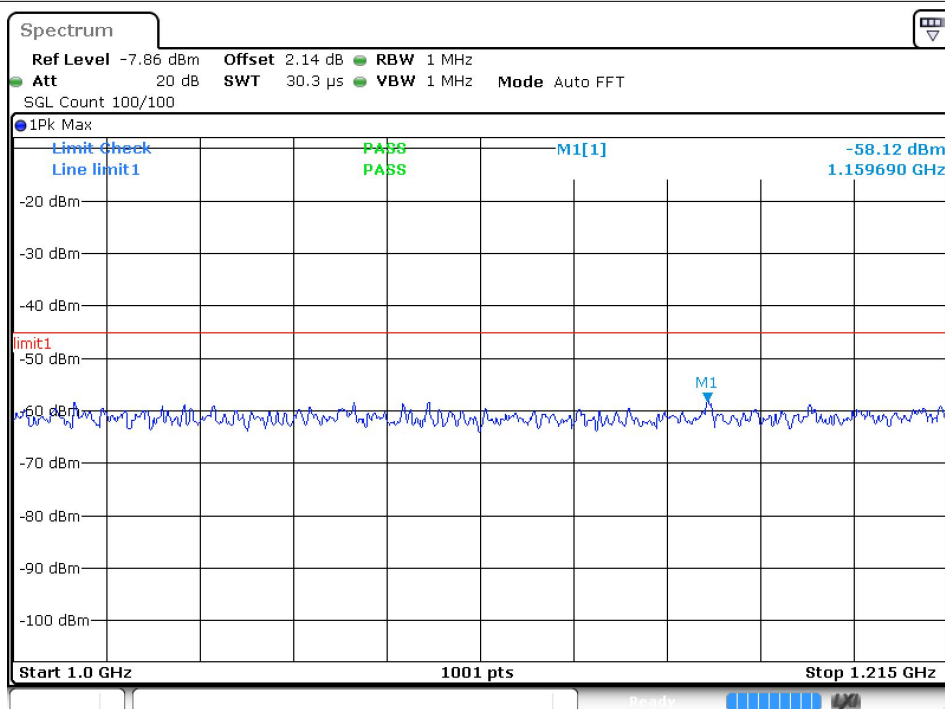


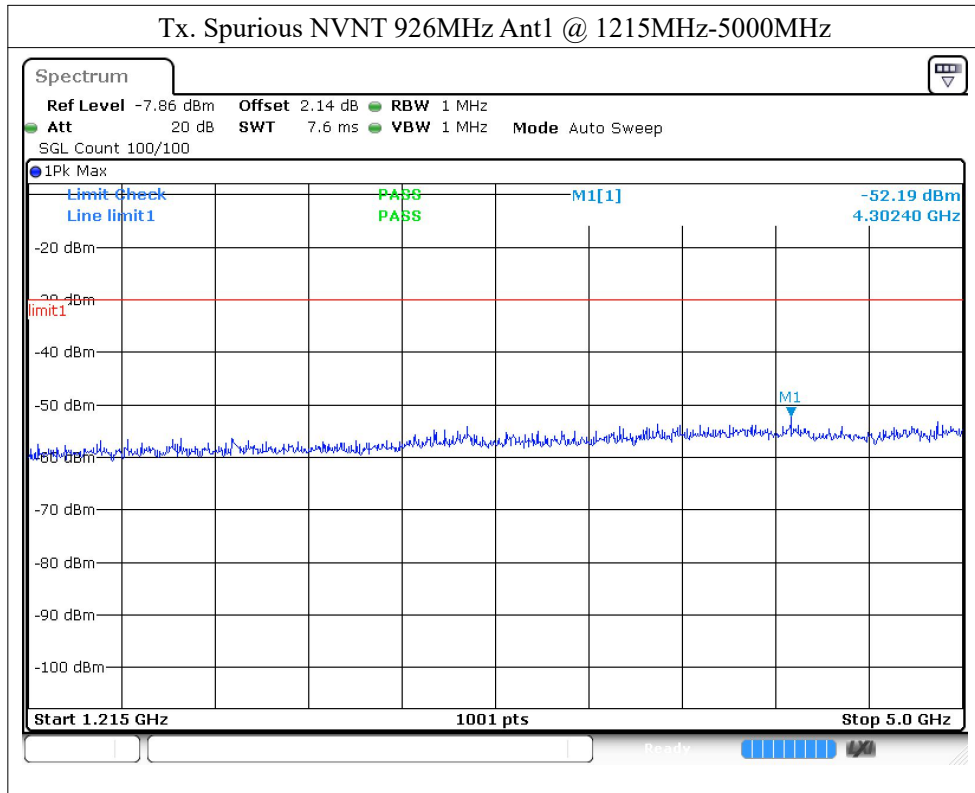


Tx. Spurious - 926MHz Ant1 @ 930MHz-1000MHz



Tx. Spurious NVNT - 926MHz Ant1 @ 1000MHz-1215MHz



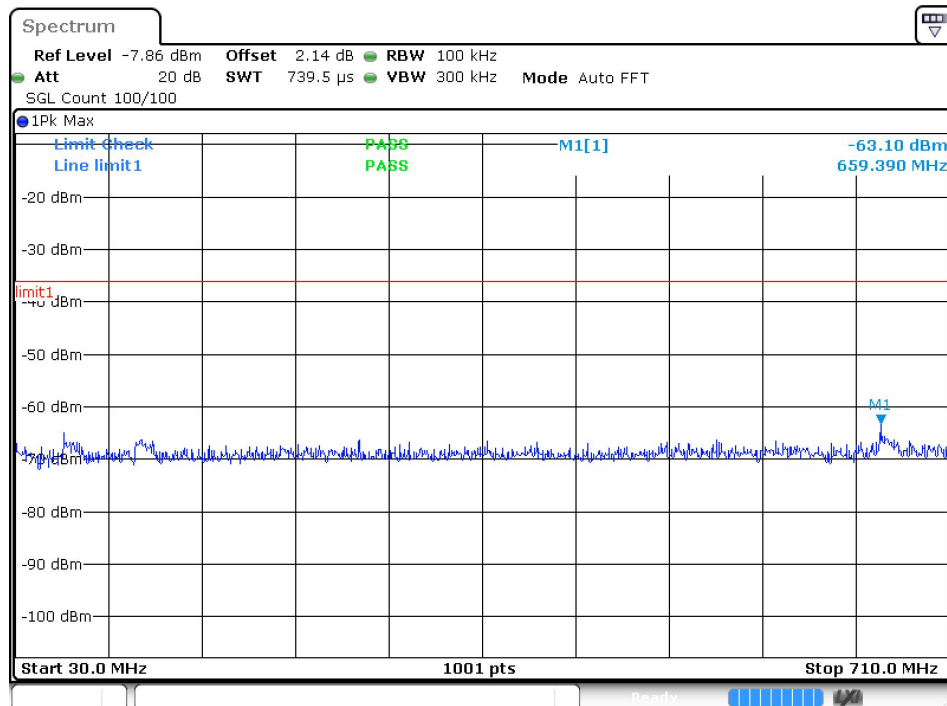


SRD 2MHz Nss1,(6.5Mbps)1TX

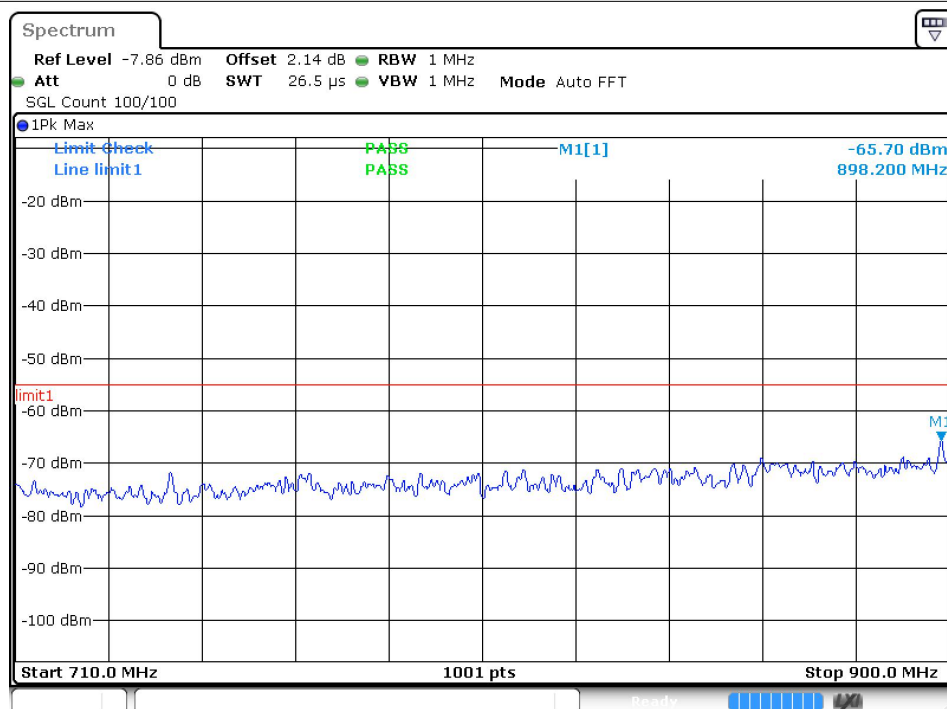
Condition	Frequency (MHz)	Range (MHz)	Emission Frequency (MHz)	Max Value (dBm)	Limit Start(dBm)	Limit Stop (dBm)	Verdict
NVNT	922.1	30-710	659.39	-63.10	-36	-36	Pass
NVNT	922.1	710-900	898.2	-65.7	-55	-55	Pass
NVNT	922.1	900-915	914.47	-60.19	-55	-55	Pass
NVNT	922.1	915-920.9	920.68	-46.44	-36	-36	Pass
NVNT	922.1	923.3-930	923.77	-46.19	-36	-36	Pass
NVNT	922.1	930-1000	930.105	-49.64	-55	-55	Pass
NVNT	922.1	1000-1215	1129.41	-57.79	-45	-45	Pass
NVNT	922.1	1215-5000	3984.7	-53.69	-30	-30	Pass
NVNT	924.3	30-710	659.39	-64.42	-36	-36	Pass
NVNT	924.3	710-900	860.61	-68.67	-55	-55	Pass
NVNT	924.3	900-915	914.27	-65.31	-55	-55	Pass
NVNT	924.3	915-923.1	922.46	-46.89	-36	-36	Pass
NVNT	924.3	925.5-930	926.9	-47.64	-36	-36	Pass
NVNT	924.3	930-1000	930.18	-58.7	-55	-55	Pass
NVNT	924.3	1000-1215	1182.89	-58.37	-45	-45	Pass
NVNT	924.3	1215-5000	4170	-53.15	-30	-30	Pass
NVNT	926.5	30-710	664.15	-63.85	-36	-36	Pass
NVNT	926.5	710-900	862.7	-68.73	-55	-55	Pass
NVNT	926.5	900-915	909.58	-63.37	-55	-55	Pass
NVNT	926.5	915-925.3	924.81	-46.17	-36	-36	Pass
NVNT	926.5	927.7-930	928.13	-46.79	-36	-36	Pass
NVNT	926.5	930-1000	930.8	-49.85	-55	-55	Pass
NVNT	926.5	1000-1215	1129.84	-58.59	-45	-45	Pass
NVNT	926.5	1215-5000	4313.7	-53.32	-30	-30	Pass
LVNT	922.1	30-710	658.77	-63.22	-36	-36	Pass
LVNT	922.1	710-900	897.52	-66.53	-55	-55	Pass
LVNT	922.1	900-915	913.99	-60.83	-55	-55	Pass
LVNT	922.1	915-920.9	920.27	-47.04	-36	-36	Pass
LVNT	922.1	923.3-930	923.32	-46.58	-36	-36	Pass
LVNT	922.1	930-1000	930.04	-50.61	-55	-55	Pass
LVNT	922.1	1000-1215	1129.03	-57.84	-45	-45	Pass
LVNT	922.1	1215-5000	3984.37	-53.82	-30	-30	Pass
LVNT	924.3	30-710	658.57	-64.53	-36	-36	Pass
LVNT	924.3	710-900	860.59	-69.34	-55	-55	Pass
LVNT	924.3	900-915	913.29	-65.42	-55	-55	Pass
LVNT	924.3	915-923.1	921.72	-46.96	-36	-36	Pass
LVNT	924.3	925.5-930	925.95	-47.94	-36	-36	Pass
LVNT	924.3	930-1000	929.74	-59.64	-55	-55	Pass
LVNT	924.3	1000-1215	1182.71	-59.28	-45	-45	Pass
LVNT	924.3	1215-5000	4169.23	-53.22	-30	-30	Pass
LVNT	926.5	30-710	663.21	-64.70	-36	-36	Pass
LVNT	926.5	710-900	862.55	-69.42	-55	-55	Pass
LVNT	926.5	900-915	909.21	-63.61	-55	-55	Pass

LVNT	926.5	915-925.3	923.84	-46.23	-36	-36	Pass
LVNT	926.5	927.7-930	927.57	-47.36	-36	-36	Pass
LVNT	926.5	930-1000	930.51	-50.02	-55	-55	Pass
LVNT	926.5	1000-1215	1129.39	-59.20	-45	-45	Pass
LVNT	926.5	1215-5000	4312.93	-53.85	-30	-30	Pass
HVNT	922.1	30-710	658.86	-63.73	-36	-36	Pass
HVNT	922.1	710-900	897.80	-66.14	-55	-55	Pass
HVNT	922.1	900-915	913.93	-60.90	-55	-55	Pass
HVNT	922.1	915-920.9	919.86	-46.46	-36	-36	Pass
HVNT	922.1	923.3-930	923.71	-46.44	-36	-36	Pass
HVNT	922.1	930-1000	929.38	-50.54	-55	-55	Pass
HVNT	922.1	1000-1215	1129.39	-58.19	-45	-45	Pass
HVNT	922.1	1215-5000	3984.25	-54.17	-30	-30	Pass
HVNT	924.3	30-710	658.80	-65.24	-36	-36	Pass
HVNT	924.3	710-900	859.93	-69.14	-55	-55	Pass
HVNT	924.3	900-915	914.12	-65.69	-55	-55	Pass
HVNT	924.3	915-923.1	921.48	-47.44	-36	-36	Pass
HVNT	924.3	925.5-930	926.14	-47.79	-36	-36	Pass
HVNT	924.3	930-1000	929.95	-59.19	-55	-55	Pass
HVNT	924.3	1000-1215	1182.81	-59.32	-45	-45	Pass
HVNT	924.3	1215-5000	4169.90	-53.77	-30	-30	Pass
HVNT	926.5	30-710	664.01	-64.50	-36	-36	Pass
HVNT	926.5	710-900	862.29	-68.75	-55	-55	Pass
HVNT	926.5	900-915	909.05	-64.09	-55	-55	Pass
HVNT	926.5	915-925.3	923.84	-46.84	-36	-36	Pass
HVNT	926.5	927.7-930	927.47	-47.52	-36	-36	Pass
HVNT	926.5	930-1000	930.48	-50.04	-55	-55	Pass
HVNT	926.5	1000-1215	1129.32	-58.87	-45	-45	Pass
HVNT	926.5	1215-5000	4312.83	-53.40	-30	-30	Pass

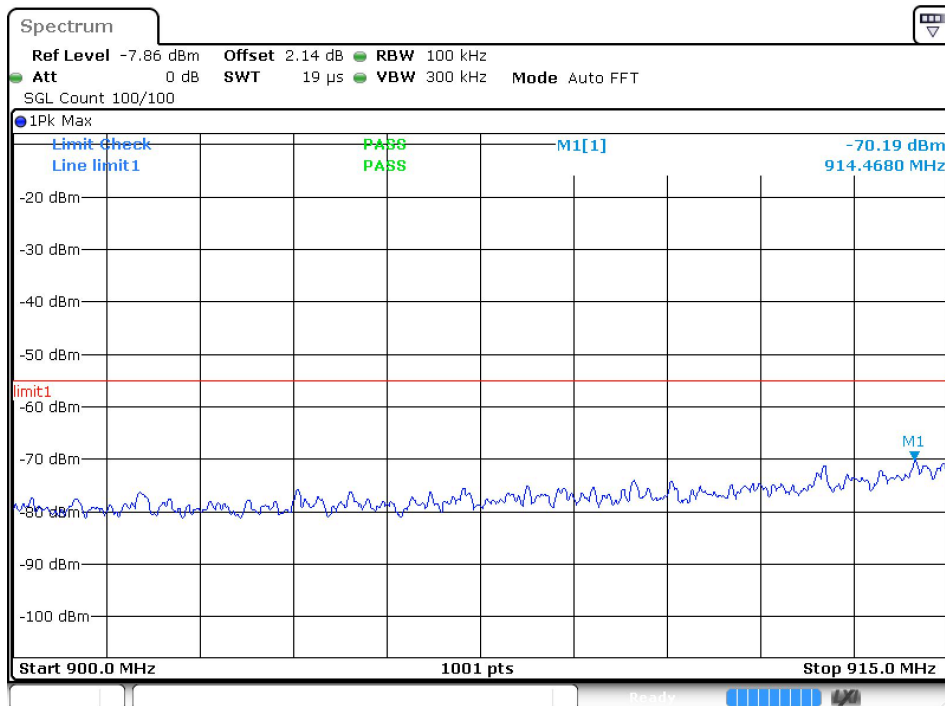
Tx. Spurious NVNT 922.1MHz Ant1 @ 30MHz-710MHz



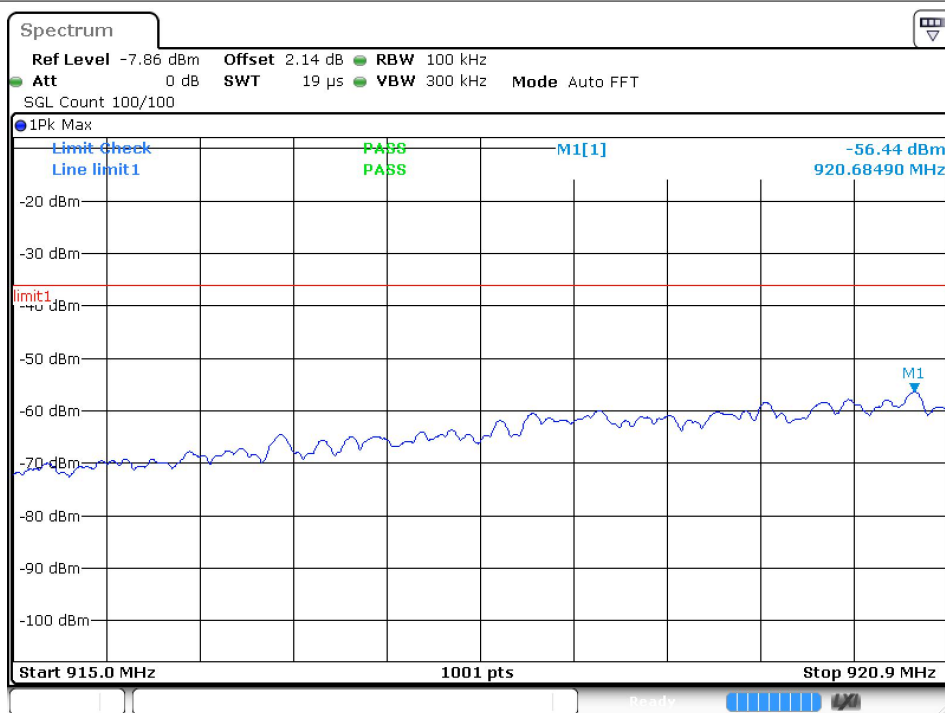
Tx. Spurious-922.1MHz Ant1 @ 710MHz-900MHz



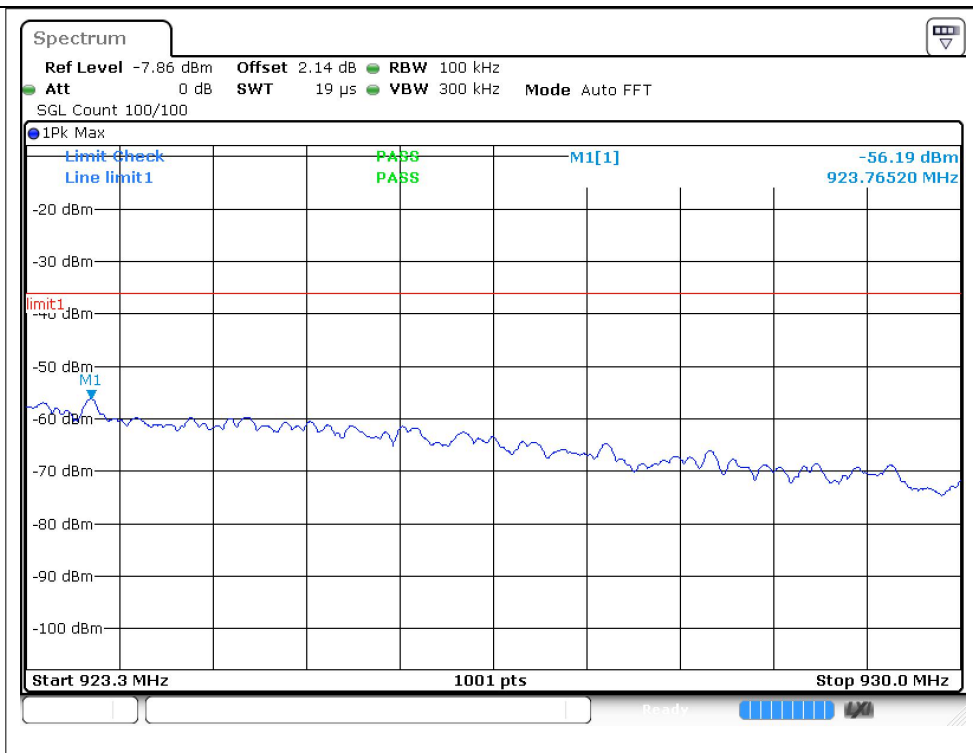
Tx. Spurious-922.1MHz Ant1 @ 900MHz-915MHz



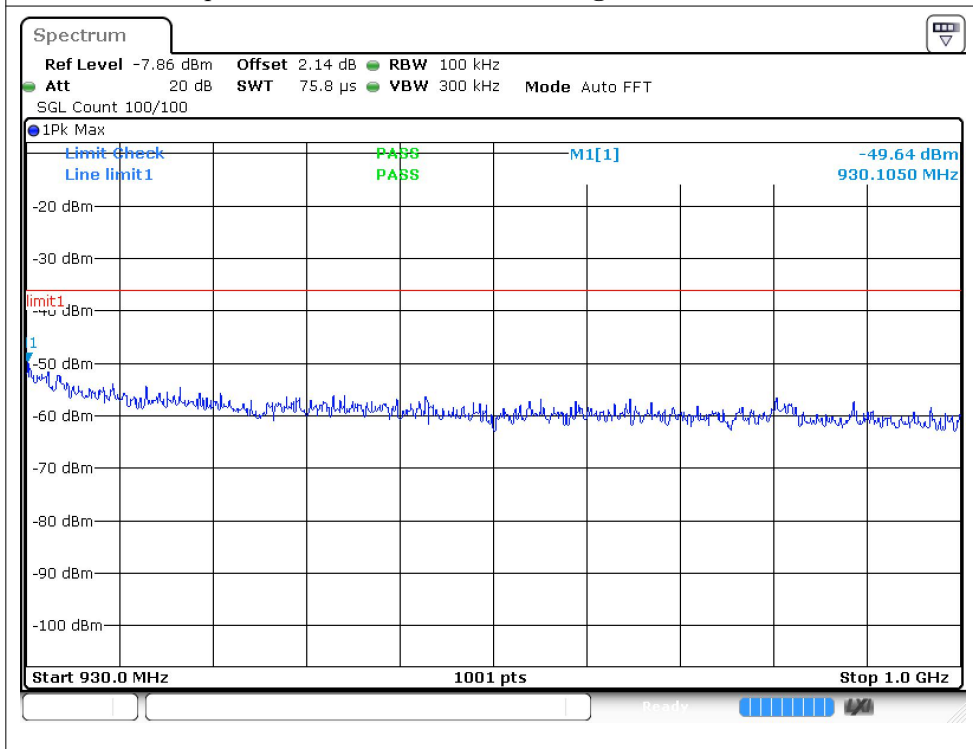
Tx. Spurious -922.1MHz Ant1 @ 915MHz-920.9MHz



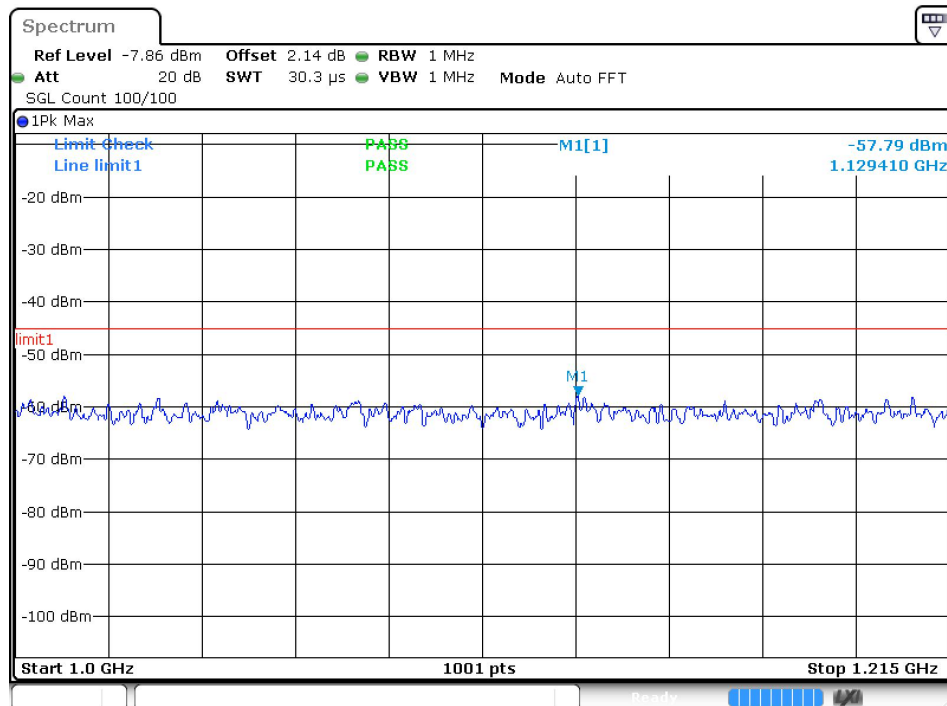
Tx. Spurious - 922.1MHz Ant1 @ 923.3MHz-930MHz



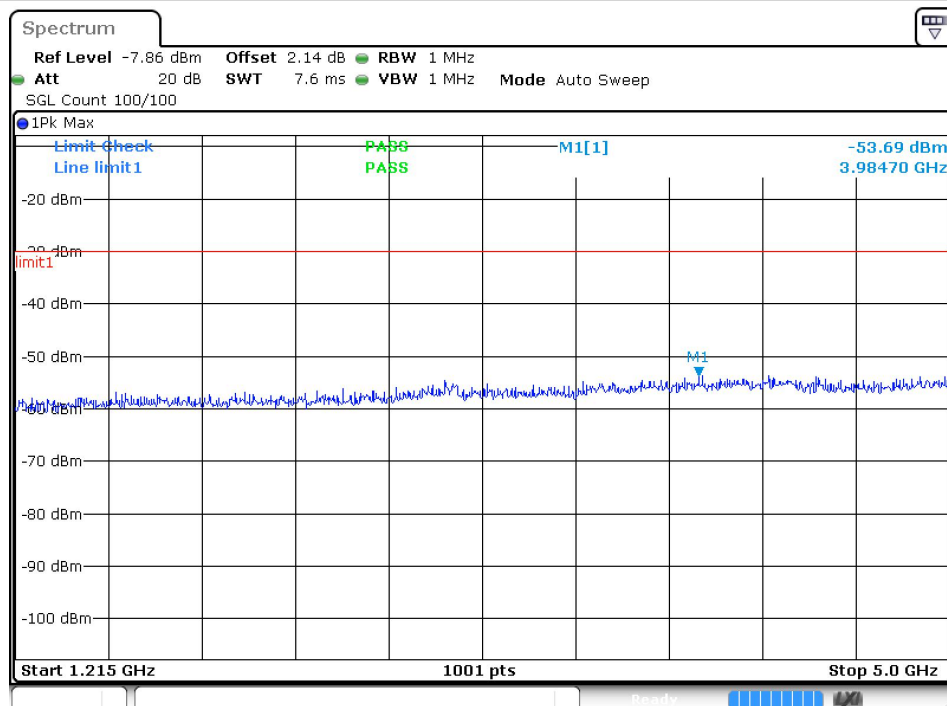
Tx. Spurious NVNT - 922.1MHz Ant1 @ 930MHz-1000MHz

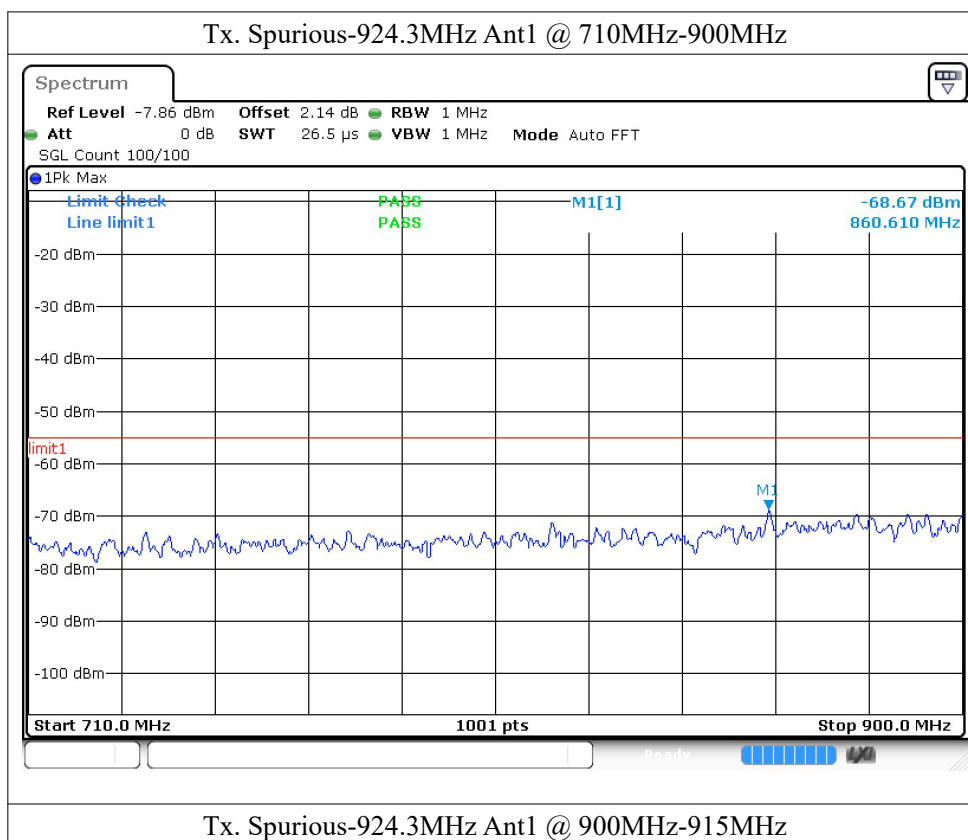
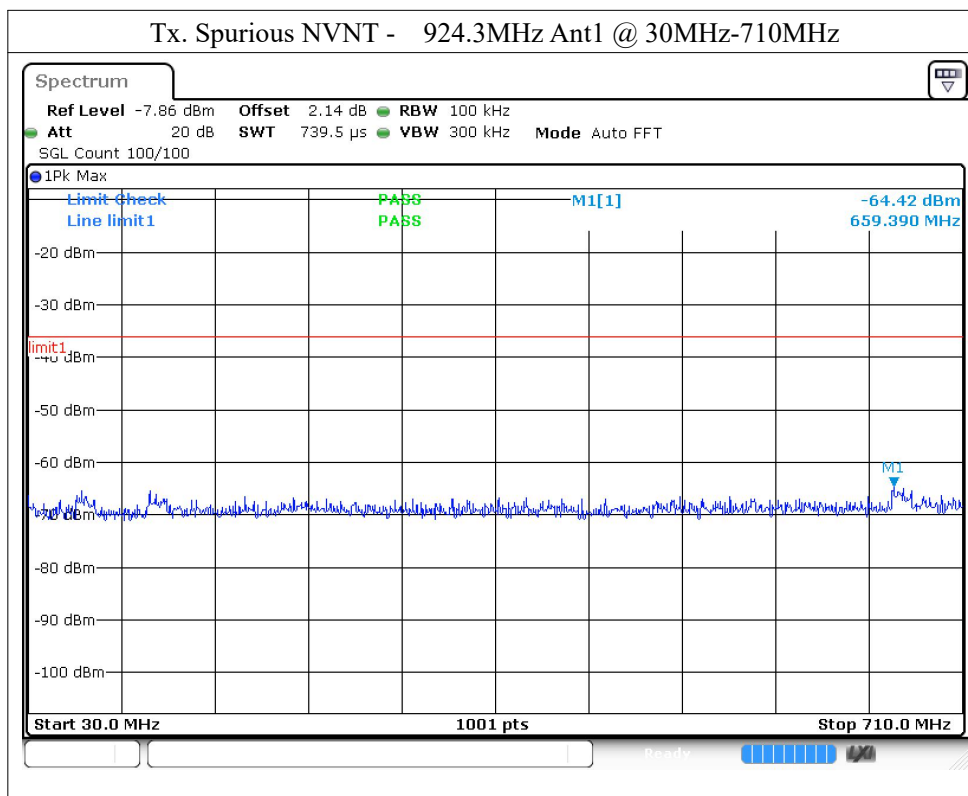


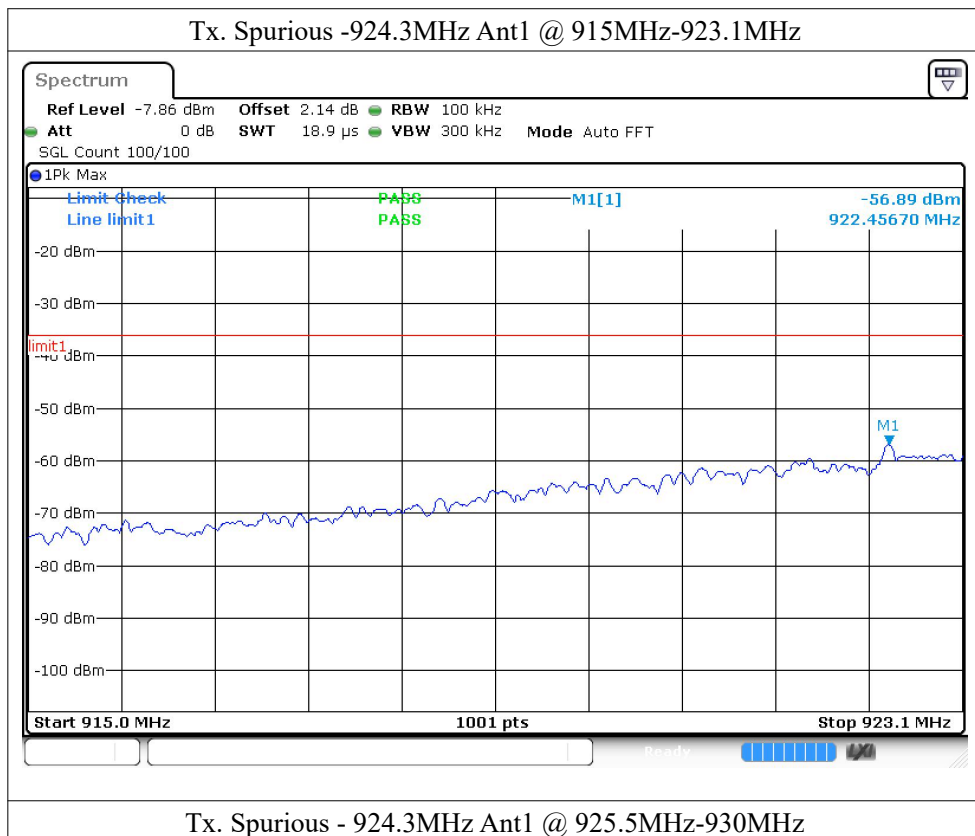
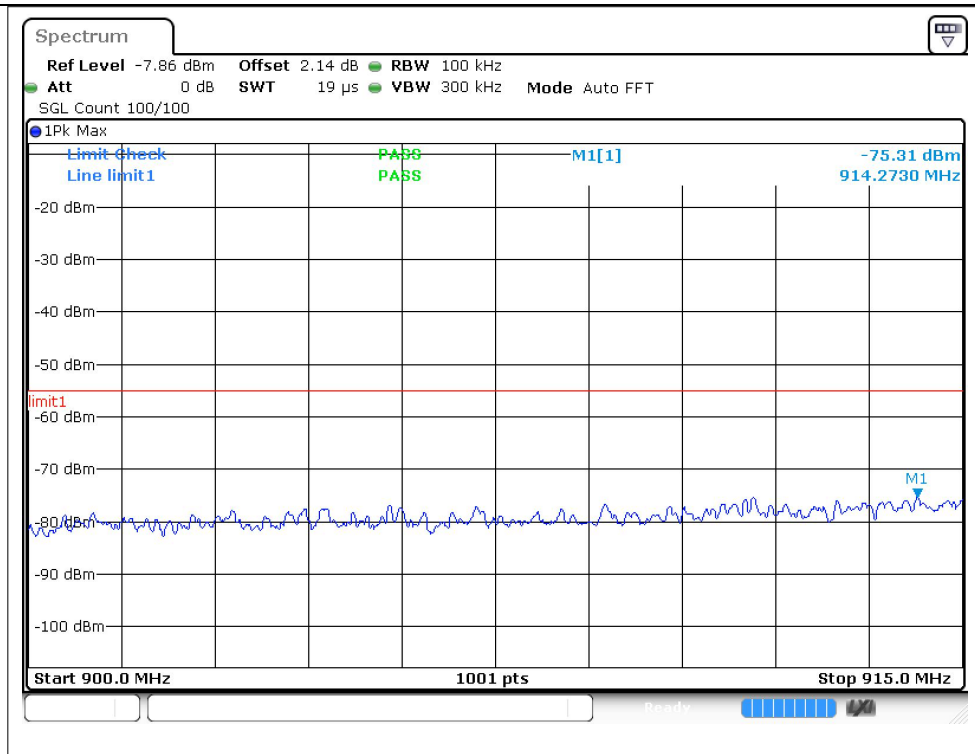
Tx. Spurious NVNT - 922.1MHz Ant1 @ 1000MHz-1215MHz

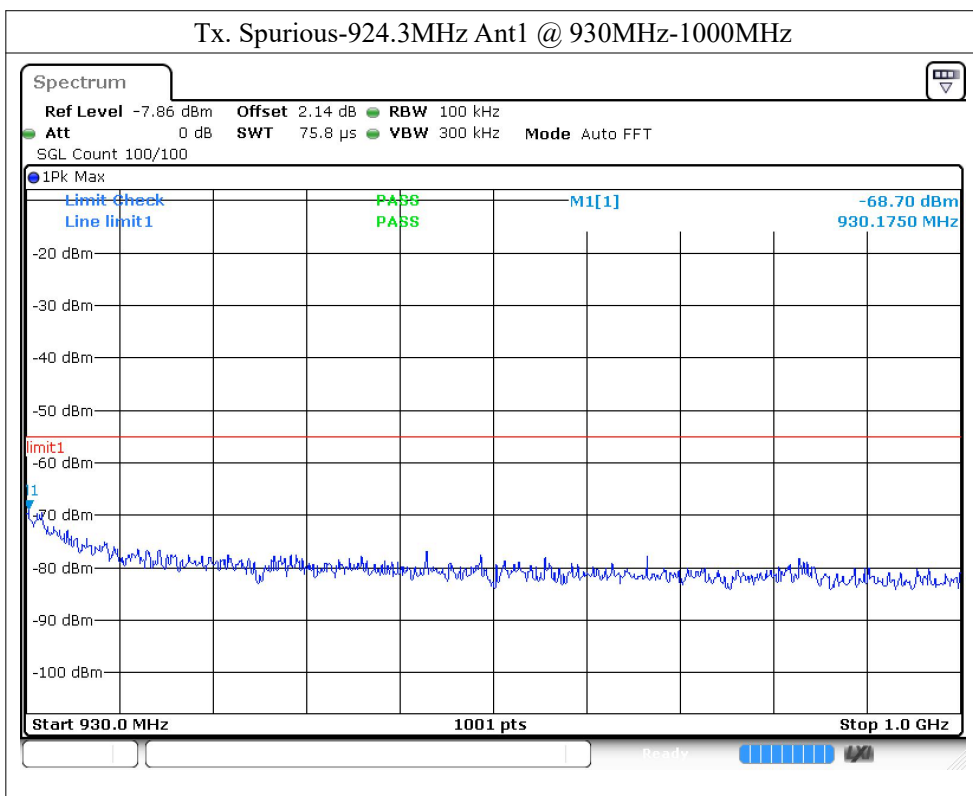
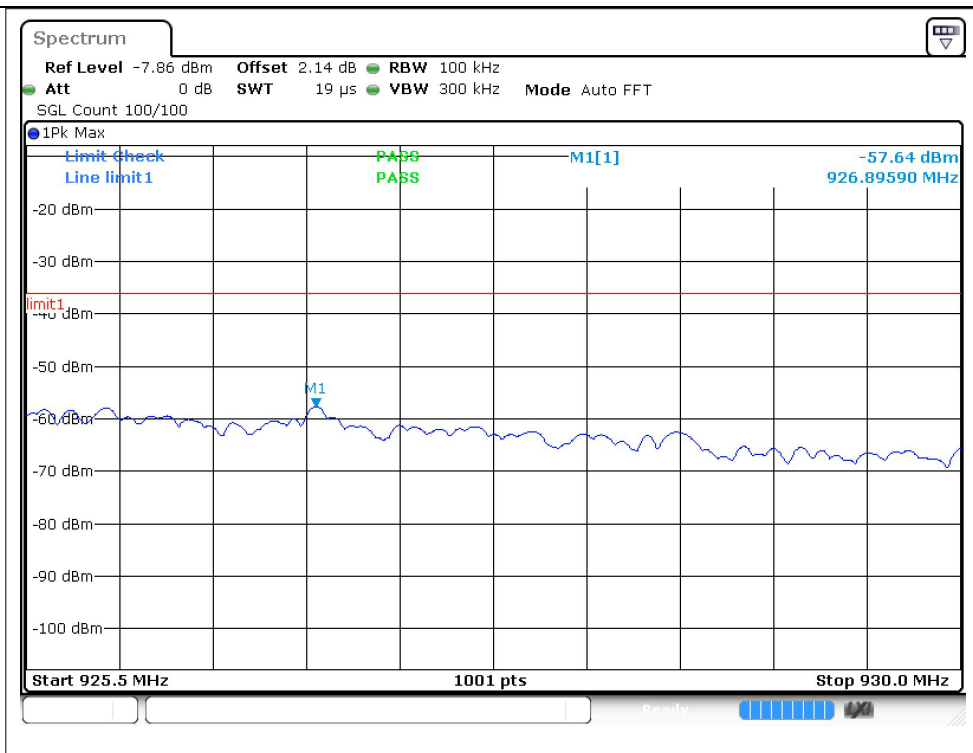


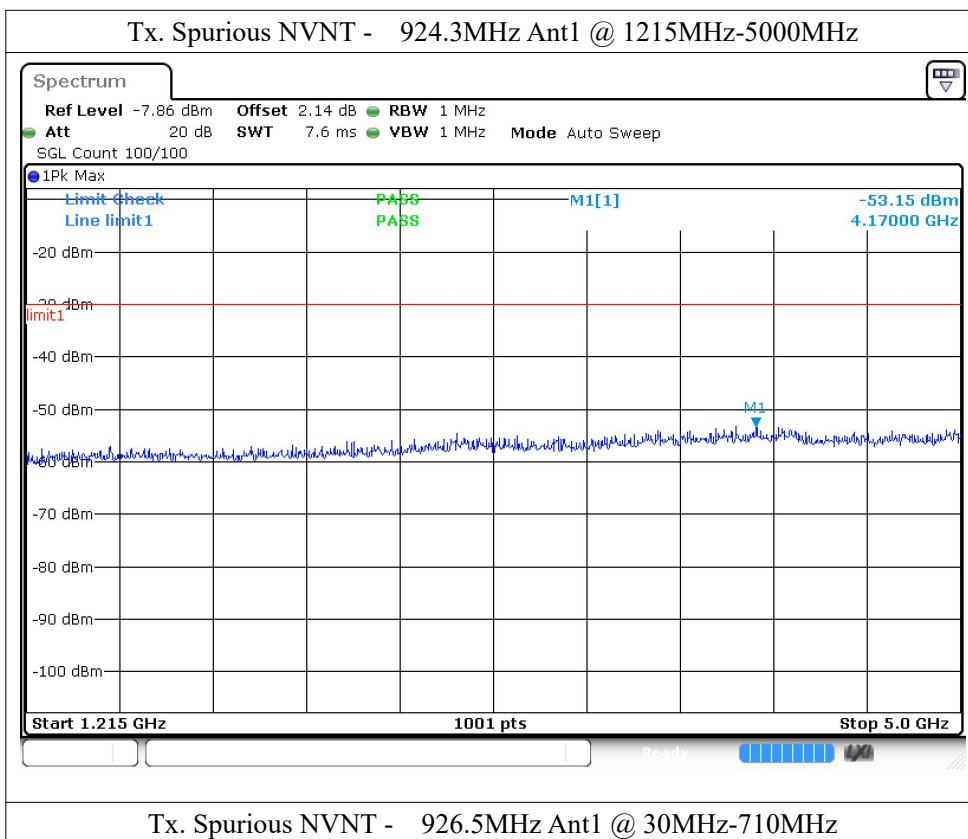
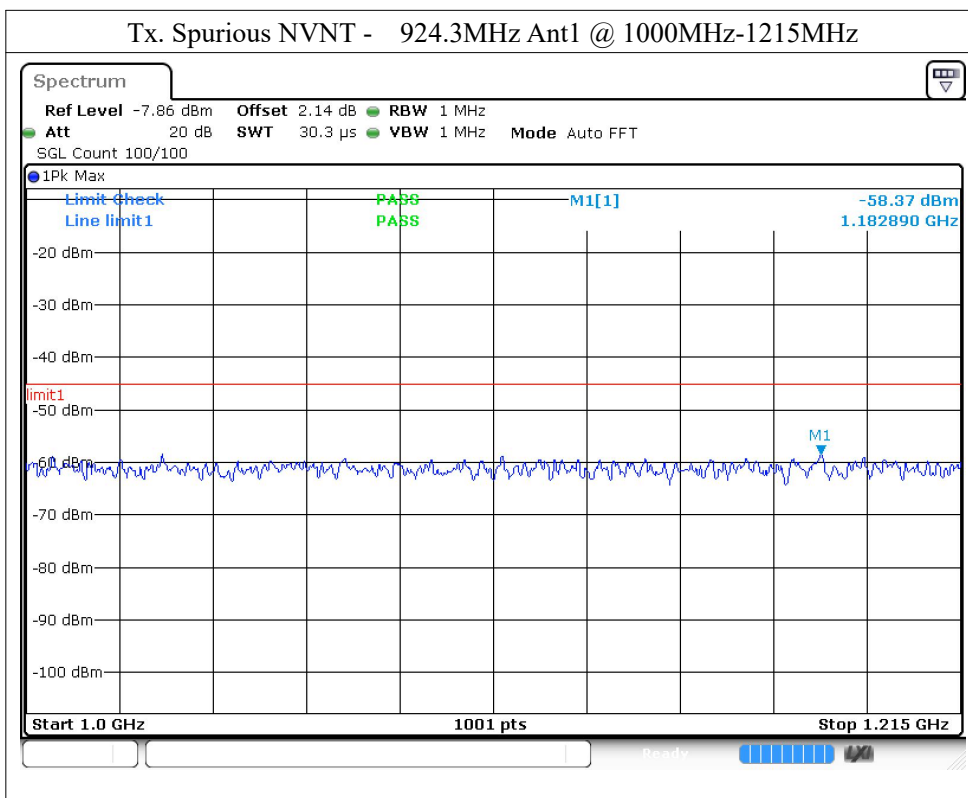
Tx. Spurious NVNT - 922.1MHz Ant1 @ 1215MHz-5000MHz

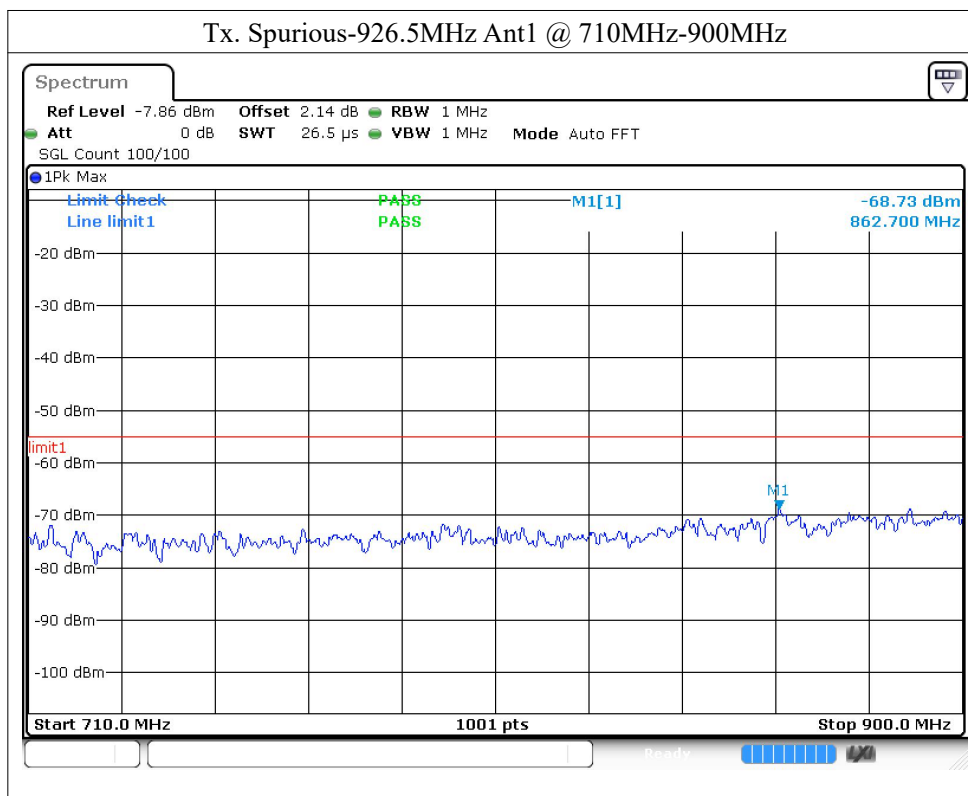
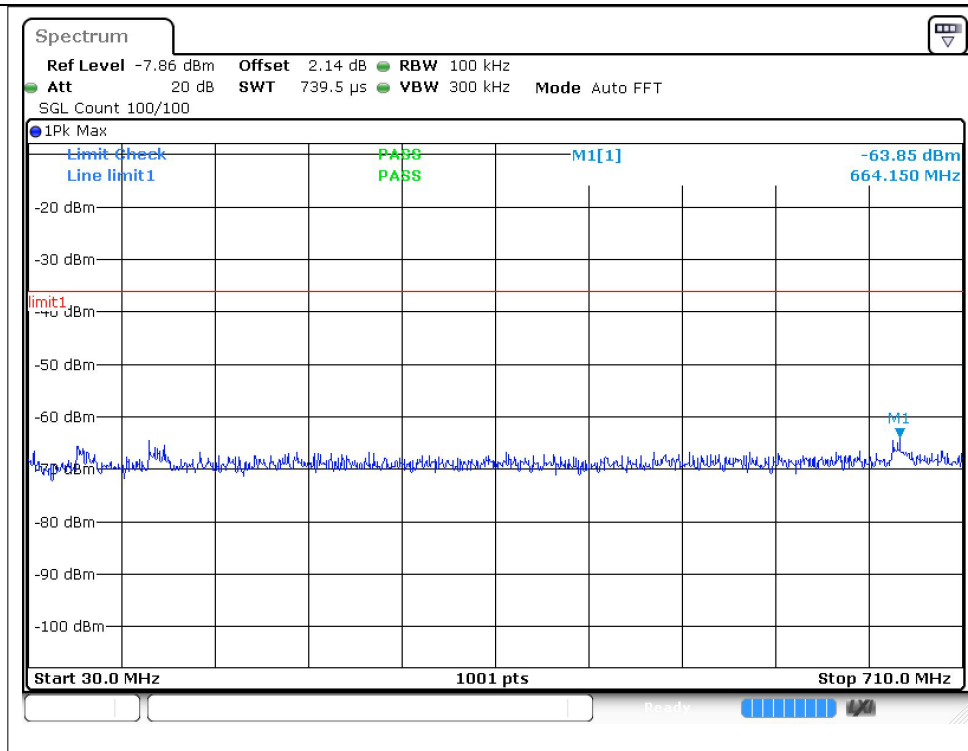


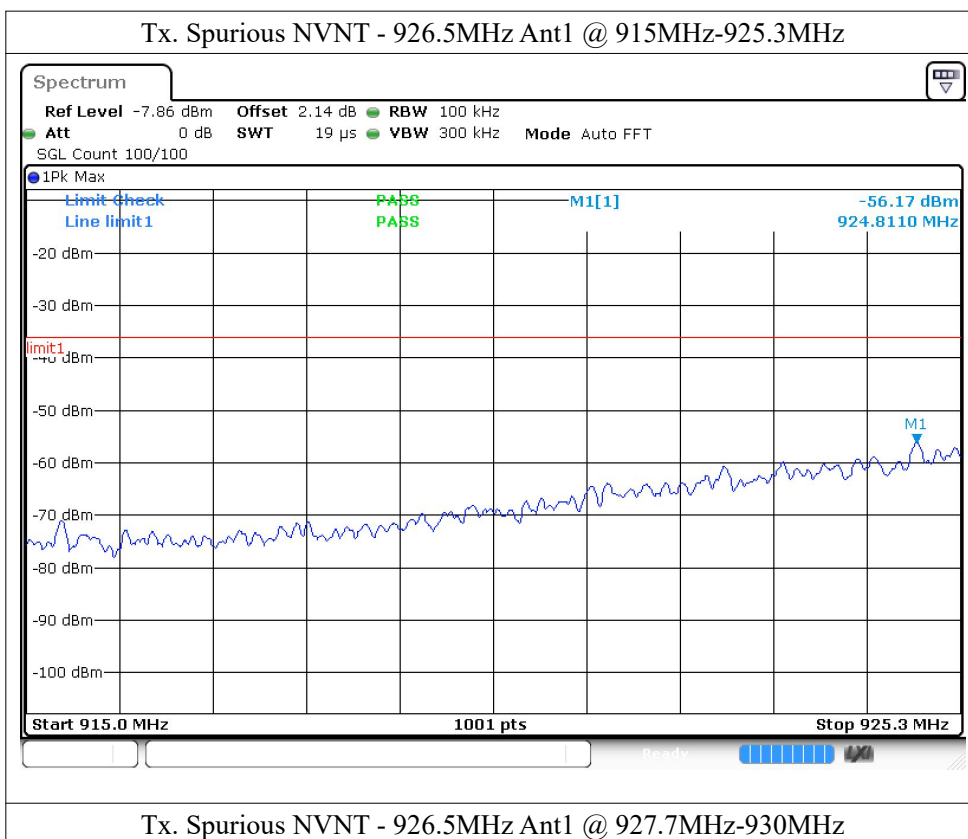
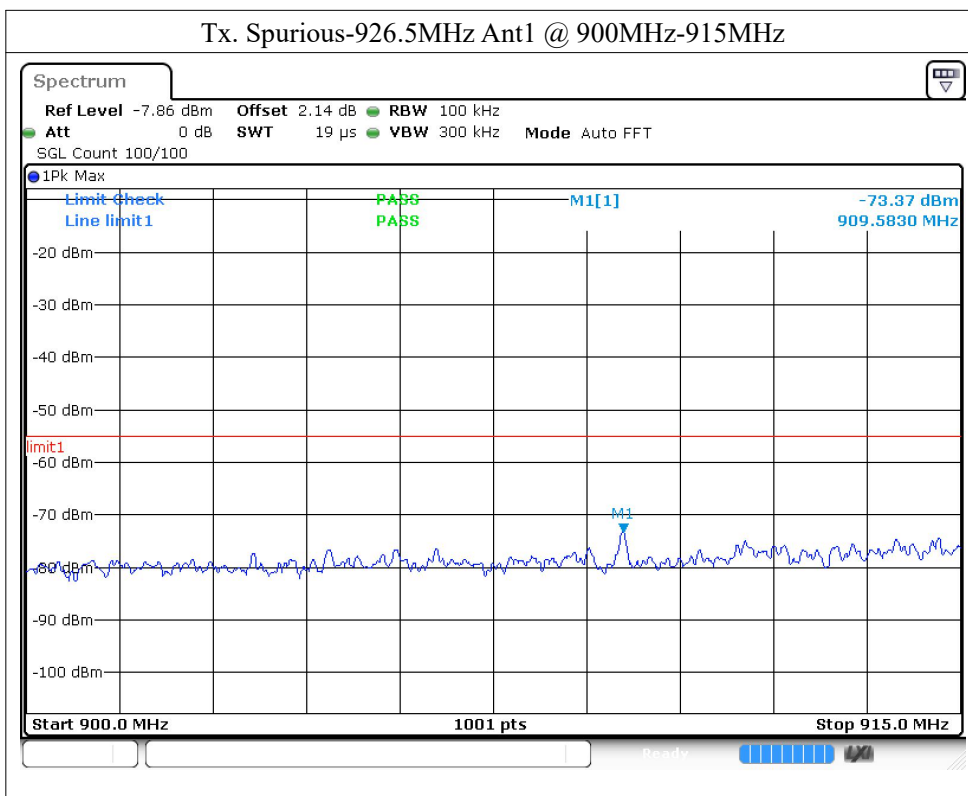


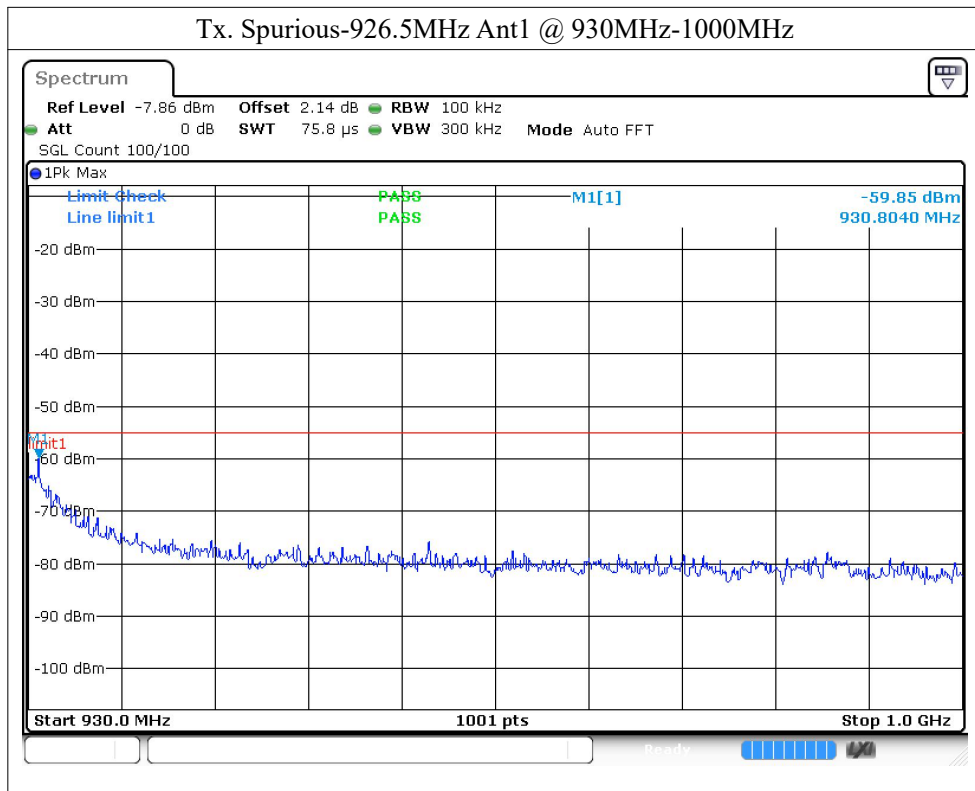
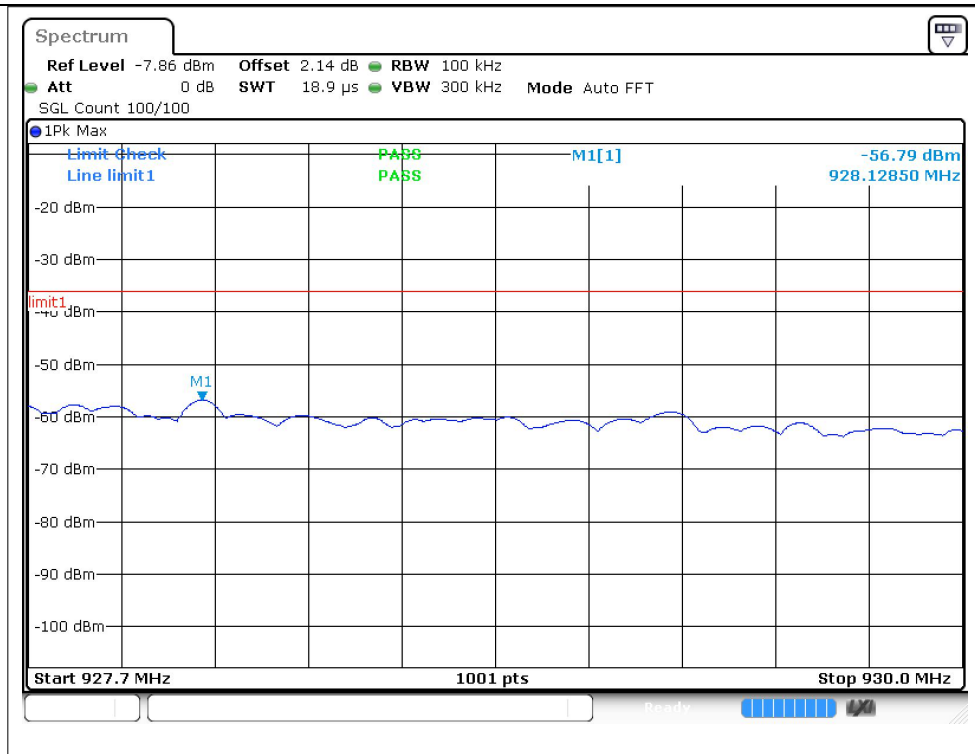


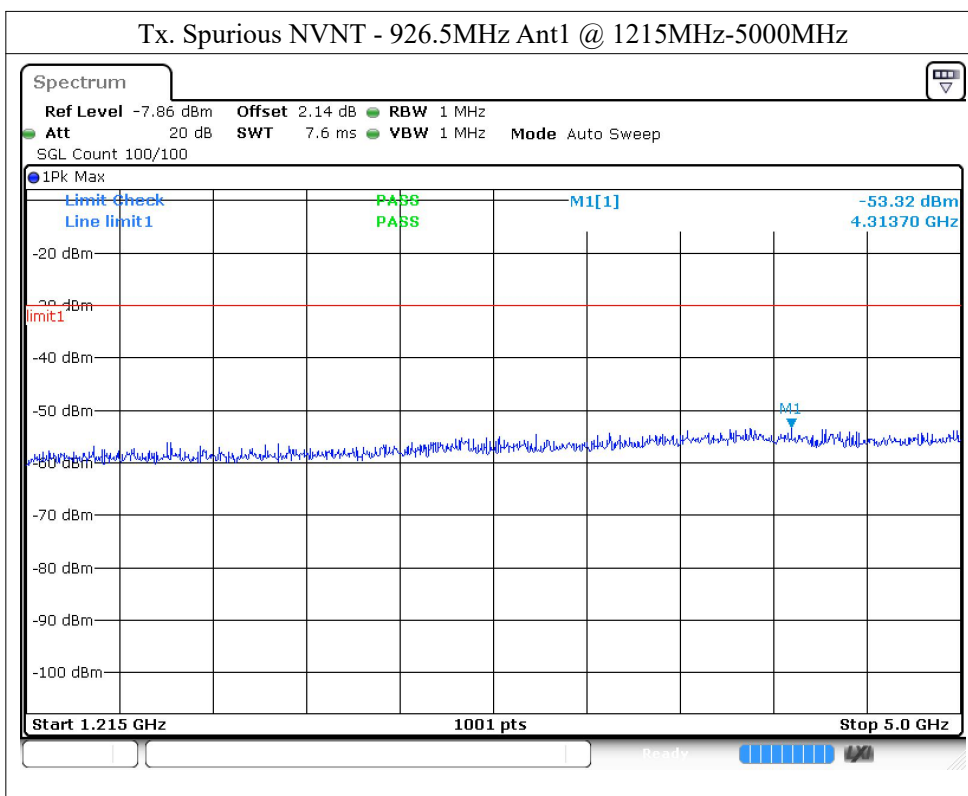
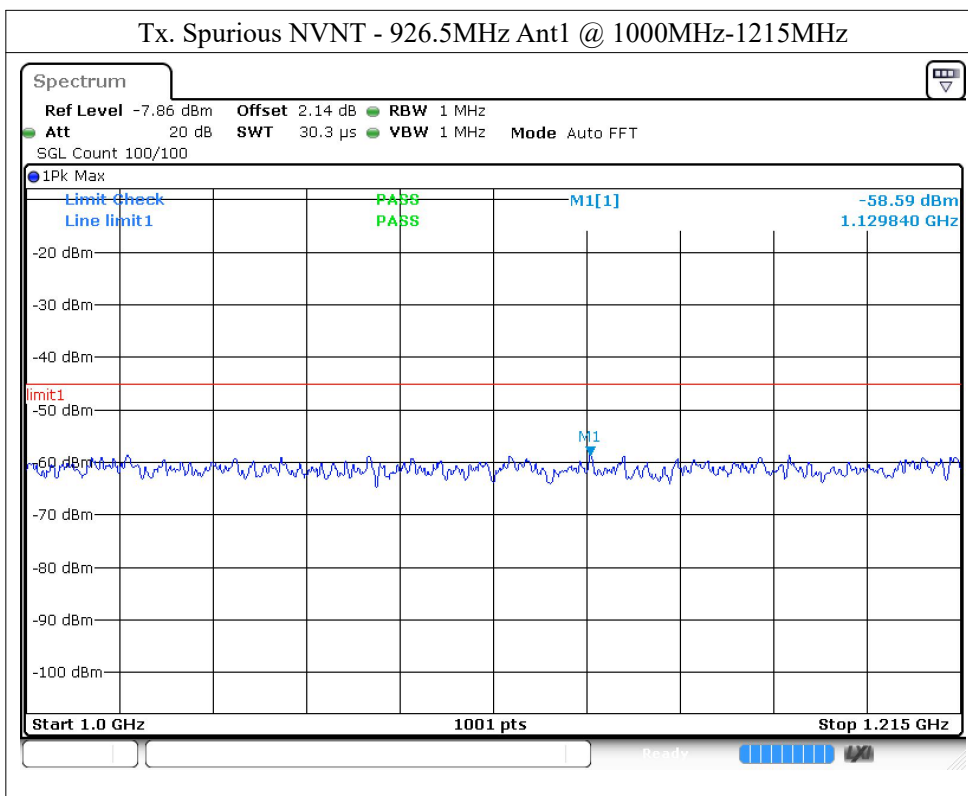








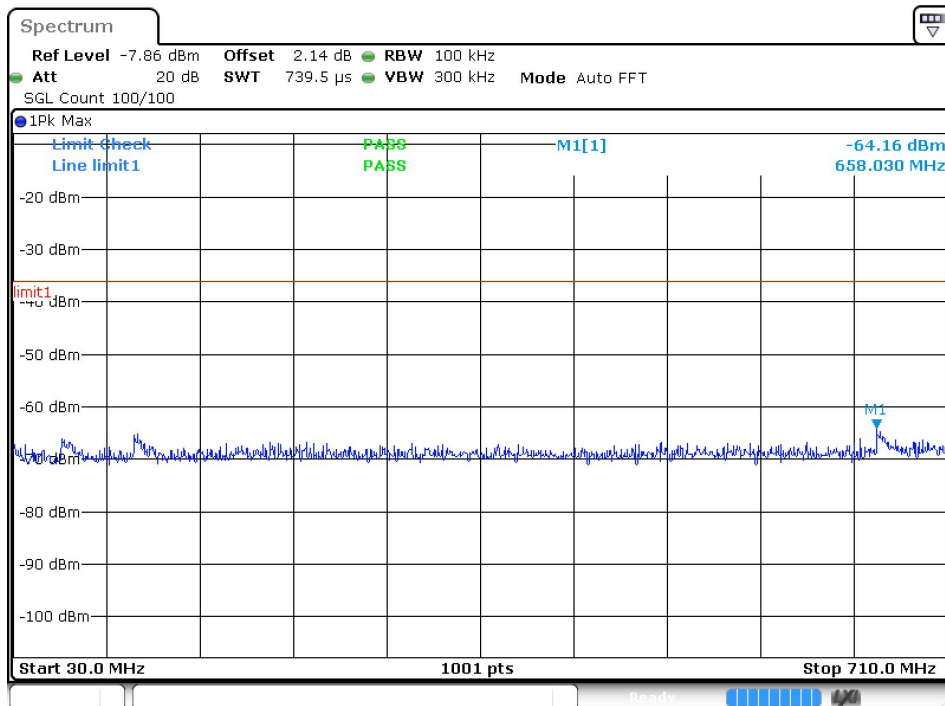




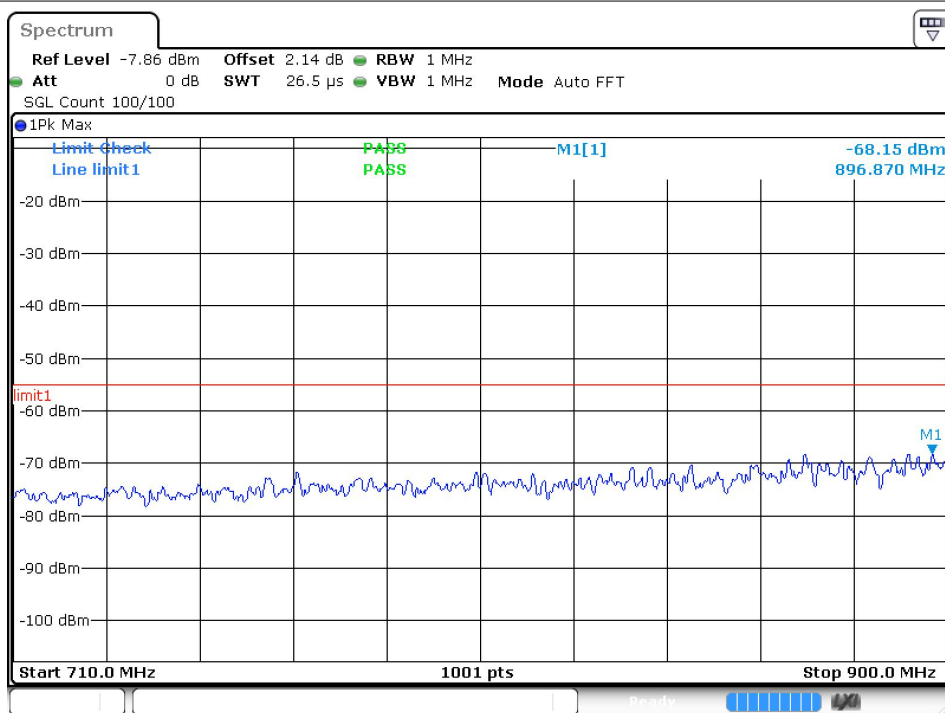
SRD_4MHz_Nss1 ,(13.5Mbps)_1TX

Condition	Frequency (MHz)	Range (MHz)	Emission Frequency (MHz)	Max Value (dBm)	Limit Start(dBm)	Limit Stop (dBm)	Verdict
NVNT	924.5	30-710	658.03	-64.16	-36	-36	Pass
NVNT	924.5	710-900	896.87	-68.15	-55	-55	Pass
NVNT	924.5	900-915	914.9	-61.64	-55	-55	Pass
NVNT	924.5	915-922.3	922.06	-48.54	-36	-36	Pass
NVNT	924.5	926.7-930	927.23	-48.95	-36	-36	Pass
NVNT	924.5	930-1000	930.25	-56.95	-55	-55	Pass
NVNT	924.5	1000-1215	1175.59	-57.44	-45	-45	Pass
NVNT	924.5	1215-5000	4260.8	-53.06	-30	-30	Pass
LVNT	924.5	30-710	657.31	-64.91	-36	-36	Pass
LVNT	924.5	710-900	896.26	-68.20	-55	-55	Pass
LVNT	924.5	900-915	914.70	-61.92	-55	-55	Pass
LVNT	924.5	915-922.3	921.45	-48.60	-36	-36	Pass
LVNT	924.5	926.7-930	927.15	-49.56	-36	-36	Pass
LVNT	924.5	930-1000	929.42	-57.65	-55	-55	Pass
LVNT	924.5	1000-1215	1175.05	-58.35	-45	-45	Pass
LVNT	924.5	1215-5000	4260.34	-53.07	-30	-30	Pass
HVNT	924.5	30-710	657.92	-65.05	-36	-36	Pass
HVNT	924.5	710-900	896.51	-68.62	-55	-55	Pass
HVNT	924.5	900-915	914.88	-62.60	-55	-55	Pass
HVNT	924.5	915-922.3	921.58	-49.49	-36	-36	Pass
HVNT	924.5	926.7-930	926.60	-49.71	-36	-36	Pass
HVNT	924.5	930-1000	930.14	-57.02	-55	-55	Pass
HVNT	924.5	1000-1215	1174.63	-57.47	-45	-45	Pass
HVNT	924.5	1215-5000	4259.84	-53.77	-30	-30	Pass

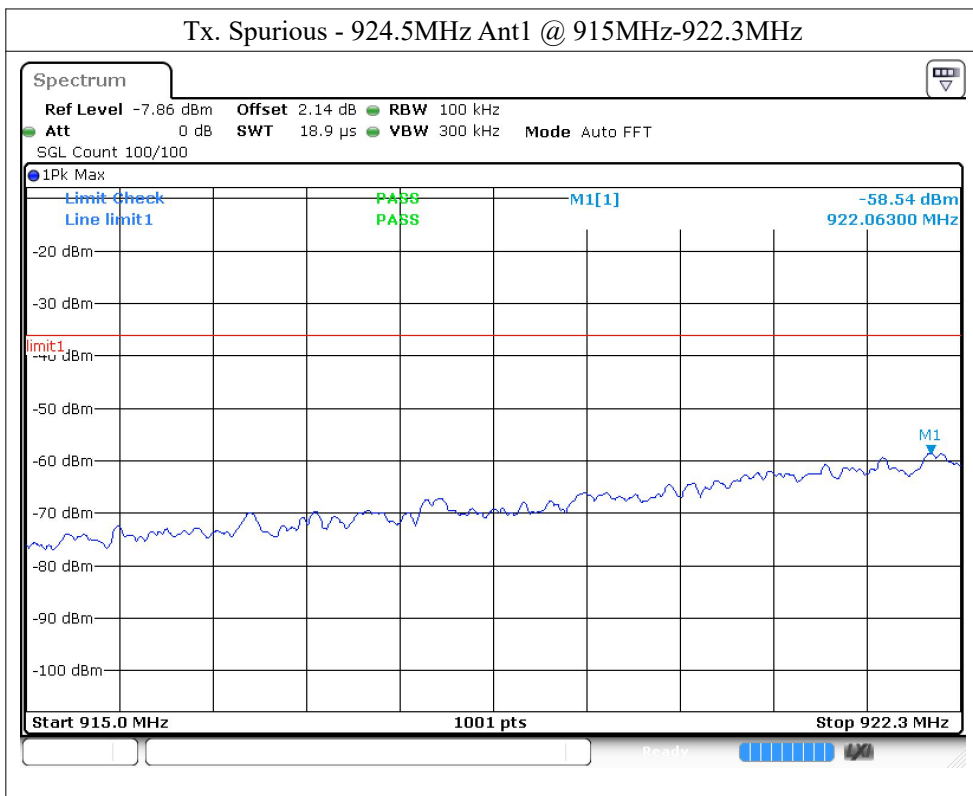
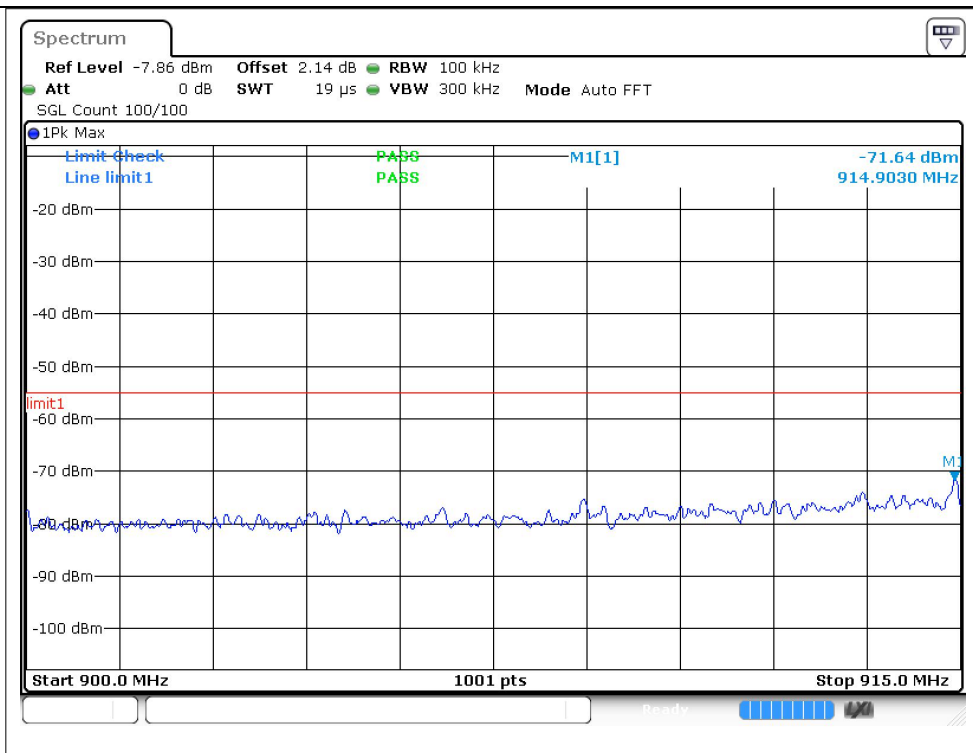
Tx. Spurious NVNT - 924.5MHz Ant1 @ 30MHz-710MHz

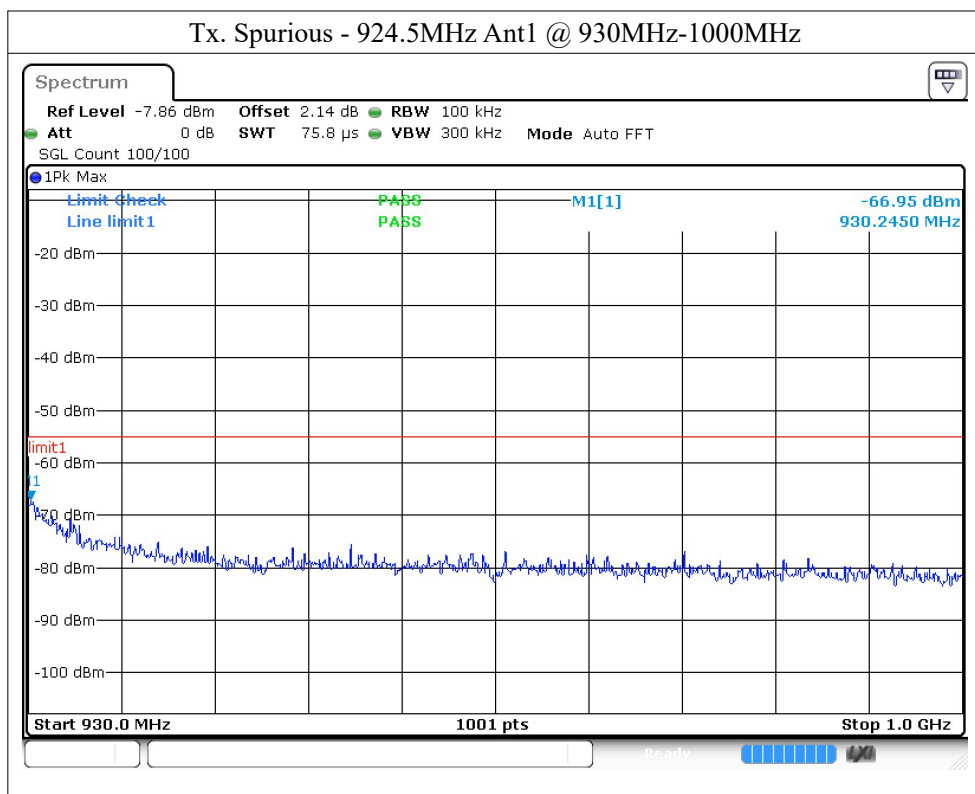
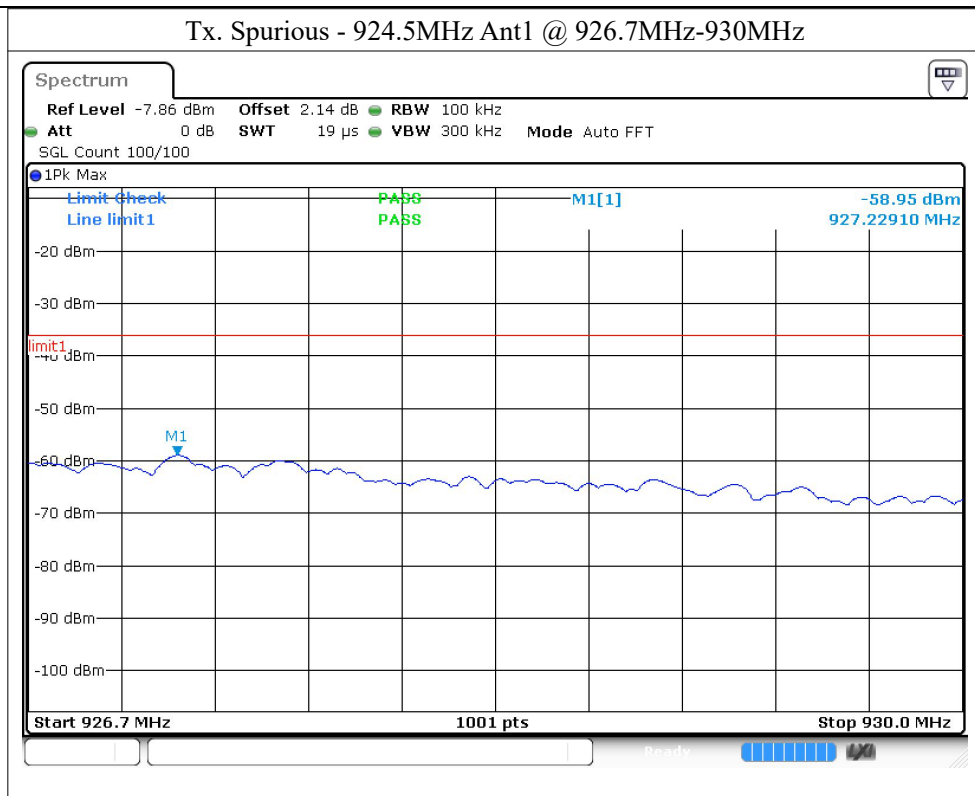


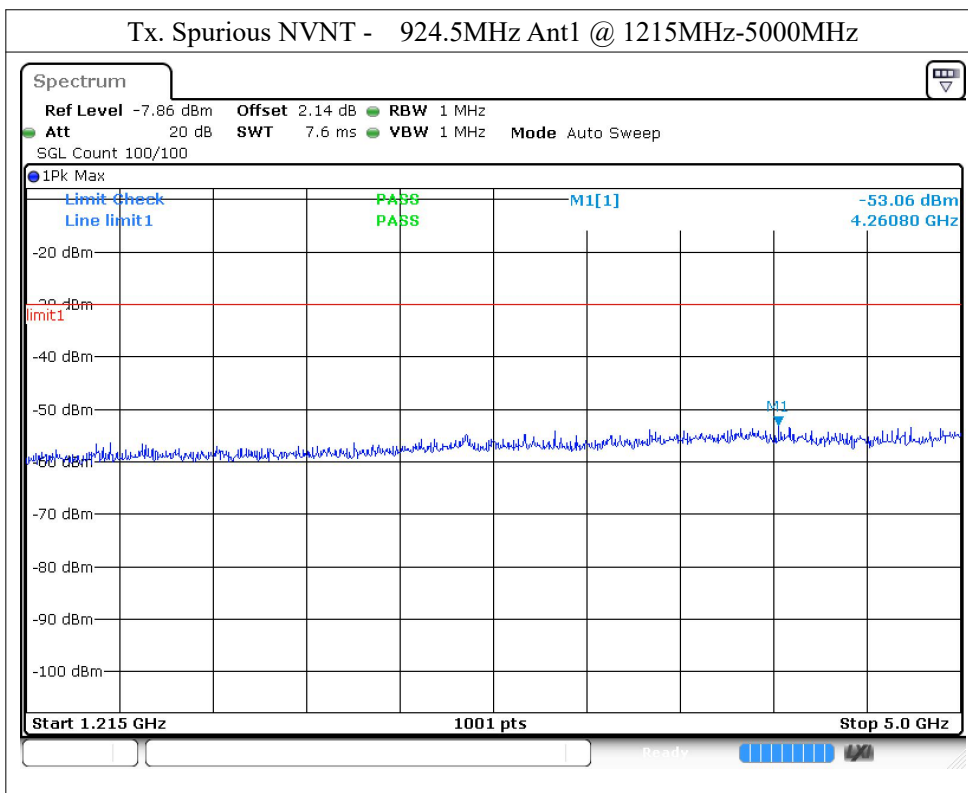
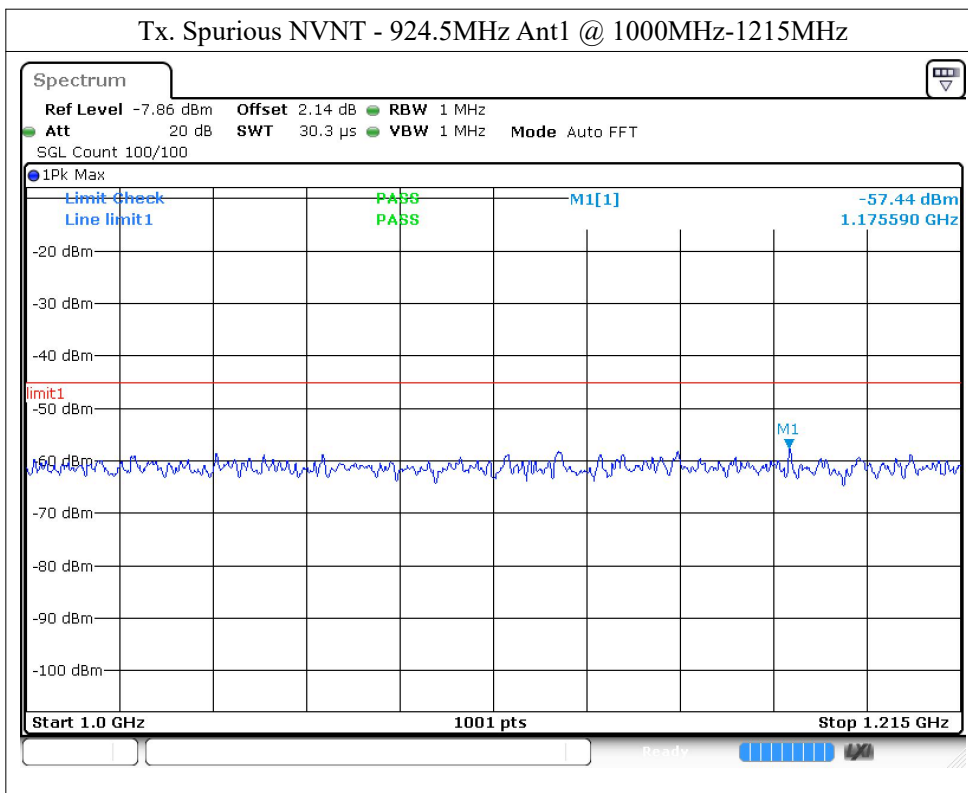
Tx. Spurious - 924.5MHz Ant1 @ 710MHz-900MHz



Tx. Spurious - 924.5MHz Ant1 @ 900MHz-915MHz







6.5 Spurious Emissions for Receiver

6.5.1 Limits of Spurious Emissions for Receiver

Frequency Band	Limit on Secondary Radiated Emissions, etc. (Antenna input)	Reference bandwidth
$f \leq 710 \text{ MHz}$	-54 dBm	100kHz
$710 \text{ MHz} < f \leq 900 \text{ MHz}$	-55 dBm	1MHz
$900 \text{ MHz} < f \leq 915 \text{ MHz}$	-55 dBm	100kHz
$915 \text{ MHz} < f \leq 930 \text{ MHz}$	-54 dBm	100kHz
$930 \text{ MHz} < f \leq 1000 \text{ MHz}$	-55 dBm	100kHz
$1000 \text{ MHz} < f$	-47 dBm	1MHz

6.5.2 Test Procedure

(1) Connected the EUT's antenna port to the Spectrum Analyzer by suitable attenuator, set the Spectrum Analyzer as below:

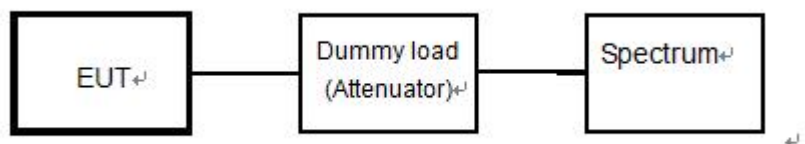
Detector: Peak.

Trace Mode: Max Hold.

(2) When the trace is complete, search the frequency of peak power

(3) All the emissions from 30 MHz to 5 GHz were measured and record.

6.5.3 TEST SETUP LAYOUT



6.5.4 Environmental Conditions

Temperature:	22 °C
Relative Humidity:	64%
ATM Pressure:	1010 mbar

6.5.5 TEST RESULT

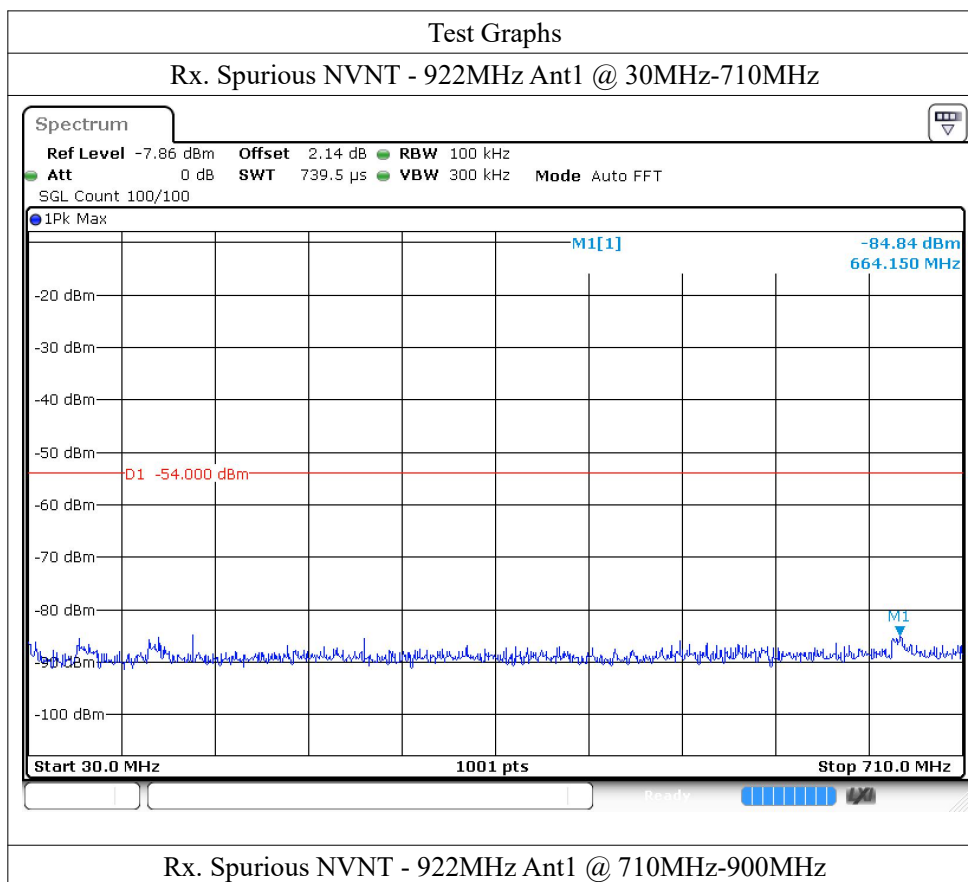
Note: The report shows the worst test diagram for NV testing.

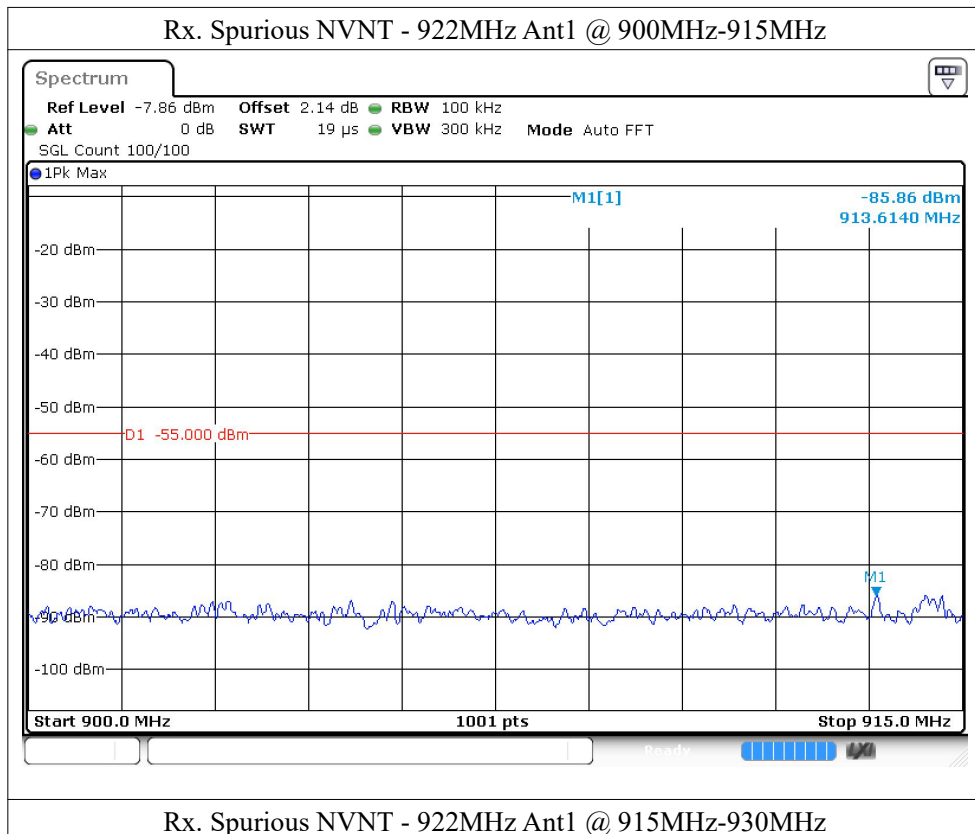
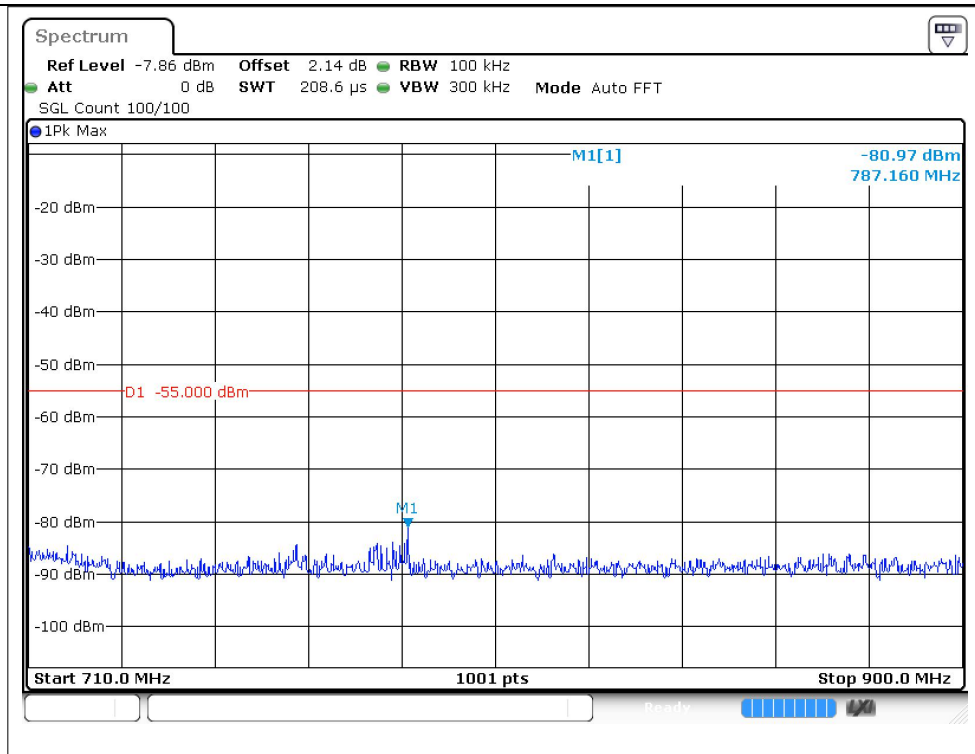
SRD 1 MHz Nss1,(3Mbps)1RX

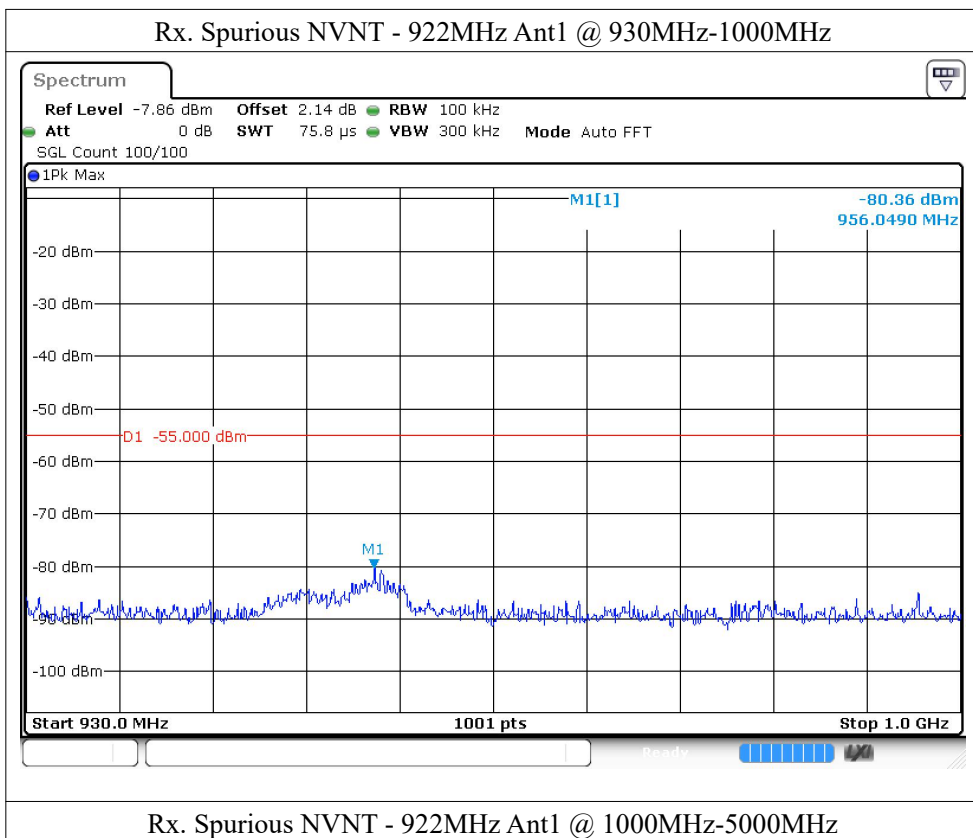
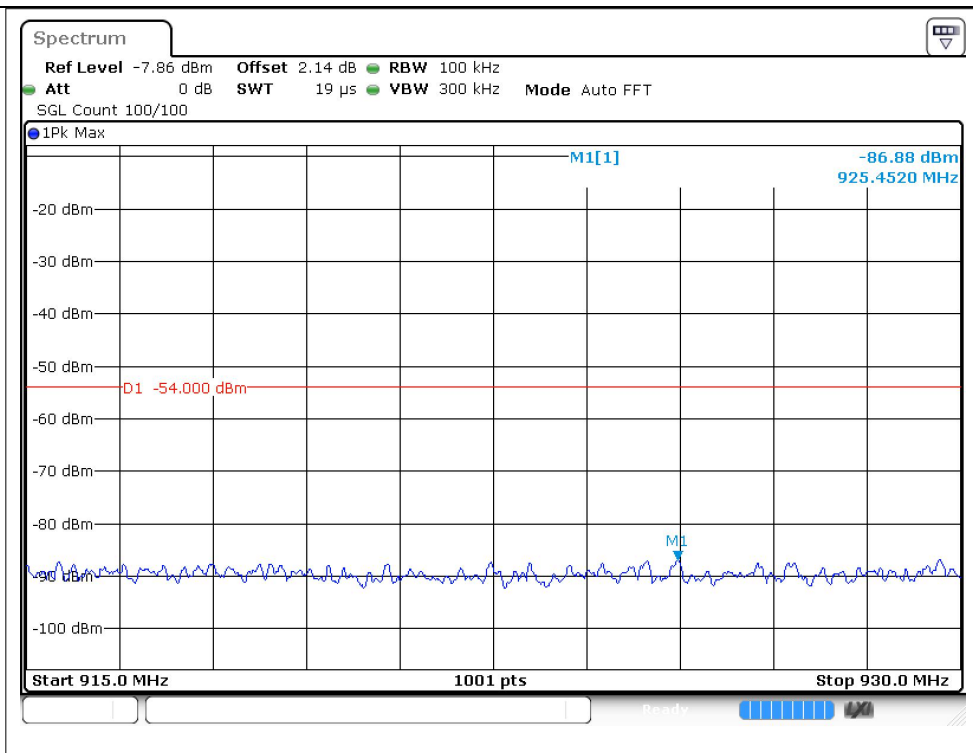
Condition	Frequency (MHz)	Range (MHz)	Emission Frequency (MHz)	Max Value (dBm)	Limit Start(dBm)	Limit Stop (dBm)	Verdict
NVNT	922	30-710	664.15	-84.84	-54	-54	Pass
NVNT	922	710-900	787.16	-80.97	-55	-55	Pass
NVNT	922	900-915	913.61	-85.86	-55	-55	Pass
NVNT	922	915-930	925.45	-86.88	-54	-54	Pass
NVNT	922	930-1000	956.05	-80.36	-55	-55	Pass
NVNT	922	1000-5000	4786.2	-72.93	-47	-47	Pass
NVNT	924	30-710	149.9	-84.58	-54	-54	Pass
NVNT	924	710-900	710.28	-84.3	-55	-55	Pass
NVNT	924	900-915	901.84	-86.15	-55	-55	Pass
NVNT	924	915-930	917.34	-85.99	-54	-54	Pass
NVNT	924	930-1000	954.44	-83.01	-55	-55	Pass
NVNT	924	1000-5000	3911.1	-72.71	-47	-47	Pass
NVNT	926	30-710	149.9	-84.64	-54	-54	Pass
NVNT	926	710-900	711.61	-84.27	-55	-55	Pass
NVNT	926	900-915	914.63	-85.94	-55	-55	Pass
NVNT	926	915-930	916.63	-86.31	-54	-54	Pass
NVNT	926	930-1000	956.54	-82.59	-55	-55	Pass
NVNT	926	1000-5000	2416.6	-69.27	-47	-47	Pass
LVNT	922	30-710	663.75	-85.51	-54	-54	Pass
LVNT	922	710-900	786.29	-81.85	-55	-55	Pass
LVNT	922	900-915	913.45	-86.83	-55	-55	Pass
LVNT	922	915-930	925.25	-87.14	-54	-54	Pass
LVNT	922	930-1000	955.67	-80.46	-55	-55	Pass
LVNT	922	1000-5000	4785.32	-73.82	-47	-47	Pass
LVNT	924	30-710	149.48	-85.30	-54	-54	Pass
LVNT	924	710-900	710.23	-84.96	-55	-55	Pass
LVNT	924	900-915	900.91	-86.34	-55	-55	Pass
LVNT	924	915-930	916.51	-86.03	-54	-54	Pass
LVNT	924	930-1000	953.71	-83.64	-55	-55	Pass
LVNT	924	1000-5000	3910.38	-73.35	-47	-47	Pass
LVNT	926	30-710	149.04	-84.91	-54	-54	Pass
LVNT	926	710-900	711.21	-84.30	-55	-55	Pass
LVNT	926	900-915	913.70	-86.10	-55	-55	Pass
LVNT	926	915-930	915.64	-86.54	-54	-54	Pass
LVNT	926	930-1000	956.30	-83.19	-55	-55	Pass
LVNT	926	1000-5000	2415.81	-69.83	-47	-47	Pass
HVNT	922	30-710	663.86	-85.08	-54	-54	Pass
HVNT	922	710-900	786.55	-81.76	-55	-55	Pass
HVNT	922	900-915	913.09	-86.46	-55	-55	Pass

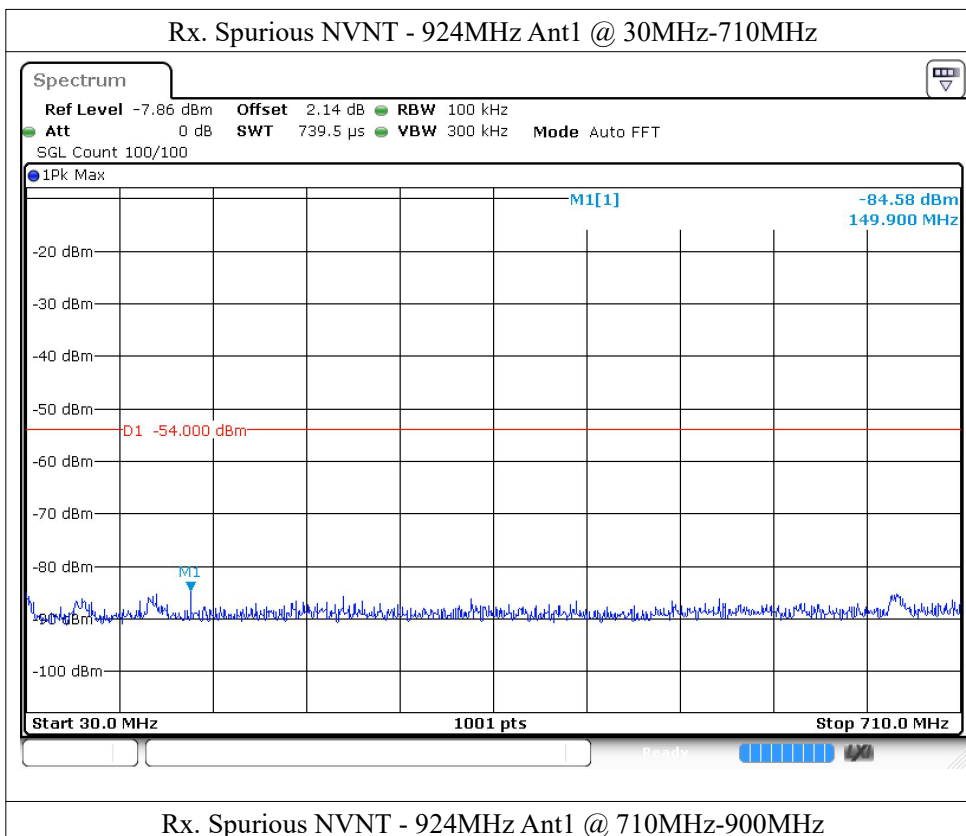
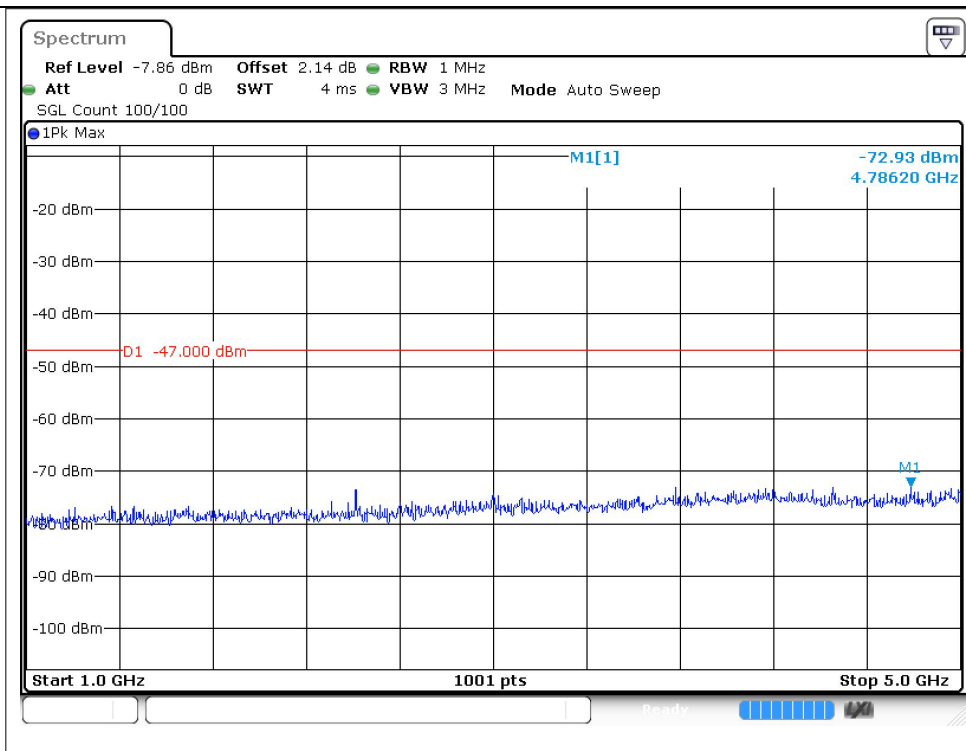
HVNT	922	915-930	925.03	-87.02	-54	-54	Pass
HVNT	922	930-1000	955.51	-80.62	-55	-55	Pass
HVNT	922	1000-5000	4785.90	-73.91	-47	-47	Pass
HVNT	924	30-710	148.94	-84.88	-54	-54	Pass
HVNT	924	710-900	709.64	-84.81	-55	-55	Pass
HVNT	924	900-915	901.64	-86.75	-55	-55	Pass
HVNT	924	915-930	917.00	-86.43	-54	-54	Pass
HVNT	924	930-1000	954.07	-83.87	-55	-55	Pass
HVNT	924	1000-5000	3910.53	-73.39	-47	-47	Pass
HVNT	926	30-710	149.84	-85.24	-54	-54	Pass
HVNT	926	710-900	711.57	-84.96	-55	-55	Pass
HVNT	926	900-915	913.83	-86.93	-55	-55	Pass
HVNT	926	915-930	916.29	-87.12	-54	-54	Pass
HVNT	926	930-1000	956.01	-82.87	-55	-55	Pass
HVNT	926	1000-5000	2415.98	-69.96	-47	-47	Pass

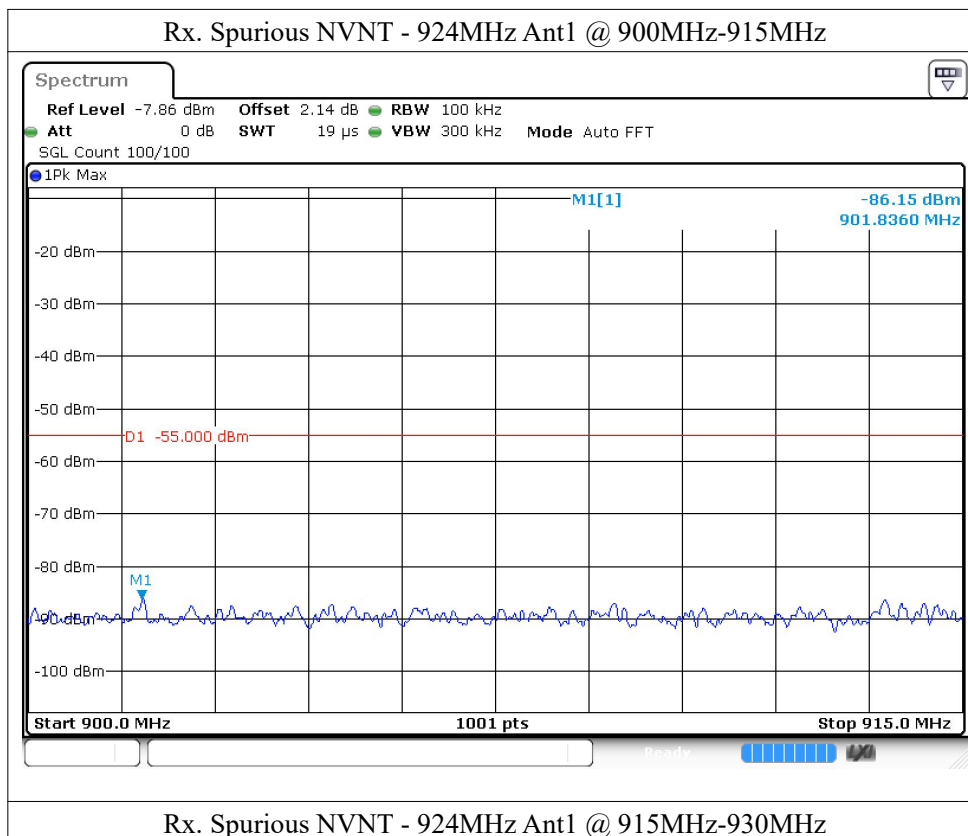
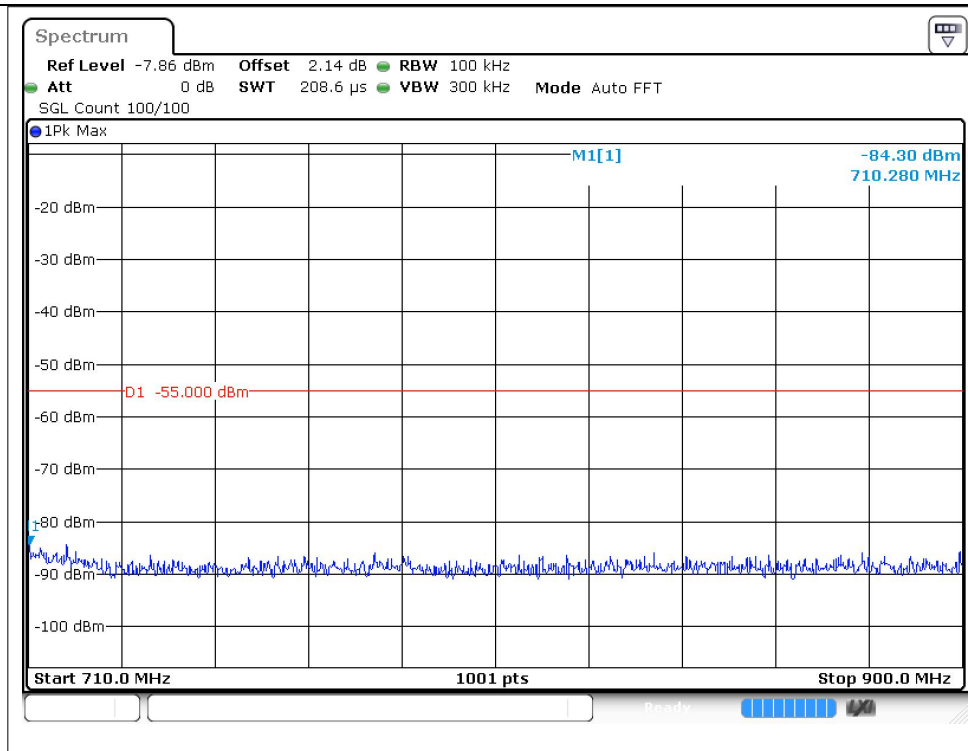
NV

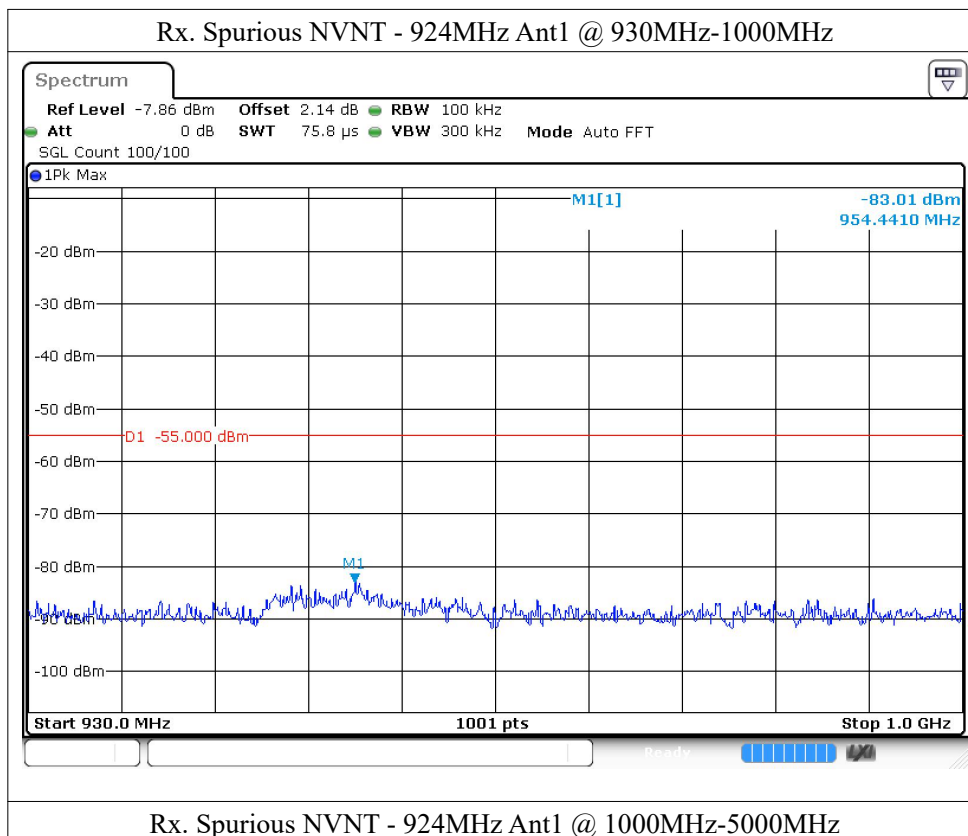
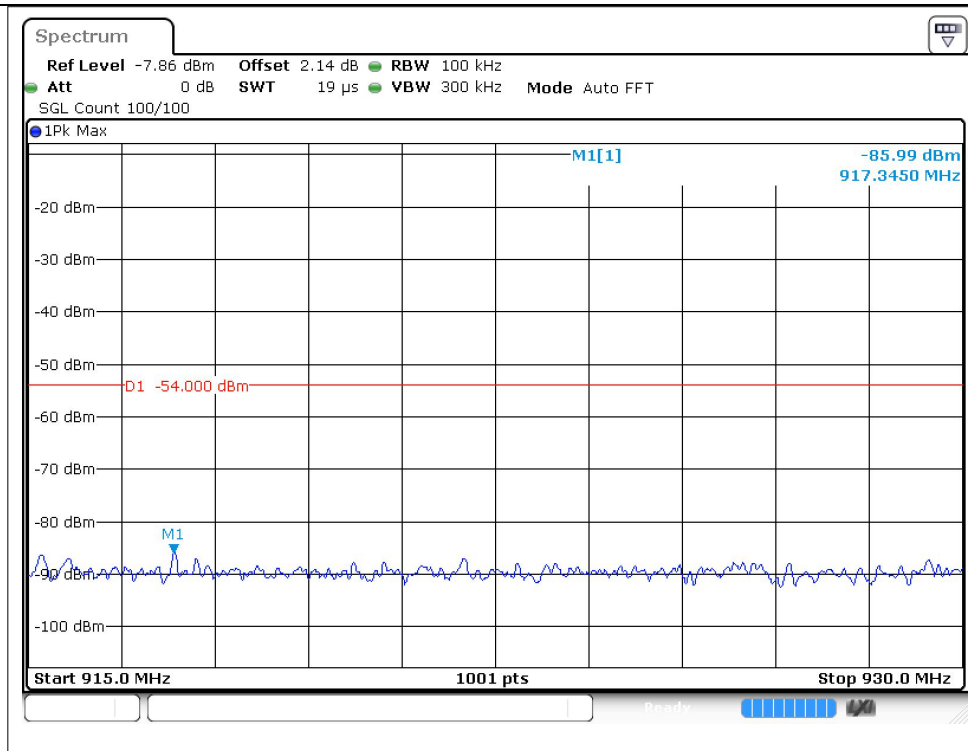


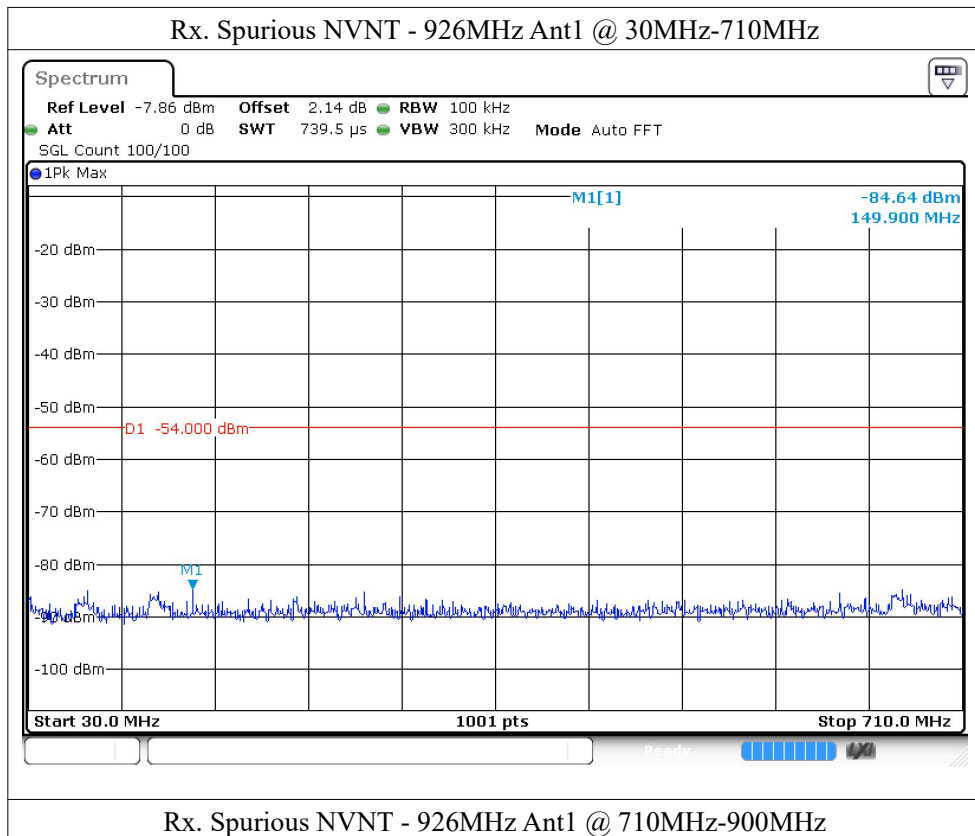
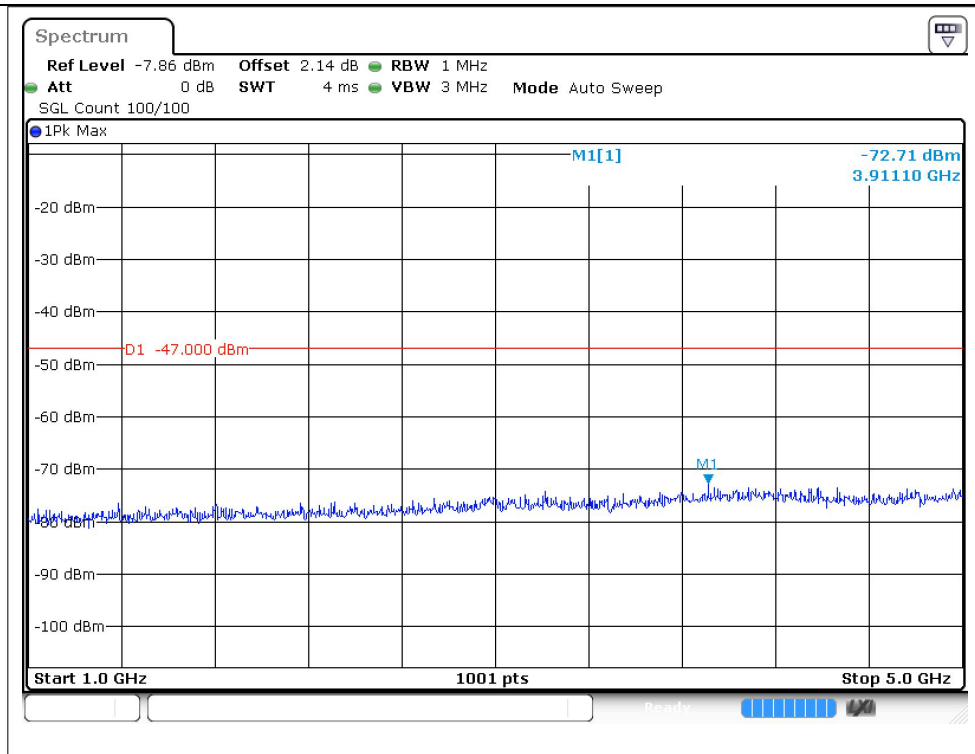


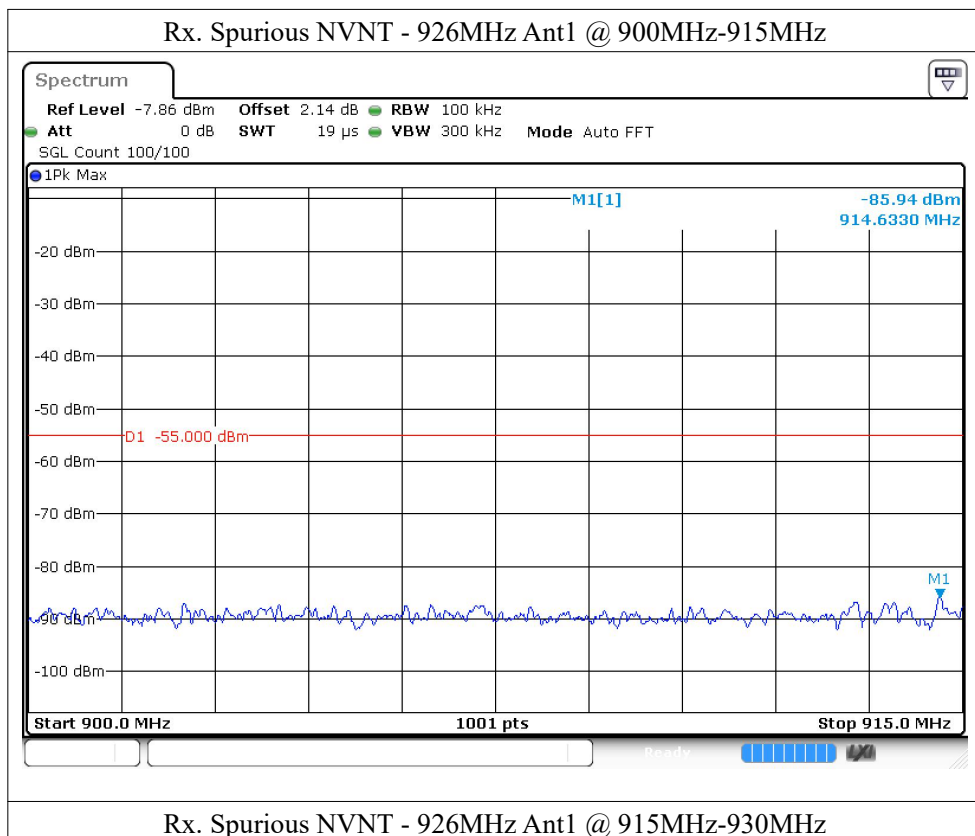
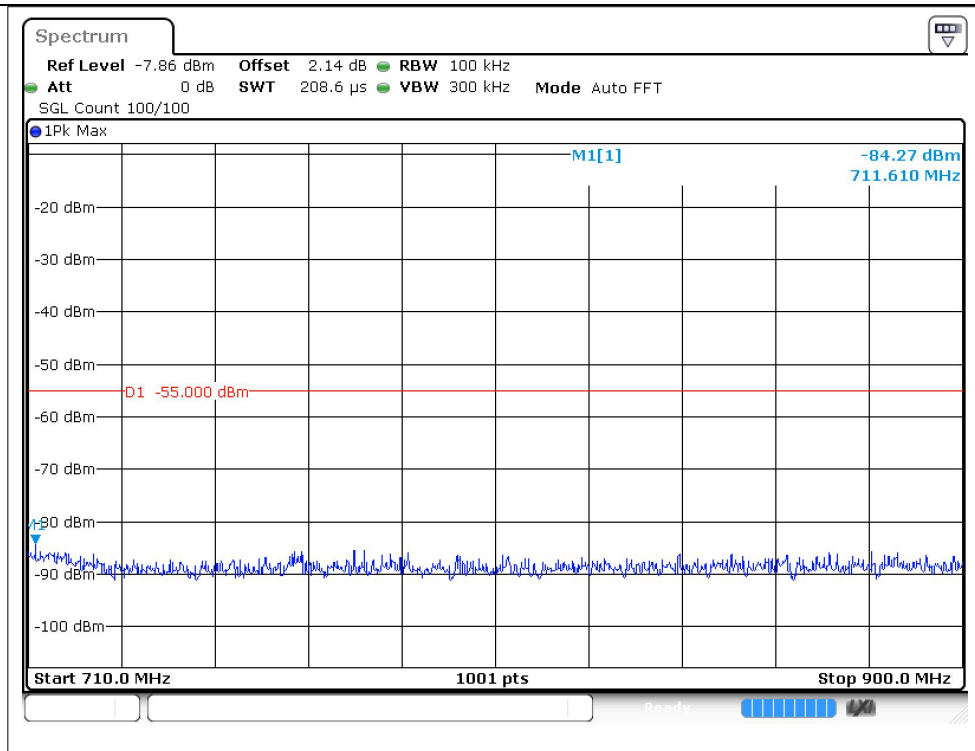


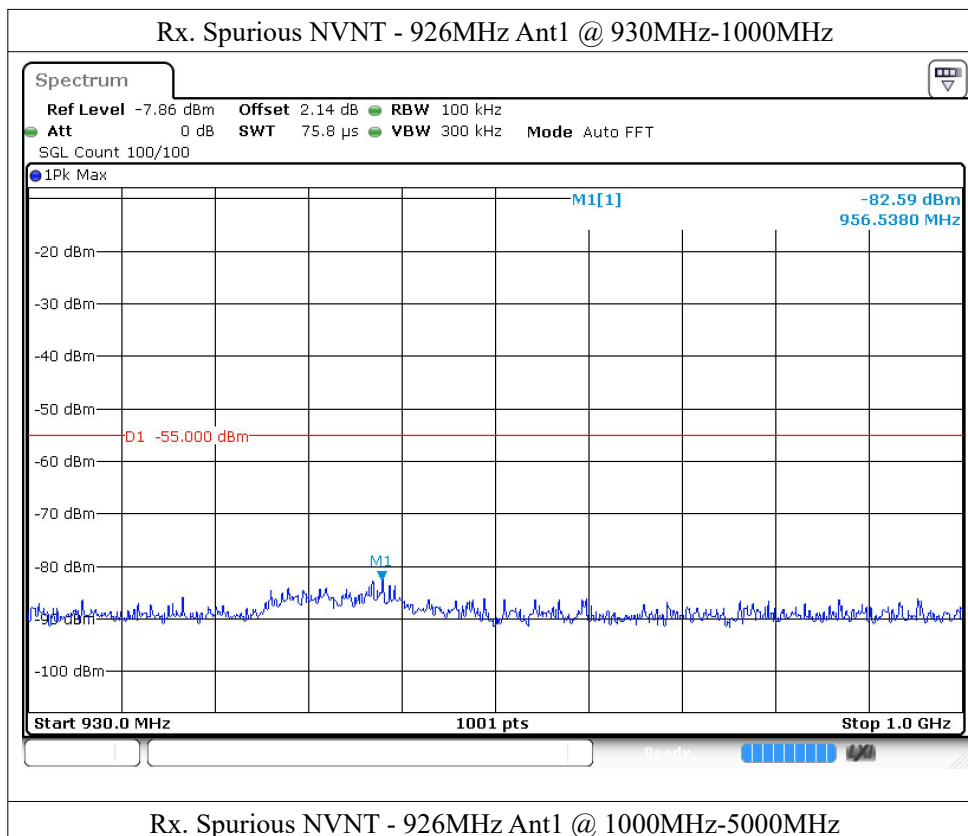
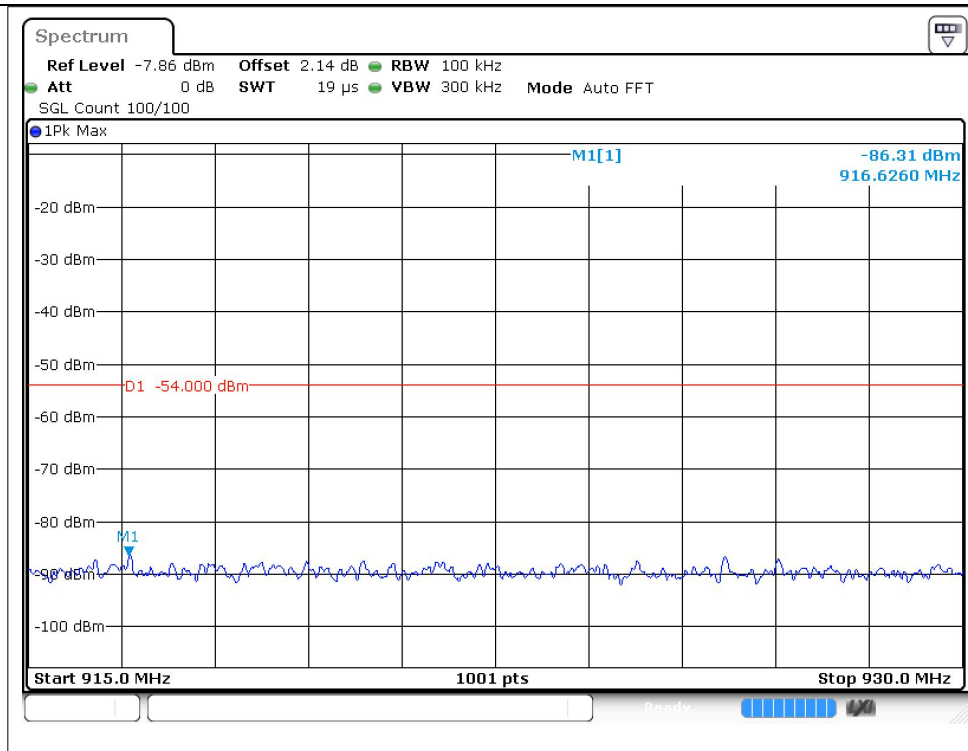


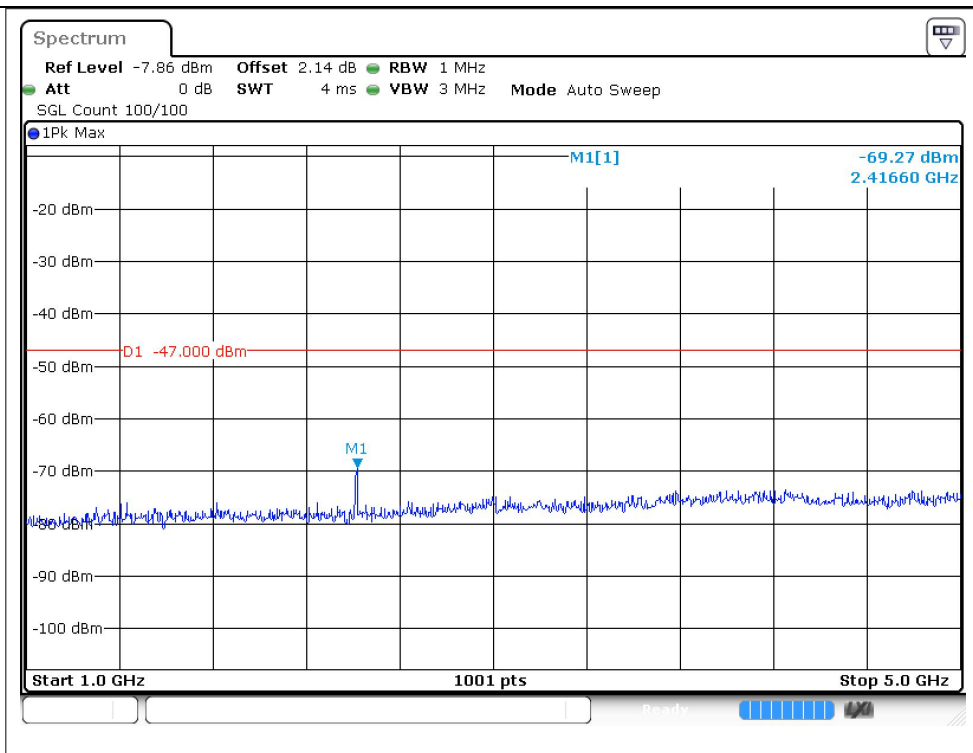










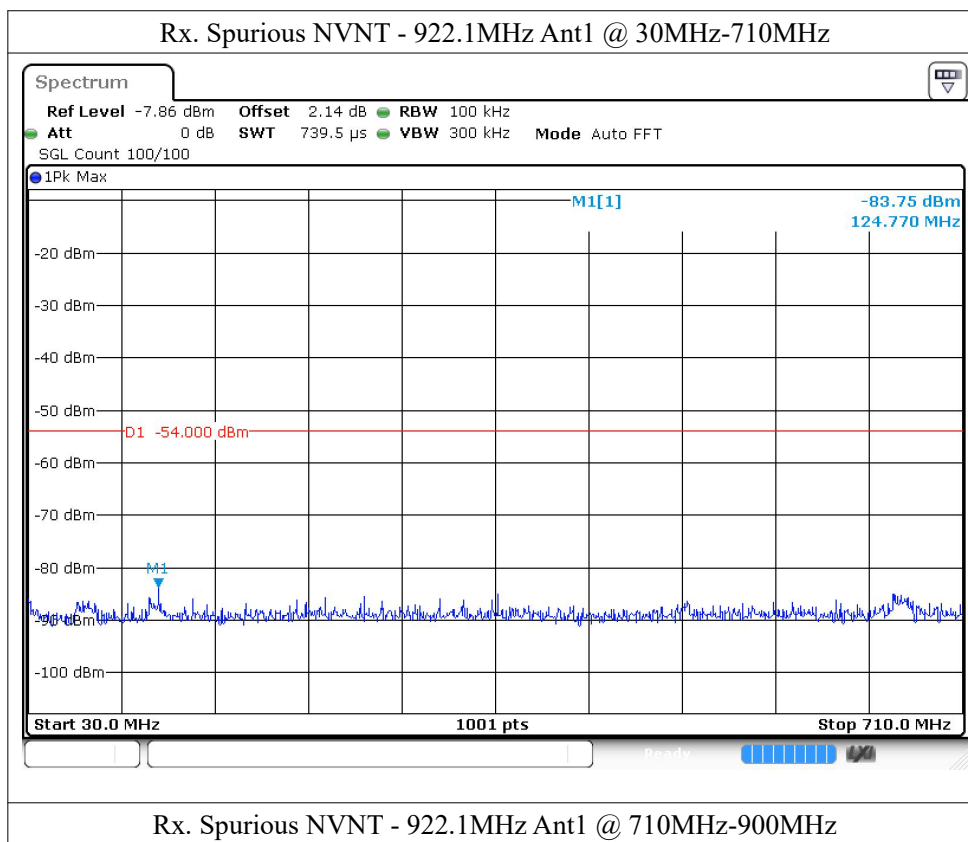


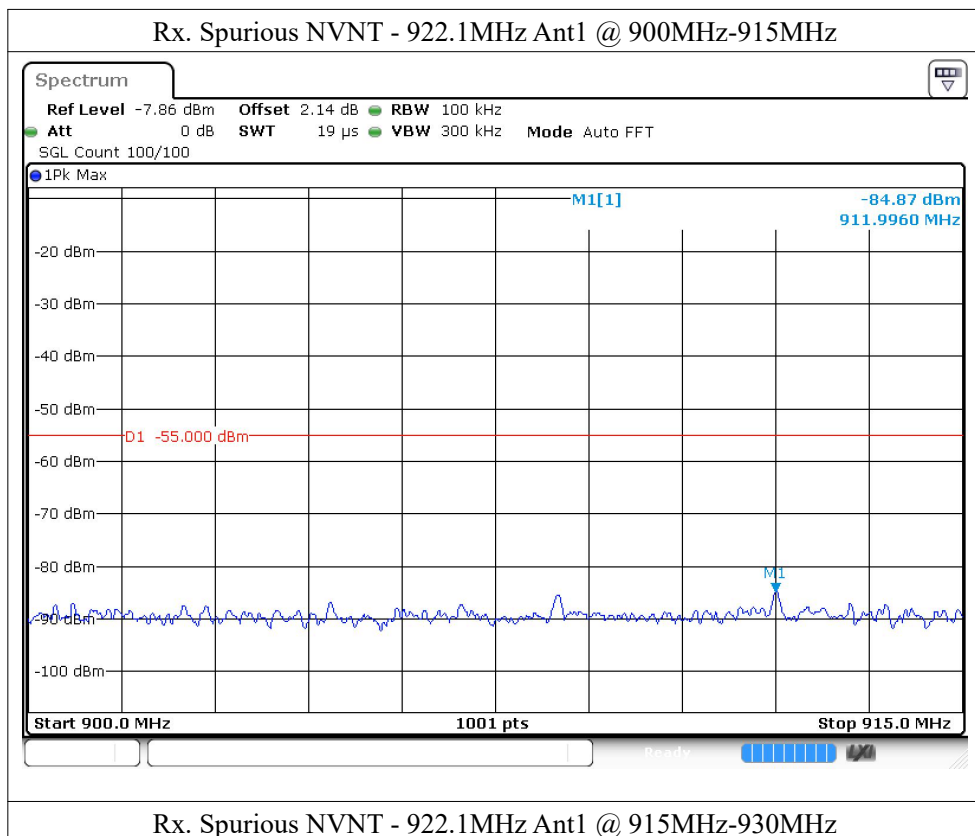
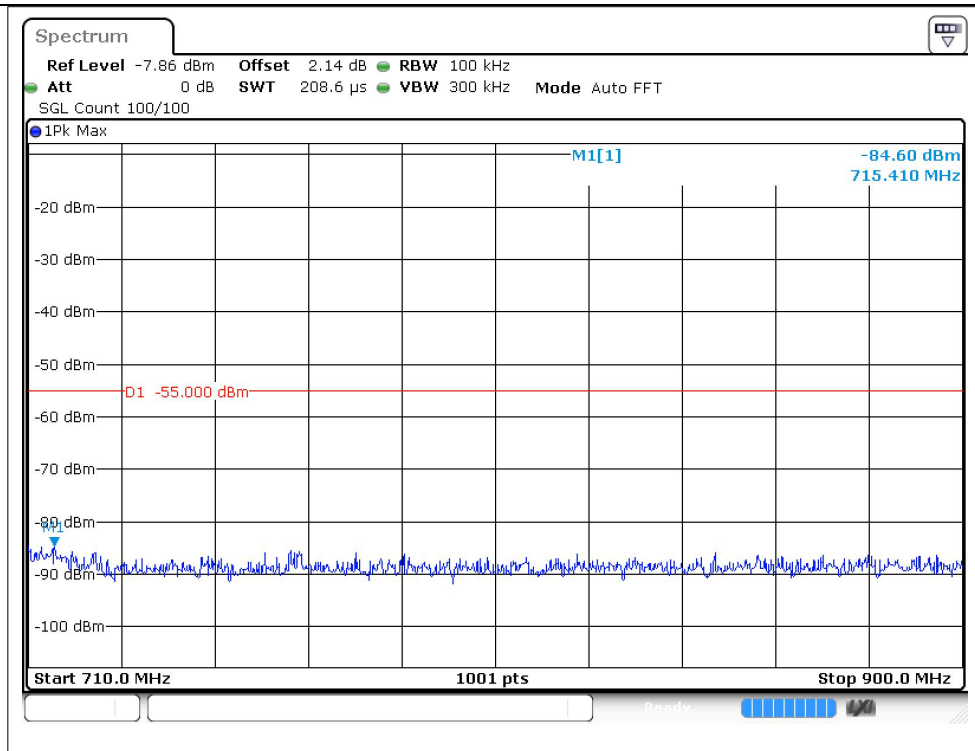
SRD 2MHz Nss1,(6.5Mbps)1RX

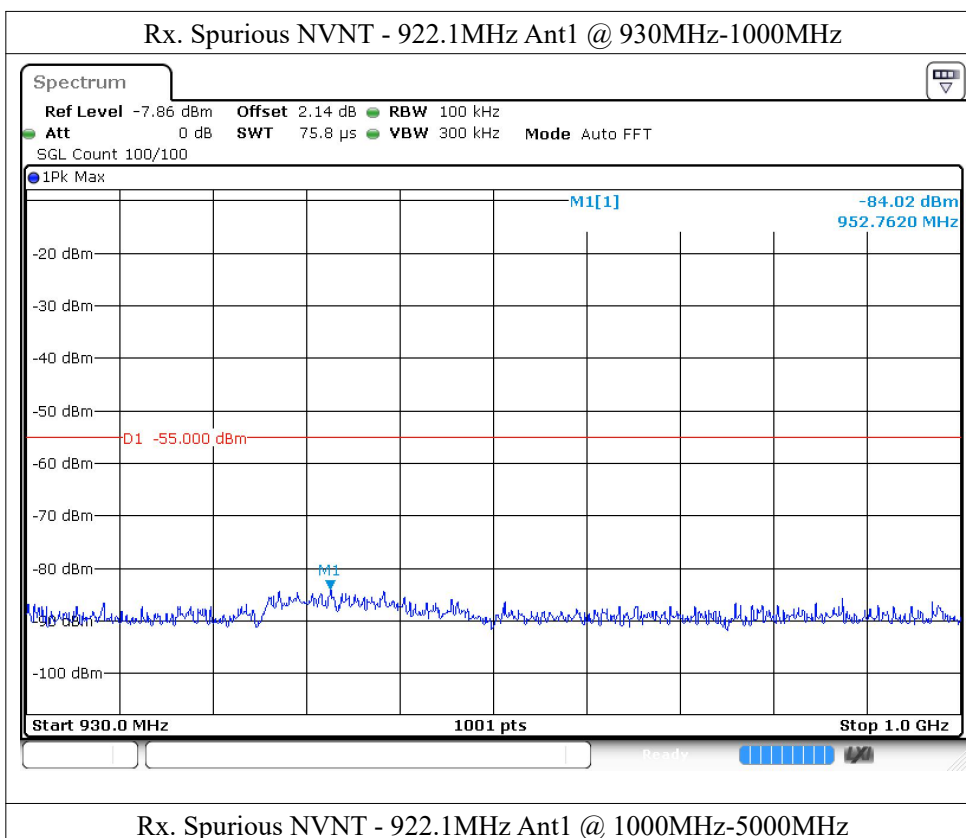
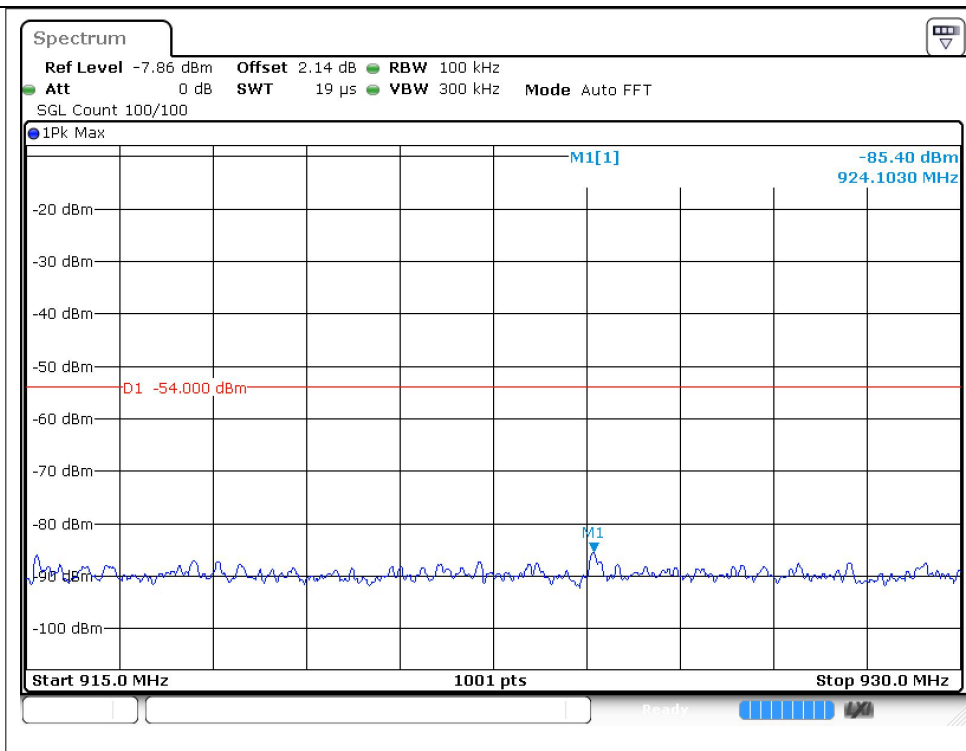
Condition	Frequency (MHz)	Range (MHz)	Emission Frequency (MHz)	Max Value (dBm)	Limit Start(dBm)	Limit Stop (dBm)	Verdict
NVNT	922.1	30-710	124.77	-83.75	-54	-54	Pass
NVNT	922.1	710-900	715.41	-84.6	-55	-55	Pass
NVNT	922.1	900-915	912	-84.87	-55	-55	Pass
NVNT	922.1	915-930	924.1	-85.4	-54	-54	Pass
NVNT	922.1	930-1000	952.76	-84.02	-55	-55	Pass
NVNT	922.1	1000-5000	2412.6	-68.09	-47	-47	Pass
NVNT	924.3	30-710	658.71	-84.58	-54	-54	Pass
NVNT	924.3	710-900	764.57	-83.58	-55	-55	Pass
NVNT	924.3	900-915	906.09	-86.61	-55	-55	Pass
NVNT	924.3	915-930	922.47	-86.43	-54	-54	Pass
NVNT	924.3	930-1000	950.1	-82.52	-55	-55	Pass
NVNT	924.3	1000-5000	2412.6	-69.94	-47	-47	Pass
NVNT	926.5	30-710	660.07	-83.8	-54	-54	Pass
NVNT	926.5	710-900	763.81	-84.02	-55	-55	Pass
NVNT	926.5	900-915	900.89	-85.93	-55	-55	Pass
NVNT	926.5	915-930	915.05	-86.25	-54	-54	Pass
NVNT	926.5	930-1000	956.33	-82.98	-55	-55	Pass
NVNT	926.5	1000-5000	2416.6	-70.66	-47	-47	Pass
LVNT	922.1	30-710	123.90	-84.56	-54	-54	Pass
LVNT	922.1	710-900	715.27	-84.62	-55	-55	Pass
LVNT	922.1	900-915	911.16	-84.94	-55	-55	Pass
LVNT	922.1	915-930	924.06	-85.95	-54	-54	Pass
LVNT	922.1	930-1000	952.21	-84.52	-55	-55	Pass
LVNT	922.1	1000-5000	2411.82	-69.07	-47	-47	Pass
LVNT	924.3	30-710	657.88	-85.48	-54	-54	Pass
LVNT	924.3	710-900	763.57	-84.25	-55	-55	Pass
LVNT	924.3	900-915	905.41	-86.63	-55	-55	Pass
LVNT	924.3	915-930	922.03	-87.09	-54	-54	Pass
LVNT	924.3	930-1000	949.41	-82.69	-55	-55	Pass
LVNT	924.3	1000-5000	2411.85	-69.98	-47	-47	Pass
LVNT	926.5	30-710	659.55	-84.25	-54	-54	Pass
LVNT	926.5	710-900	763.68	-84.28	-55	-55	Pass
LVNT	926.5	900-915	899.97	-86.77	-55	-55	Pass
LVNT	926.5	915-930	914.34	-86.51	-54	-54	Pass
LVNT	926.5	930-1000	955.53	-83.94	-55	-55	Pass
LVNT	926.5	1000-5000	2416.24	-71.53	-47	-47	Pass
HVNT	922.1	30-710	123.99	-84.19	-54	-54	Pass
HVNT	922.1	710-900	715.27	-85.59	-55	-55	Pass
HVNT	922.1	900-915	911.87	-85.09	-55	-55	Pass
HVNT	922.1	915-930	923.79	-86.28	-54	-54	Pass
HVNT	922.1	930-1000	952.02	-84.69	-55	-55	Pass
HVNT	922.1	1000-5000	2412.47	-68.95	-47	-47	Pass
HVNT	924.3	30-710	657.73	-84.91	-54	-54	Pass

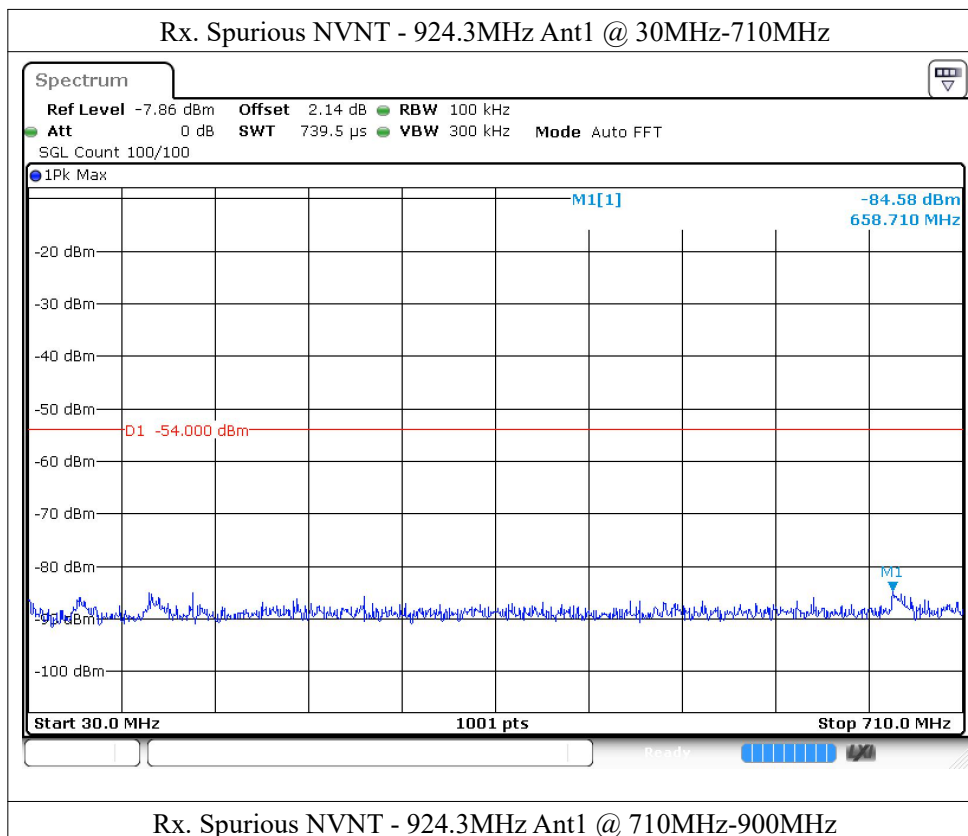
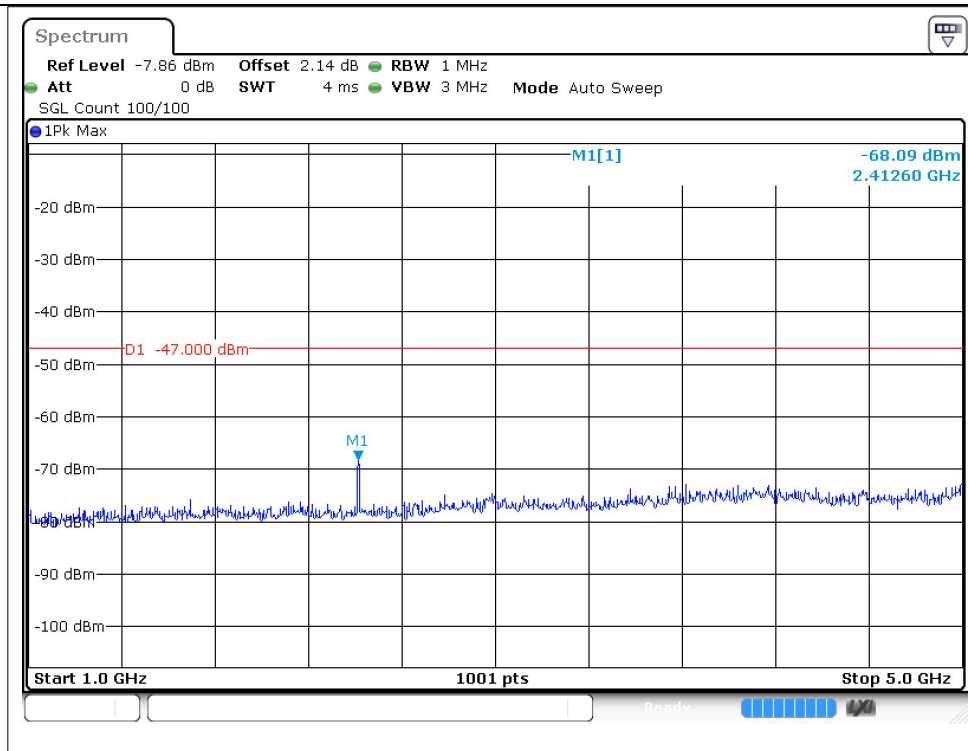
HVNT	924.3	710-900	764.03	-84.54	-55	-55	Pass
HVNT	924.3	900-915	905.84	-86.70	-55	-55	Pass
HVNT	924.3	915-930	922.01	-86.93	-54	-54	Pass
HVNT	924.3	930-1000	949.95	-83.00	-55	-55	Pass
HVNT	924.3	1000-5000	2412.58	-69.99	-47	-47	Pass
HVNT	926.5	30-710	659.70	-84.56	-54	-54	Pass
HVNT	926.5	710-900	763.09	-84.41	-55	-55	Pass
HVNT	926.5	900-915	900.49	-86.26	-55	-55	Pass
HVNT	926.5	915-930	914.59	-86.48	-54	-54	Pass
HVNT	926.5	930-1000	956.12	-83.73	-55	-55	Pass
HVNT	926.5	1000-5000	2415.77	-70.70	-47	-47	Pass

NV

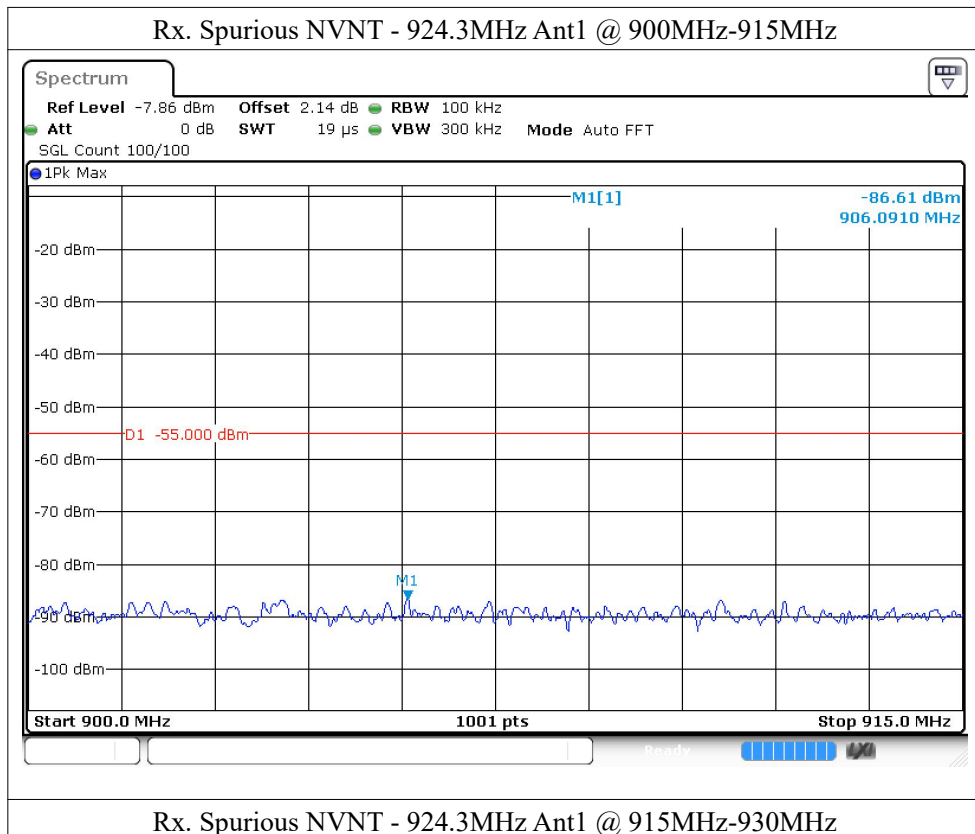
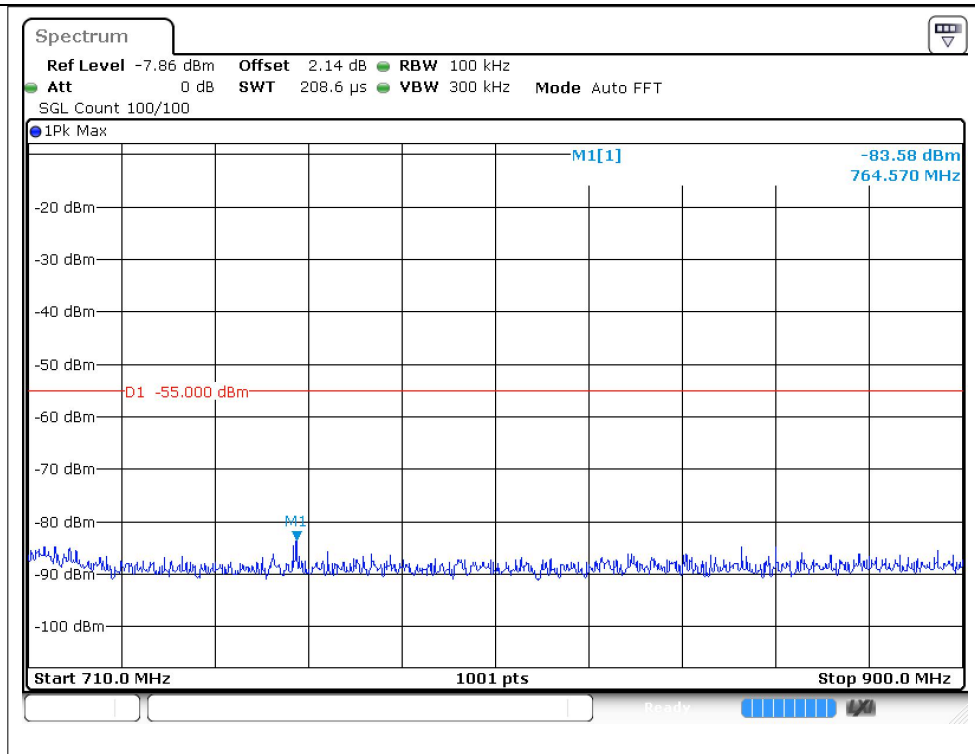


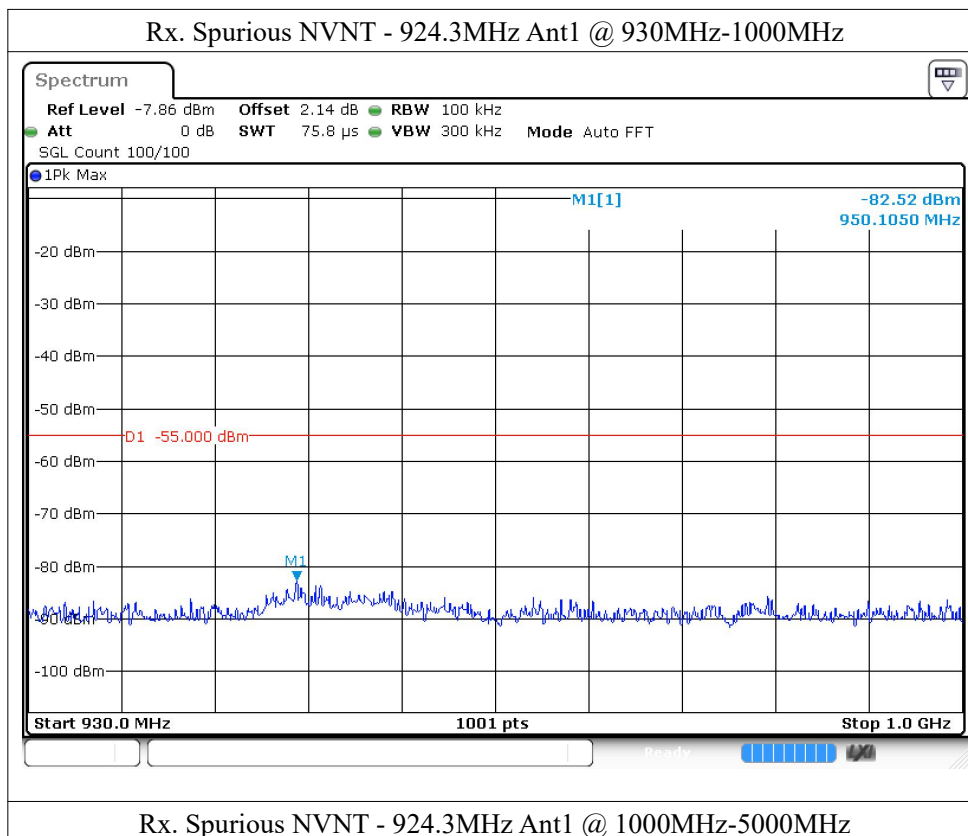
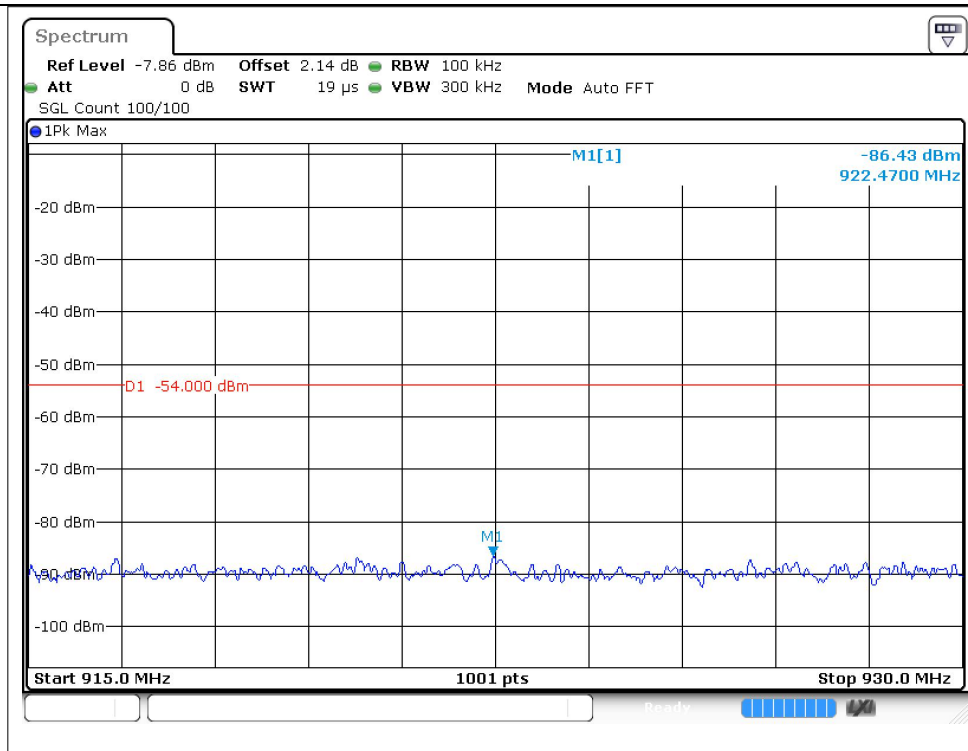


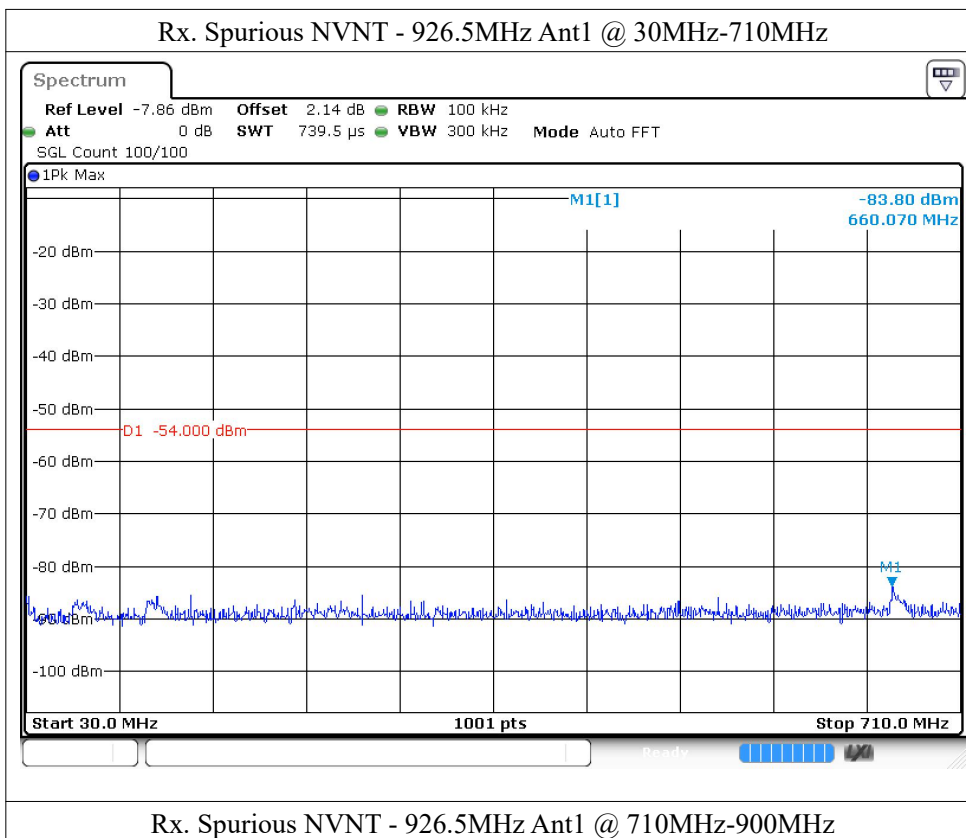
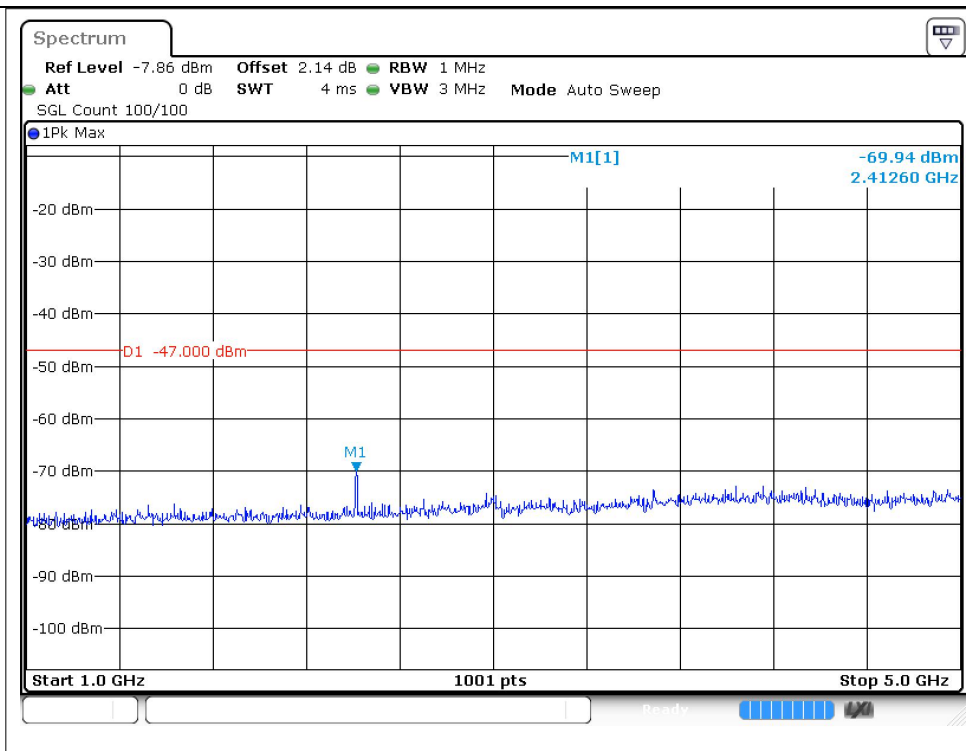


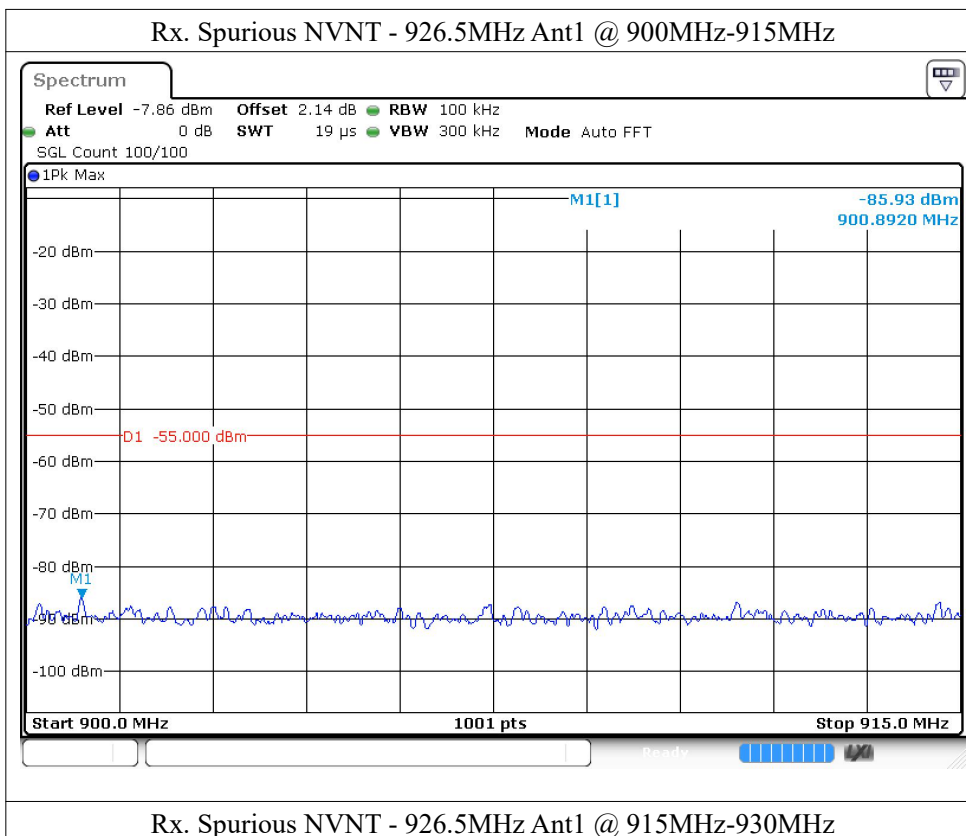
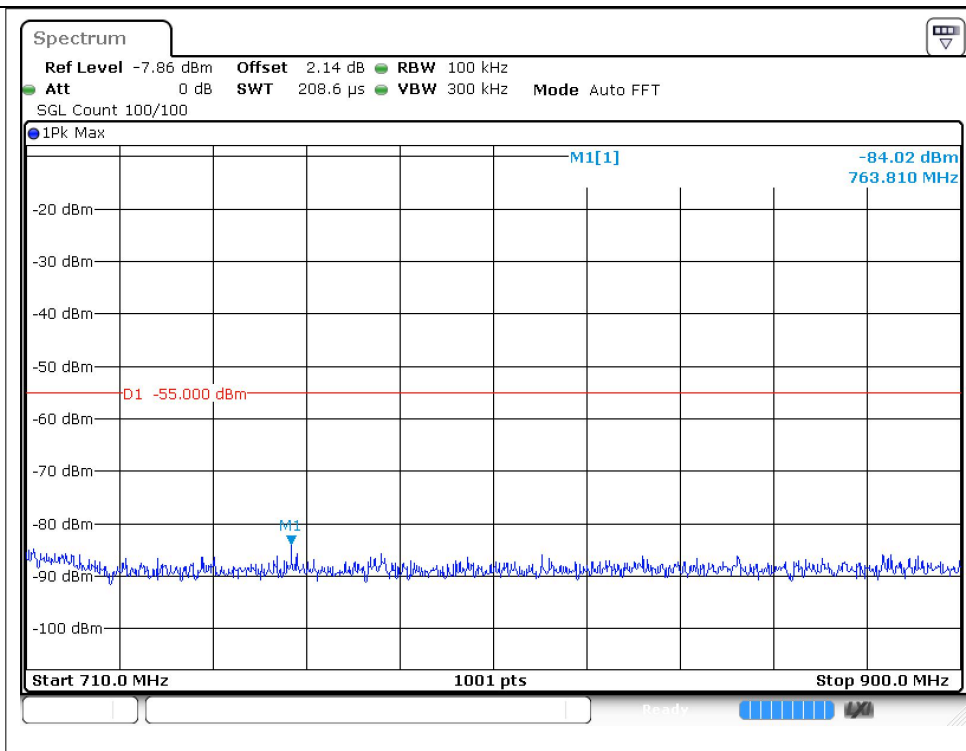


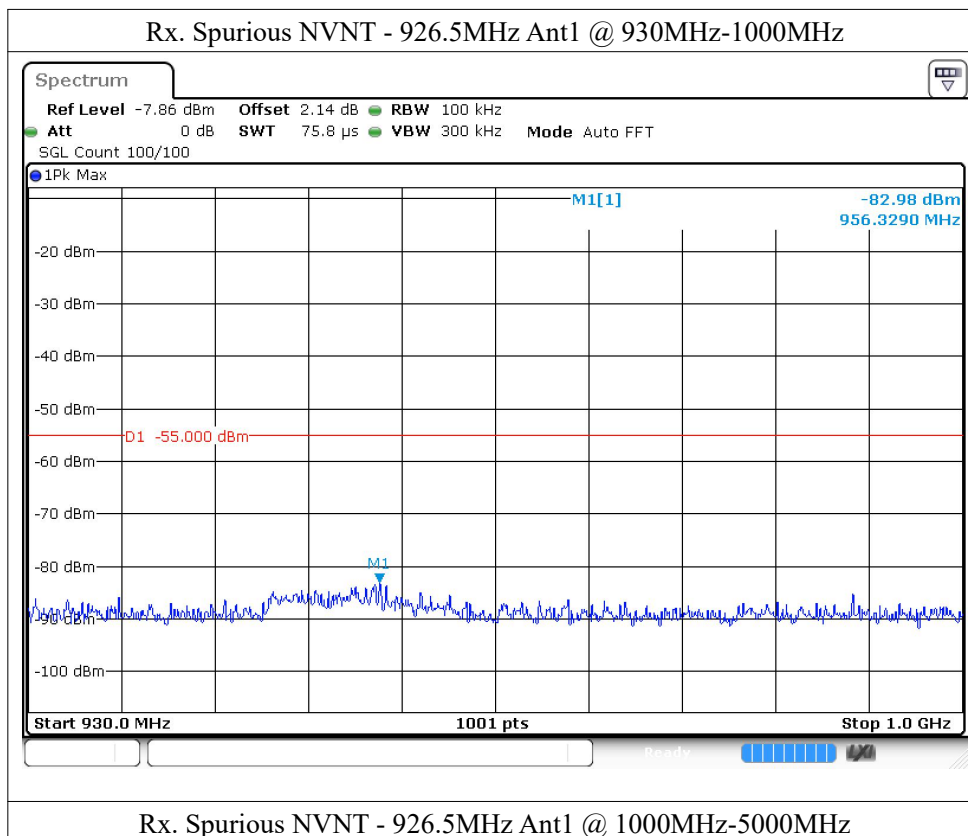
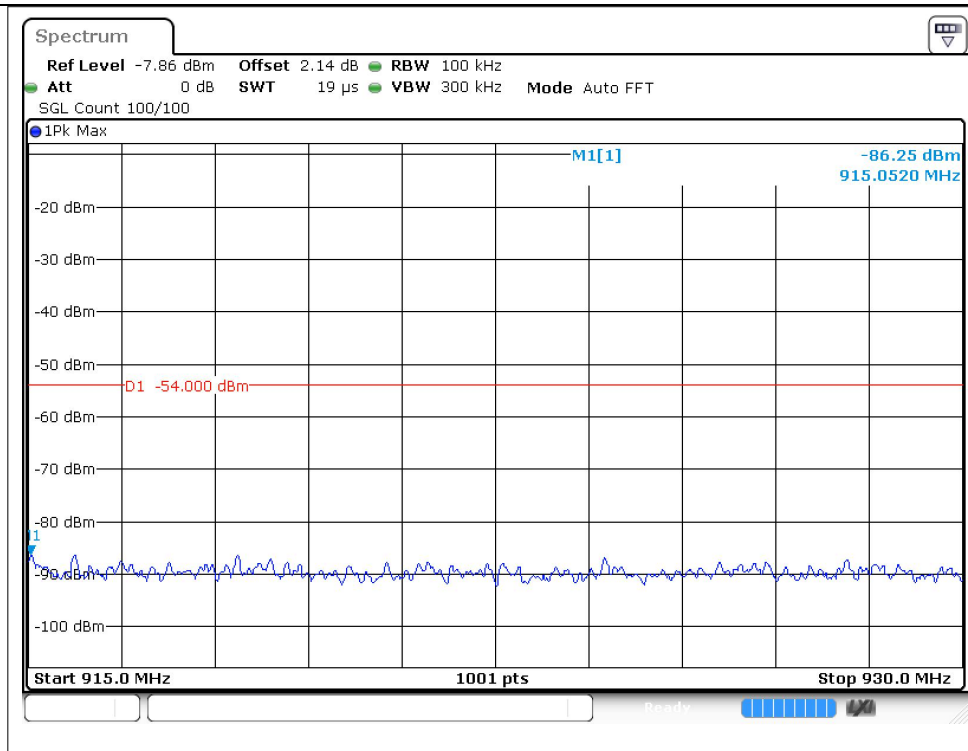
Rx. Spurious NVNT - 924.3MHz Ant1 @ 710MHz-900MHz

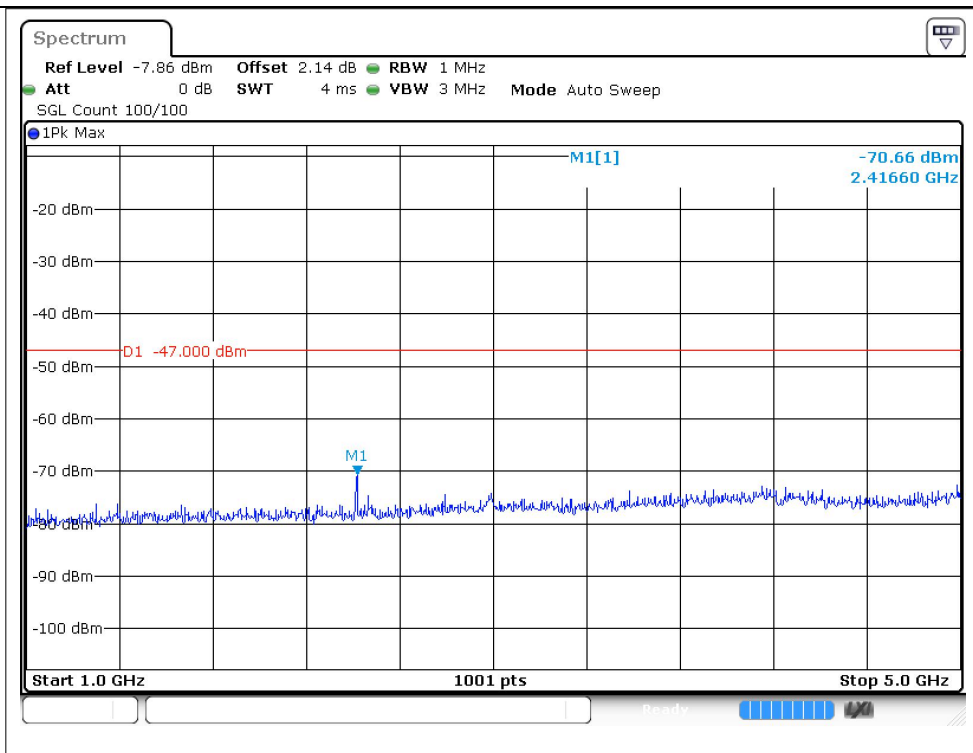








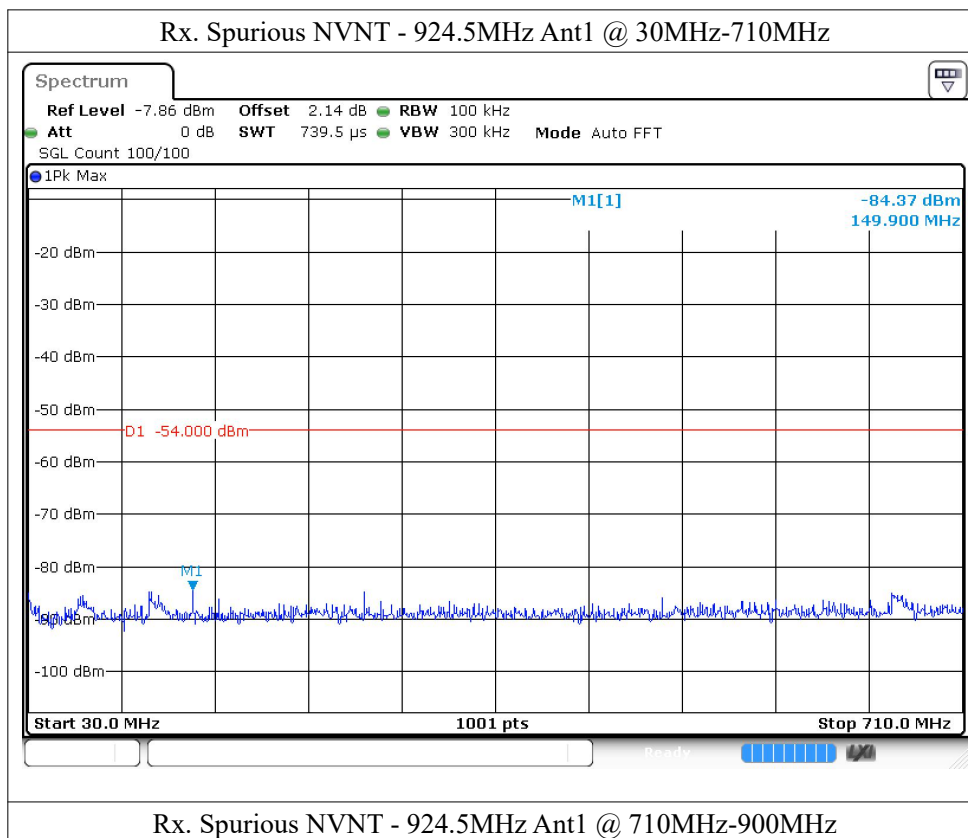


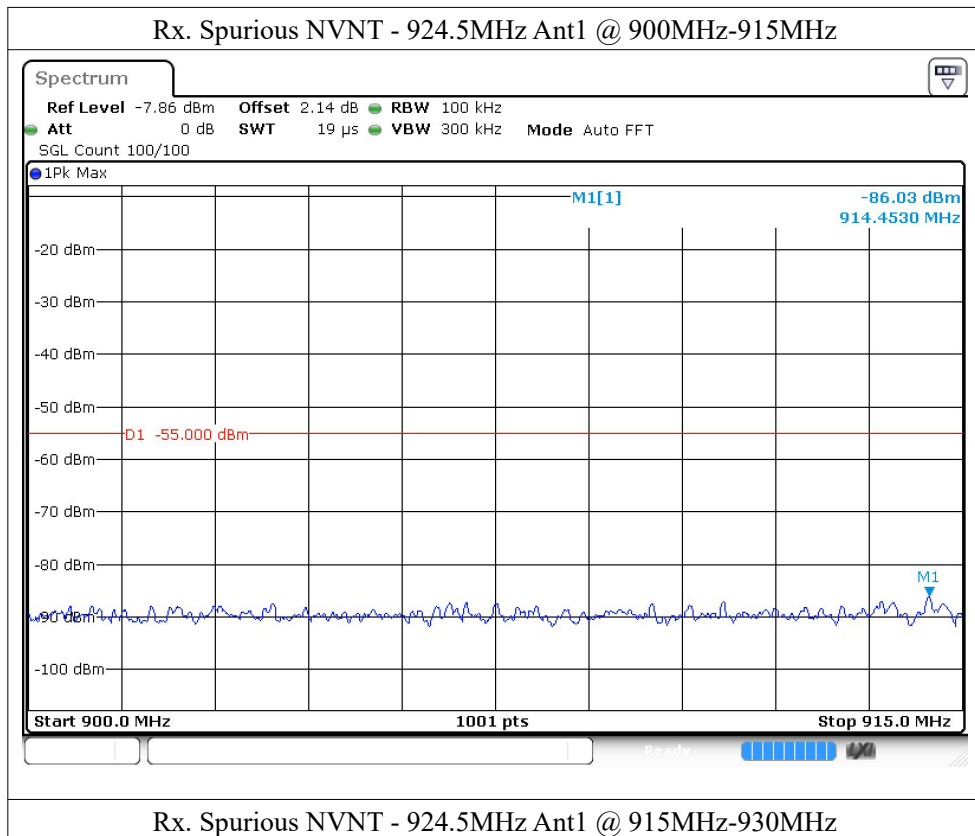
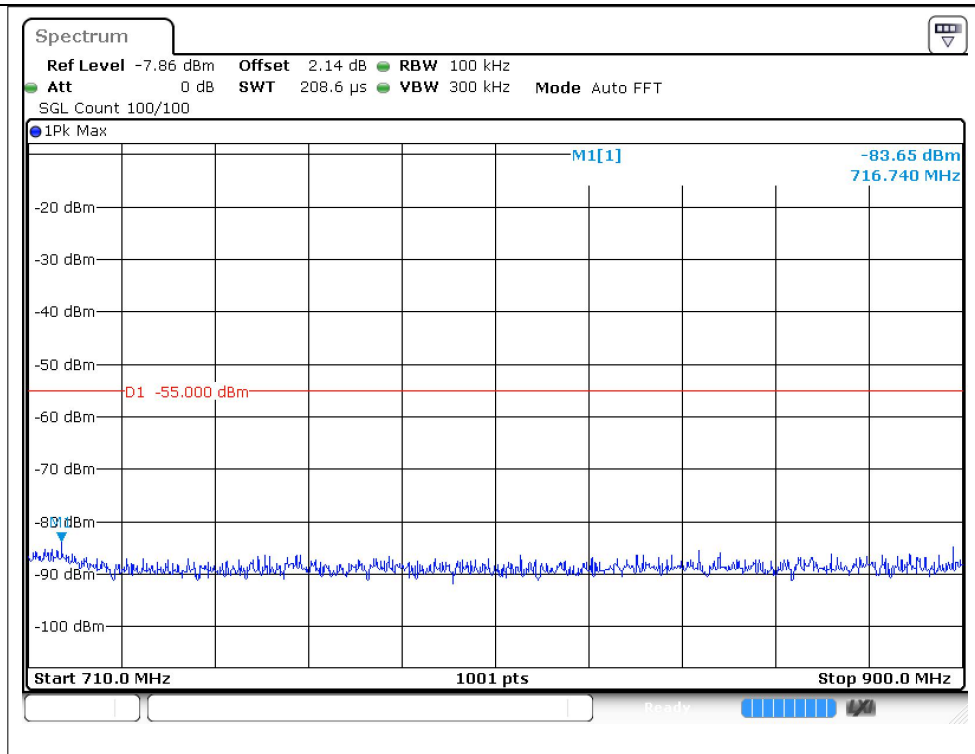


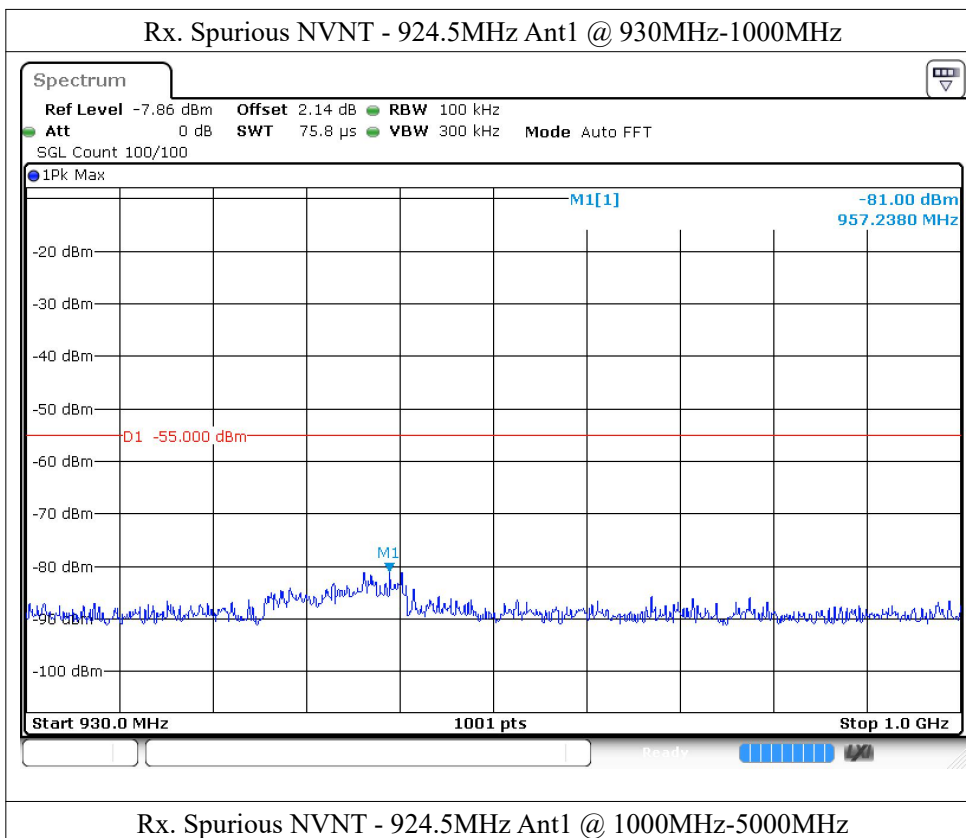
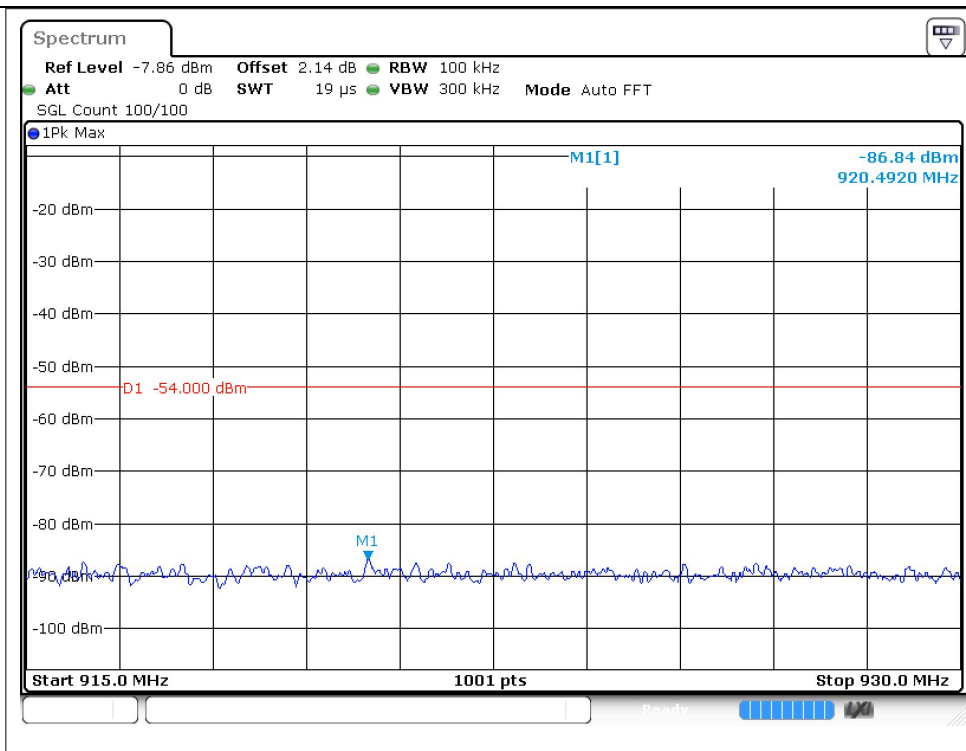
SRD_4MHz_Nss1 ,(13.5Mbps)_1RX

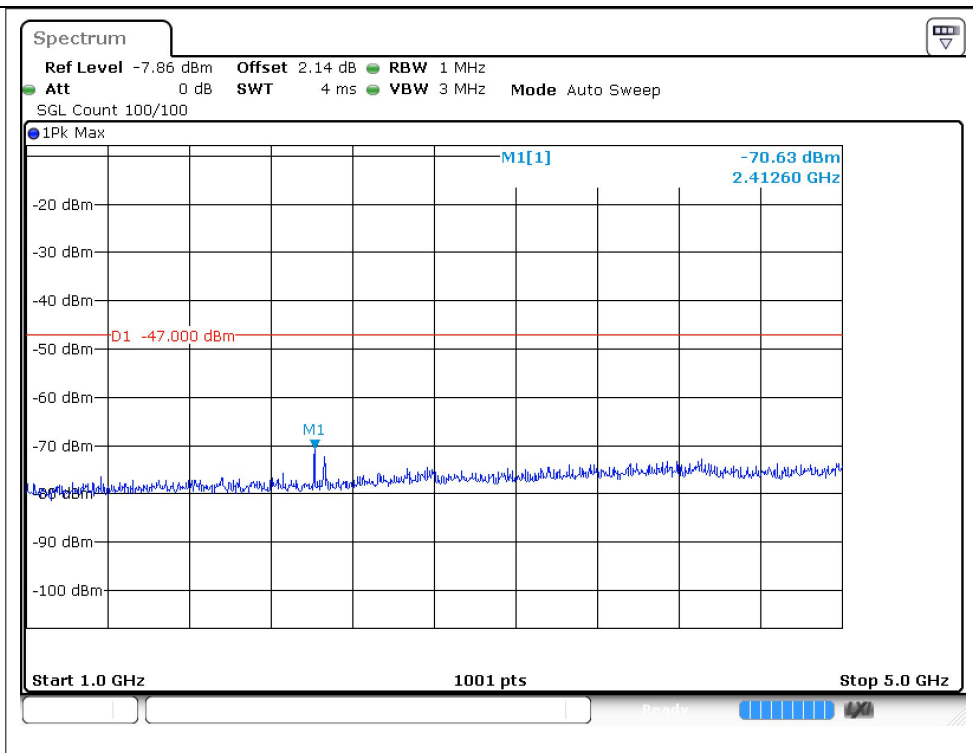
Condition	Frequency (MHz)	Range (MHz)	Emission Frequency (MHz)	Max Value (dBm)	Limit Start(dBm)	Limit Stop (dBm)	Verdict
NVNT	924.5	30-710	149.9	-84.37	-54	-54	Pass
NVNT	924.5	710-900	716.74	-83.65	-55	-55	Pass
NVNT	924.5	900-915	914.45	-86.03	-55	-55	Pass
NVNT	924.5	915-930	920.49	-86.84	-54	-54	Pass
NVNT	924.5	930-1000	957.24	-81	-55	-55	Pass
NVNT	924.5	1000-5000	2412.6	-70.63	-47	-47	Pass
LVNT	924.5	30-710	149.59	-84.40	-54	-54	Pass
LVNT	924.5	710-900	716.69	-84.47	-55	-55	Pass
LVNT	924.5	900-915	914.19	-86.85	-55	-55	Pass
LVNT	924.5	915-930	919.57	-87.20	-54	-54	Pass
LVNT	924.5	930-1000	956.43	-81.46	-55	-55	Pass
LVNT	924.5	1000-5000	2412.51	-71.19	-47	-47	Pass
HVNT	924.5	30-710	149.45	-85.06	-54	-54	Pass
HVNT	924.5	710-900	716.73	-83.97	-55	-55	Pass
HVNT	924.5	900-915	914.04	-86.25	-55	-55	Pass
HVNT	924.5	915-930	920.10	-87.06	-54	-54	Pass
HVNT	924.5	930-1000	957.10	-81.41	-55	-55	Pass
HVNT	924.5	1000-5000	2412.03	-70.91	-47	-47	Pass

NV









6.6 Transmission Time Control

6.6.1 Limits of Transmission Time Control

Frequency (MHz)	Total transmission time	Transmission Time Restriction	Carrier Sensing Function *1
920.5 ~ 923.5MHz	N/A	Transmission ON : within 4 sec. Transmission OFF : over 0.05 sec. *When retransmitting within 4 sec. after discharging a radio wave first, it can transmit without a suspend time.(It restricts to the retransmission within applicable time by the transmission after performing the carrier sense for less than 128 μs or more.)	Feeder input point : *Antenna power *20mW or less : -80dBm *Antenna power *more than 20mW : -80dBm - (antenna power XXmW - 20mW) Receive bandwidth : 200 x n kHz Judgment time : 5ms or more
920.5 ~ 928.1MHz	360sec. / hour or less (except Acknowledge)	Transmission ON : within 0.4 sec. Transmission OFF : over 0.002 sec. Transmission ON : within 0.006 sec. Transmission OFF : No requirement	Feeder input point : *Antenna power *20mW or less : -80dBm *Antenna power *more than 20mW : -80dBm - (antenna power XXmW - 20mW) Receive bandwidth : 200 x n kHz Judgment time : 128μs or more
915.9 ~ 928.1MHz	3.6sec. / hour or less	Transmission ON : within 0.1 sec. Transmission OFF : over 0.1 sec. *When retransmitting within 0.1sec. after discharging a radio wave first, it can transmit without a suspend time.(It restricts to the retransmission within applicable time.)	No requirement

*1 "Carrier sensing function" is unnecessary for the following transmissions.

* Acknowledge (A response to a "request from another radio equipment" with a transmission starts within 0.002 seconds and transmission time within 0.005 seconds(when 1 unit channel use : 0.05 seconds).)

"request from another radio equipment" : limited to when carrier sensing function was performed on the radio channel to transmit.

6.6.2 Test Procedure

(1) Set the standard signal generator as follow:

Carrier frequency: Center of reception frequency band of the EUT

Output level: -80 dBm.

(2) Set the spectrum analyzer as follow:

Carrier frequency: Center of reception frequency band to be used

Frequency sweep width: 3 MHz

Resolution bandwidth: 100 kHz

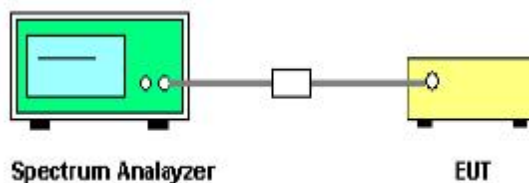
Video bandwidth: 300 kHz

Y-axis scale: 10 dB/Div

Trigger condition: Free run

Detection mode: Positive peak

6.6.3 TEST SETUP LAYOUT



6.6.4 Environmental Conditions

Temperature:	24 °C
Relative Humidity:	64%
ATM Pressure:	1010 mbar

6.6.5 TEST RESULT

SRD 1 MHz Nss1,(3Mbps)

TRANSMISSION TIME CONTROL				
Frequency (MHz)	Sending Duration		Pause Duration	
	Test Result (ms)	Limit	Test Result (ms)	Min.Limit
922	59.4	400ms or less	279.6	2ms
924	58.02	400ms or less	279.3	2ms
926	58.11	400ms or less	279.1	2ms

Frequency (MHz)	The sum of emission time per arbitrary one hour	
	Test Result (ms)	Limit
922	59400	360sec or less
924	59410	360sec or less
926	59405	360sec or less

SRD 2MHz Nss1,(6.5Mbps)

TRANSMISSION TIME CONTROL				
Frequency (MHz)	Sending Duration		Pause Duration	
	Test Result (ms)	Limit	Test Result (ms)	Min.Limit
922.1	59.29	400ms or less	278.64	2ms
924.3	57.53	400ms or less	278.90	2ms
926.5	57.95	400ms or less	279.04	2ms

SRD_4MHz_Nss1 ,(13.5Mbps)

TRANSMISSION TIME CONTROL				
Frequency (MHz)	Sending Duration		Pause Duration	
	Test Result (ms)	Limit	Test Result (ms)	Min.Limit
924.5	59.25	400ms or less	278.63	2ms

Frequency (MHz)	The sum of emission time per arbitrary one hour	
	Test Result (ms)	Limit
924.5	59398	360sec or less

Note: 1. The number of launches declared by the client per hour does not exceed 1000.
2. All condition has been tested, list the worst case in this item.

6.7 Assigned frequency or designated frequency

6.7.1 LIMIT

It shall be within 20×10^{-6} .

6.7.2 Test Procedure

(1) Test Conditions: Spectrum Analyzer is used for measurement.

Centre Frequency: The centre frequency of the channel under test.

Resolution BW: 10 kHz.

Video BW: 30 kHz.

Span: 1 MHz.

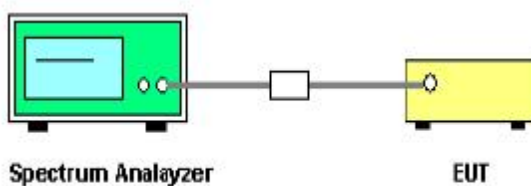
Sweep Time: Auto

Detector mode: Positive peak

Indication mode: Max hold

(2) When the trace is complete, find the peak value of the power envelope and record the frequency.

6.7.3 TEST SETUP LAYOUT



6.7.4 Environmental Conditions

Temperature:	22 °C
Relative Humidity:	64%
ATM Pressure:	1010 mbar

6.7.5 TEST RESULT

SRD 1 MHz Nss1,(3Mbps)

Test Voltage	Test Frequency (MHz)	Measured Frequency (MHz)	Frequency Error(ppm)	Limit (ppm)	P/F
NV	922	922.01	-10.85	±20	PASS
NV	924	924.01	-10.82	±20	PASS
NV	926	926.01	-10.80	±20	PASS
LV	922	921.99	10.85	±20	PASS
LV	924	923.99	10.82	±20	PASS
LV	926	926.01	-10.80	±20	PASS
HV	922	922.01	-10.85	±20	PASS
HV	924	924.01	-10.82	±20	PASS
HV	926	925.99	10.80	±20	PASS

SRD 2MHz Nss1,(6.5Mbps)

Test Voltage	Test Frequency (MHz)	Measured Frequency (MHz)	Frequency Error(ppm)	Limit (ppm)	P/F
NV	922.1	922.11	-10.84	±20	PASS
NV	924.3	924.31	-10.82	±20	PASS
NV	926.5	926.51	-10.79	±20	PASS
LV	922.1	922.09	10.84	±20	PASS
LV	924.3	924.31	-10.82	±20	PASS
LV	926.5	926.49	10.79	±20	PASS
HV	922.1	922.09	10.84	±20	PASS
HV	924.3	924.31	-10.82	±20	PASS
HV	926.5	926.51	-10.79	±20	PASS

SRD_4MHz_Nss1 ,(13.5Mbps)

Test Voltage	Test Frequency (MHz)	Measured Frequency (MHz)	Frequency Error(ppm)	Limit (ppm)	P/F
NV	924.5	924.51	-10.82	±20	PASS
LV	924.5	924.49	10.82	±20	PASS
HV	924.5	924.51	-10.82	±20	PASS

6.8 Adjacent Channel Leakage Power

6.8.1 Limit

FREQUENCIES (MHz) 915.9MHz TO 916.9MHz	LIMIT
RADIO CHANNEL BOTH ENDS	-20dBm
ADJACENT UNIT CHANNEL	-26dBm

FREQUENCIES (MHz) 920.5MHz TO 922.3MHz	LIMIT
RADIO CHANNEL BOTH ENDS	-7dBm
ADJACENT UNIT CHANNEL	-15dBm or less (*)

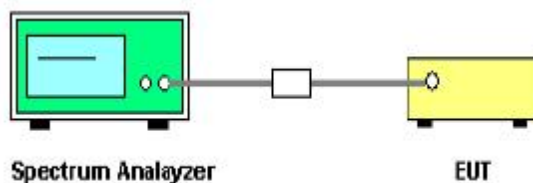
*If antenna power is reduced by 1mW or less, each adjacent channel leakage power is 26dBm or less.

FREQUENCIES (MHz) 922.3MHz TO 928.1MHz (Antenna power is 1mW or less)	LIMIT
ADJACENT UNIT CHANNEL	-26dBm or less

FREQUENCIES (MHz) 922.3MHz TO 928.1MHz (Antenna power is more than 1mW and 20mW or less.)	LIMIT
ADJACENT UNIT CHANNEL	-15dBm or less

FREQUENCIES (MHz) 928.1MHz TO 929.7MHz	LIMIT
ADJACENT UNIT CHANNEL	-26dBm or less

6.8.2 TEST SETUP LAYOUT



6.8.3 Test procedure

(1) Connected the EUT's antenna port to the Spectrum Analyzer by suitable attenuator, set the Spectrum Analyzer as below:

Centre Frequency: The centre frequency of the channel under test.

Resolution BW: 1 kHz.

Video BW: 1 kHz.

Span: 1 MHz.

Detector: Peak.

Trace Mode: Max Hold.

(2) When the trace is complete, search the frequency of peak power

Note: The cable loss and attenuator loss have been put into spectrum analyzer as amplitude offset.

6.8.4 Test result

Note: The report shows the worst test data for NV testing.

VOLTAGE	Channel	Adjacent unit channel	Measured	Limit
			dBm	dBm
NV	922MHz	Lower adjacent unit channel	-29.03	-15
		Upper adjacent unit channel	-28.02	-15
LV	922MHz	Lower adjacent unit channel	-29.01	-15
		Upper adjacent unit channel	-28.01	-15
HV	922MHz	Lower adjacent unit channel	-29.01	-15
		Upper adjacent unit channel	-28.00	-15
Conclusion: PASS				

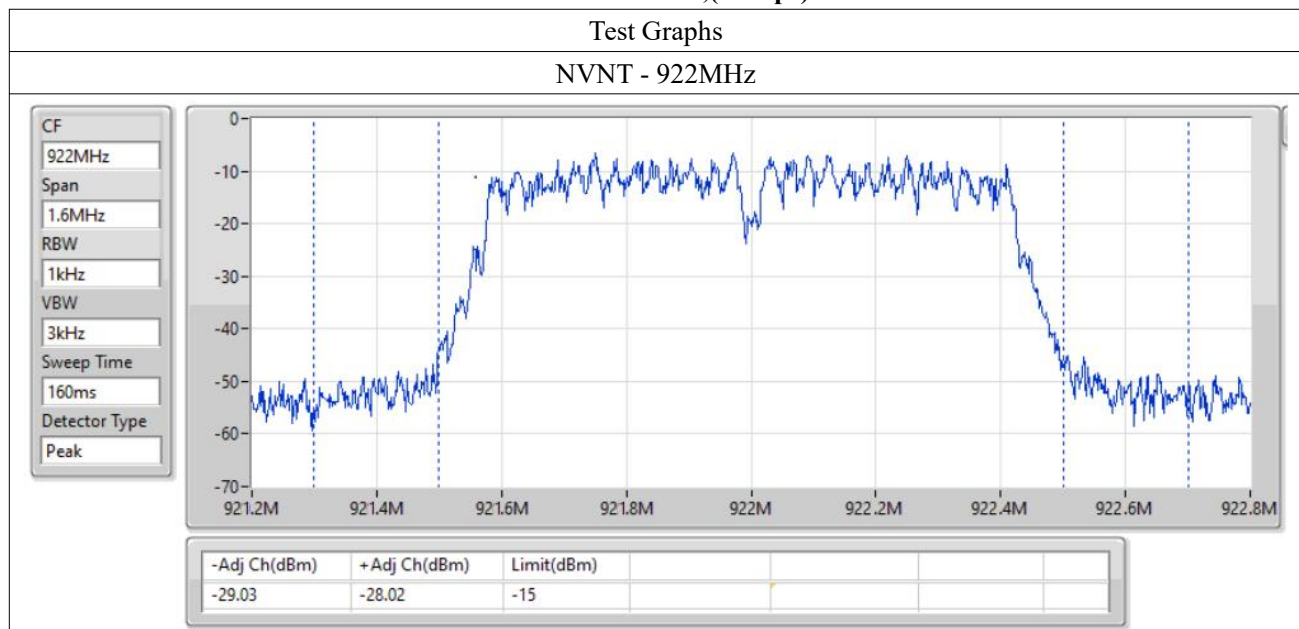
VOLTAGE	Channel	Adjacent unit channel	Measured	Limit
			dBm	dBm
NV	924MHz	Lower adjacent unit channel	-28.81	-15
		Upper adjacent	-28.52	-15

		unit channel		
LV	924MHz	Lower adjacent unit channel	-28.76	-15
		Upper adjacent unit channel	-28.50	-15
HV	924MHz	Lower adjacent unit channel	-28.76	-15
		Upper adjacent unit channel	-28.50	-15
Conclusion: PASS				

VOLTAGE	Channel	Adjacent unit channel	Measured	Limit
			dBm	dBm
Normal	926MHz	Lower adjacent unit channel	-29.40	-15
		Upper adjacent unit channel	-28.80	-15
Conclusion: PASS				

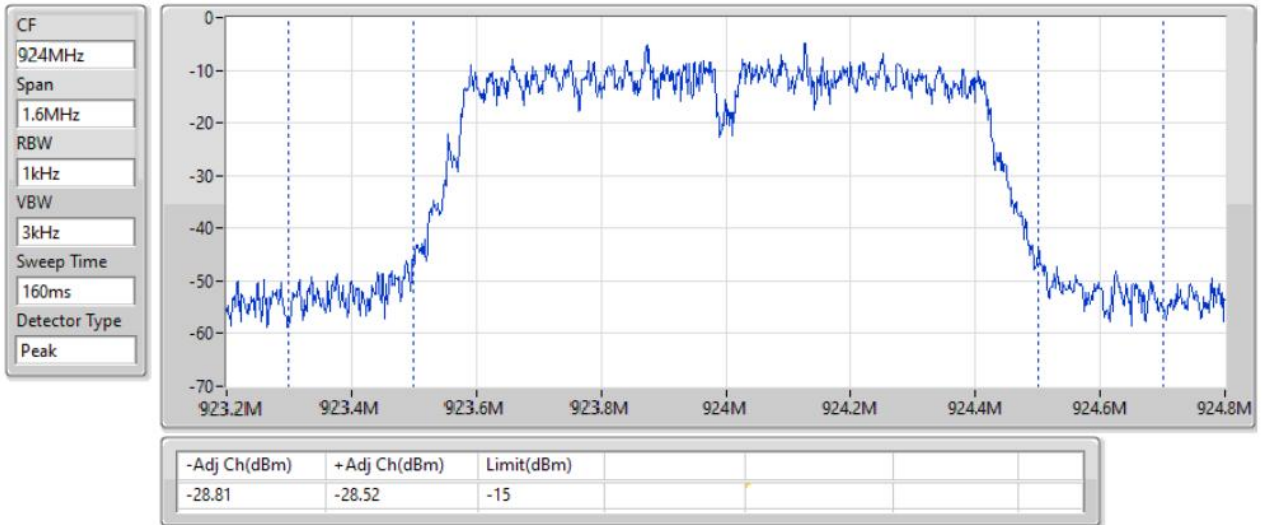
NV

SRD 1MHz Nss1,(3Mbps)



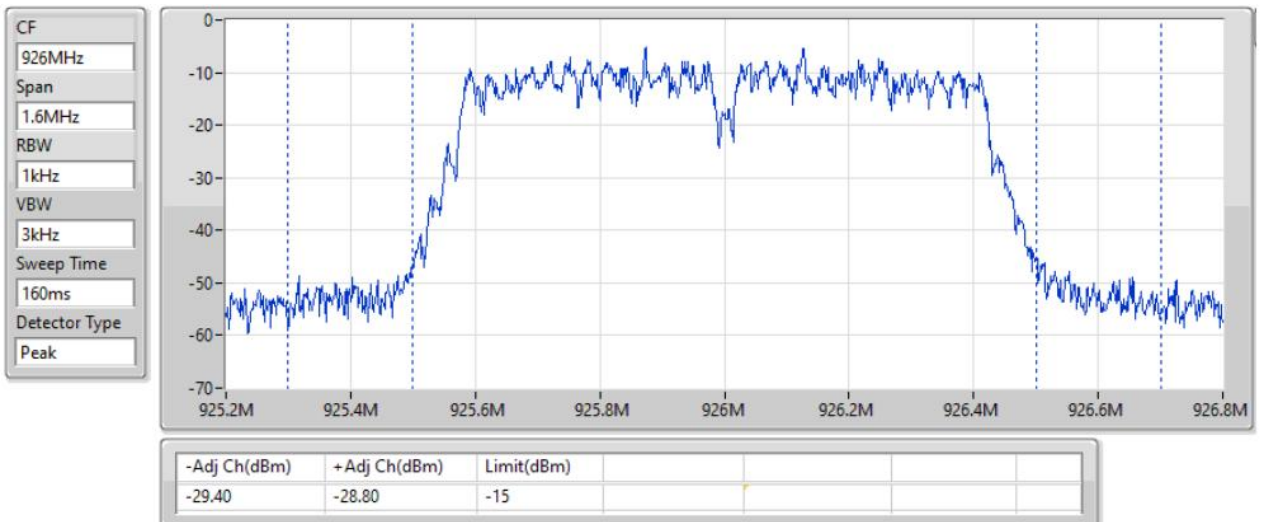
Test Graphs

NVNT - 924MHz



Test Graphs

NVNT - 926MHz



SRD 2MHz Nss1,(6.5Mbps)

VOLTAGE	Channel	Adjacent unit channel	Measured	Limit
			dBm	dBm
NV	922.1	Lower adjacent unit channel	-31.64	-15
		Upper adjacent unit channel	-30.72	-15
LV	922.1	Lower adjacent unit channel	-31.60	-15
		Upper adjacent unit channel	-30.71	-15
HV	922.1	Lower adjacent unit channel	-31.62	-15
		Upper adjacent unit channel	-30.70	-15
Conclusion: PASS				

VOLTAGE	Channel	Adjacent unit channel	Measured	Limit
			dBm	dBm
NV	924.3	Lower adjacent unit channel	-31.71	-15
		Upper adjacent unit channel	-31.21	-15
LV	924.3	Lower adjacent unit channel	-31.66	-15
		Upper adjacent unit channel	-31.20	-15
HV	924.3	Lower adjacent unit channel	-31.70	-15
		Upper adjacent unit channel	-31.19	-15
Conclusion: PASS				

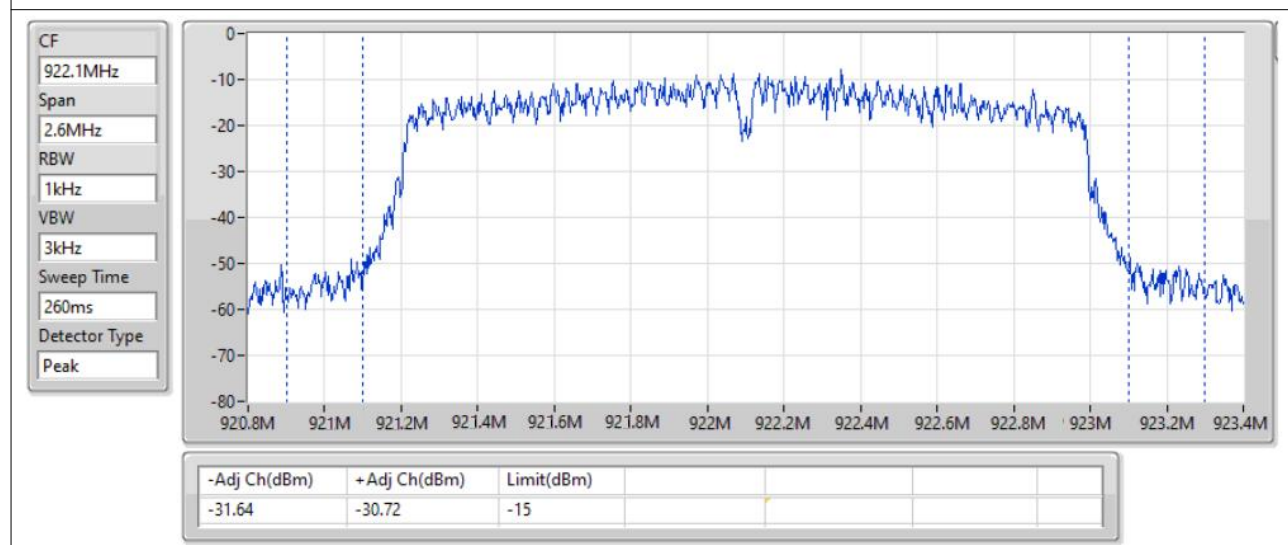
VOLTAGE	Channel	Adjacent unit channel	Measured	Limit
			dBm	dBm
NV	926.5	Lower adjacent unit channel	-31.5	-15
		Upper adjacent unit channel	-32.35	-15
LV	926.5	Lower adjacent	-30.35	-15

		unit channel		
		Upper adjacent unit channel	-31.91	-15
HV	926.5	Lower adjacent unit channel	-31.35	-15
		Upper adjacent unit channel	-32.37	-15
Conclusion: PASS				

NV

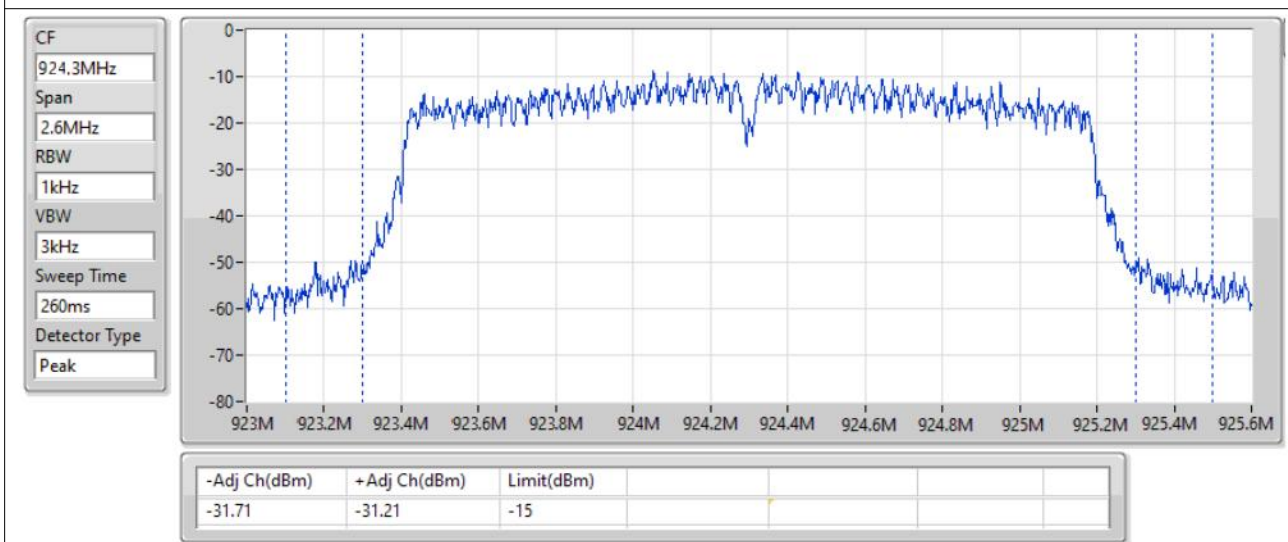
Test Graphs

NVNT - 922.1MHz



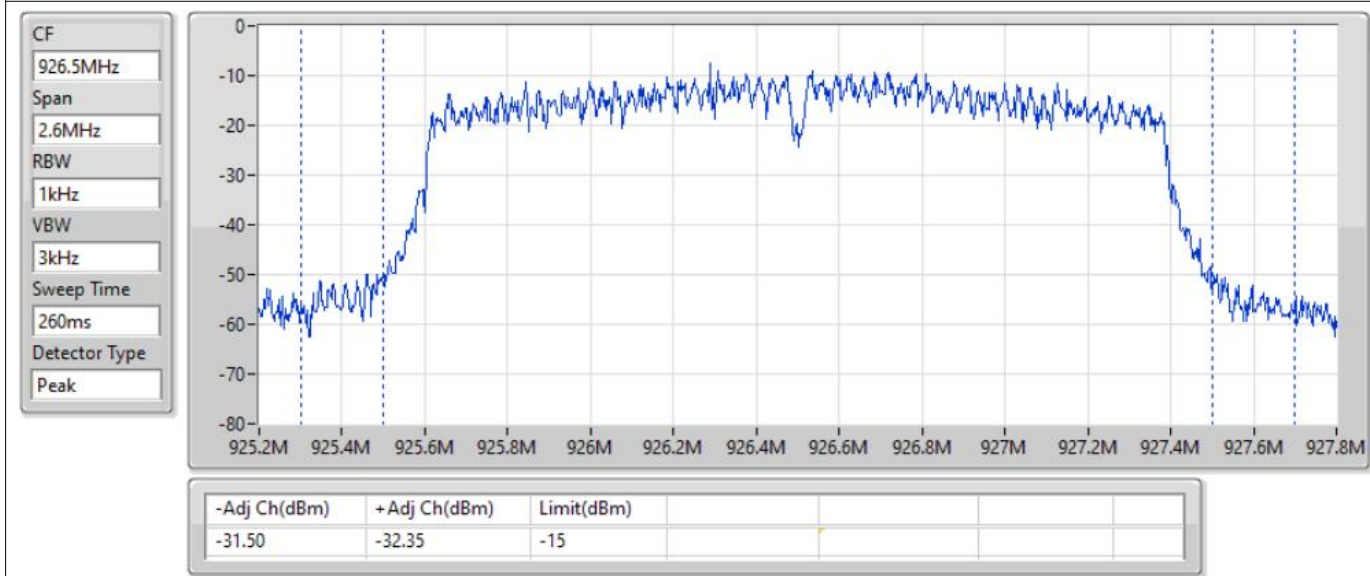
Test Graphs

NVNT - 924.3MHz



Test Graphs

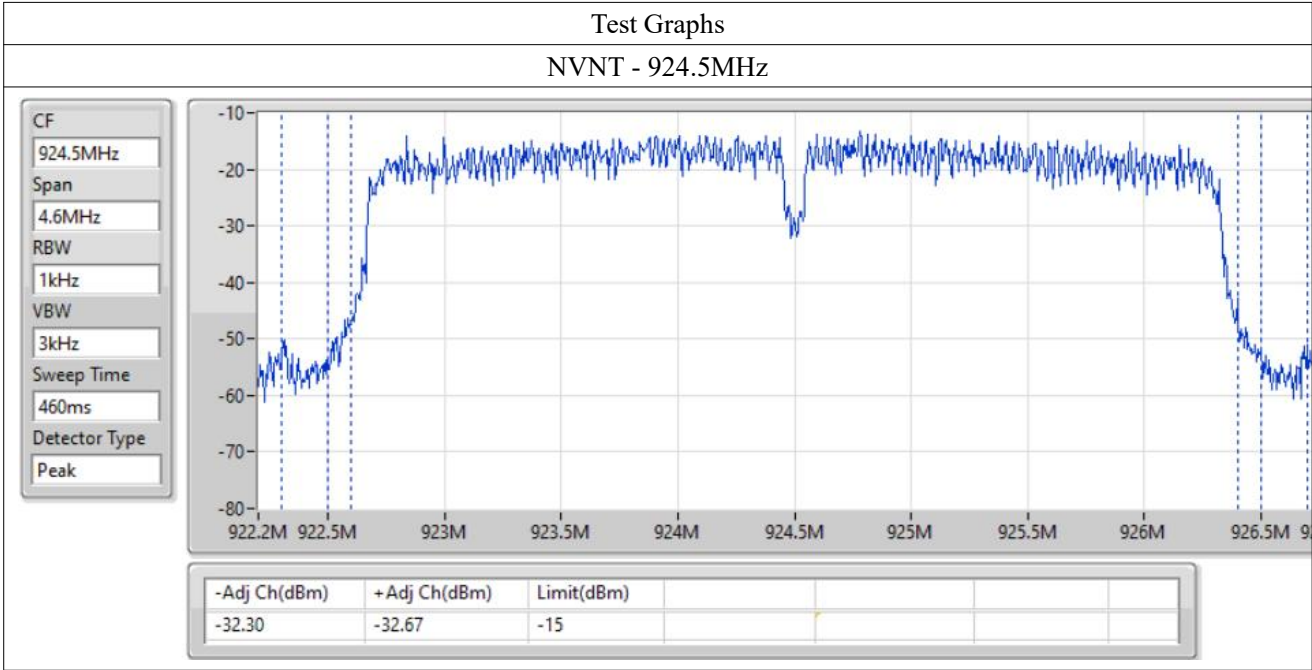
NVNT - 926.5MHz



SRD_4MHz_Nss1 ,(13.5Mbps)

VOLTAGE	Channel	Adjacent unit channel	Measured	Limit
			dBm	dBm
NV	924.5MHz	Lower adjacent unit channel	-32.30	-15
		Upper adjacent unit channel	-32.67	-15
LV	924.5MHz	Lower adjacent unit channel	-32.25	-15
		Upper adjacent unit channel	-32.53	-15
HV	924.5MHz	Lower adjacent unit channel	-32.28	-15
		Upper adjacent	-32.60	-15

		unit channel		
Conclusion: PASS				

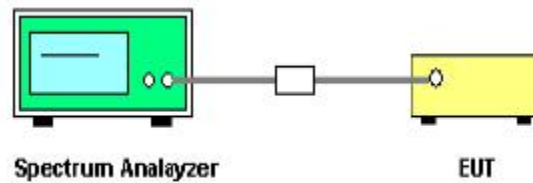


6.9 Interference prevention function

6.9.1 Limits of Interference Prevention Function

The radio equipment shall automatically transmit/receive identification codes.

6.9.2 Test Setup



6.9.3 Test Results

Environmental Conditions	
Link Mode	Test Result
Mac address	78:87:3A:1C:C6:FB
DR0-CF12/200kHz	PASS

6.10 EIRP Power

6.10.1 Limit

Limit	$\leq 16\text{dBm}$
Tolerance	+20%, -80%

6.10.2 Test Setup



6.10.3 Test Results

SRD 1 MHz Nss1,(3Mbps)1TX

Mode	Frequency MHz	Measured		Stated power	Limit
		dBm	mW	mW	mW
Tx mode	922	8.81	7.603	8	39.81
Tx mode	924	8.85	7.674	8	39.81
Tx mode	926	8.97	7.889	8	39.81
Conclusion: PASS					

SRD 2MHz Nss1,(6.5Mbps)1TX

Mode	Frequency MHz	Measured		Stated power	Limit
		dBm	mW	mW	mW
Tx mode	922.1	7.98	6.281	7	39.81
Tx mode	924.3	8.03	6.353	7	39.81
Tx mode	926.5	8.06	6.397	7	39.81
Conclusion: PASS					

SRD_4MHz_Nss1 ,(13.5Mbps)_1TX

Mode	Frequency MHz	Measured		Stated power	Limit
		dBm	mW	mW	mW
Tx mode	924.5	6.23	4.198	5	39.81
Conclusion: PASS					

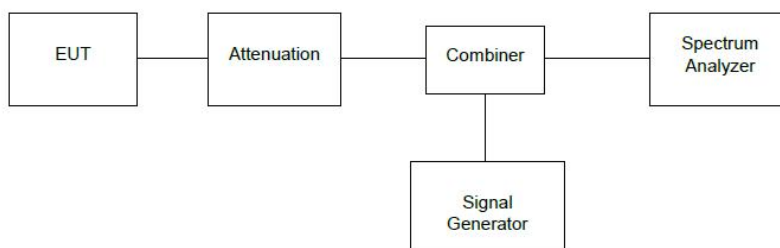
Note: All condition has been tested, list the worst case in this item.

6.11 Carrier Sense

6.11.1 Limits of Carrier Sense

Application	Antenna Power	Applied CH number	Unit CH bandwidth	CH used in a bundle	Carrier sense time
	1mW or less	1-5	200kHz	1~5 ch	None
		62-77	100kHz	1~5 ch	
V	20mW or less	24-38	200kHz	1~5 ch	5ms or more
V		33-61	200kHz	1ch	128us or more
				2ch	
				3~5ch	

6.11.2 Test Setup



6.11.3 Test Results

Carrier Sense		
Frequency (MHz)	Can transmit condition	Can not transmit condition
	EUT can transmit after 5ms that pulse signal in the off time	EUT cannot transmit anyway
922	PASS	PASS

Carrier Sense		
Frequency (MHz)	Can transmit condition	Can not transmit condition
	EUT can transmit after 128us that pulse signal in the off time	EUT cannot transmit anyway
922	PASS	PASS

Note: carrier sense level is -80dBm.

7 Photographs

7.1 EUT Constructional Details

Refer to Appendix EUT Photo.

****End of report****